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Abstract

Rydberg atoms held in optical tweezers feature electronic and vibrational degrees of freedom which can be brought into interaction in a controllable way. Therefore, these systems enable the investigation of phenomena, similar to those studied in molecular physics, but on exaggerated length and time scales. We exploit these interactions to investigate the possibility of building a quantum simulation framework for molecular systems. In our framework, we find that beyond certain coupling strengths, the vibrational motion becomes unstable, and we derive the critical values. Moreover, we derive quantum corrections to the ground state energy of a micrometer-sized diatomic and triangular atom array, which are not captured by the Born-Oppenheimer approximation. We investigate the physical implications of these corrections and find that they are linked to phonon-squeezed states. We further establish their impact on a regime where the triangular array is in a superposition of distorted triangular shapes (Jahn-Teller regime). Finally, we propose a protocol to prepare molecular states whose structure is strongly affected by electron-phonon coupling. Our results show that trapped Rydberg atom arrays indeed offer a versatile platform for the study of quantum phenomena that link to molecular physics.

Introduction

Molecular systems have continuously been the focus of intense research across a plethora of disciplines, including physics, chemistry, biology and material science, Ref. [1–4]. Because molecular phenomena are fundamental building blocks of chemical reactions, their study is crucial for addressing a wide range of topics, ranging from the origins of life on earth to significant technological advances. Since the advent of quantum mechanics in the mid 1920’s, facilitated by Schrödinger’s pioneering work, Ref. [5], it became clear that exact solutions and numerical methods on classical computers become increasingly challenging and ultimately infeasible, as the complexity of the molecular structure increases. First proposed by Yuri Manin and Richard Feynman in the early 1980’s, the notion of employing a quantum mechanical system itself to simulate another quantum system, offers a solution to overcome limitations imposed by classical methods, Ref. [6, 7]. Experimental implementations of such a quantum simulator for molecular systems may have appeared distant in those early days, however, significant advances in controlling ultracold atoms has turned this approach into tangible reality. Apart from ultracold atoms, many other platforms are explored for quantum computing and simulation tasks, Ref. [8–10]. Today, optically confined atom arrays are successfully realized in practically arbitrary, 1-d., 2-d. and 3-d. configurations, Ref. [11–13], assembled atom by atom. Due to these developments, neutral atoms are a prominent choice for quantum computing and simulation frameworks today, Ref. [14–26]. Quantum technologies build upon neutral atom arrays, typically rely on strong interatomic interactions of the atomic dipole moments. Those interactions are distance dependent. Moreover, the trapping potential experienced by each atom is state-dependent. As a result, internal (electronic) and external (motional) degrees of freedom are strongly coupled, such that the atoms move within the optical tweezers, Ref. [27]. In the realm of quantum computing, atomic motion in the optical traps is a challenge to be faced, Ref. [28–32]. This motion, however, is not solely to be viewed as a detrimental effect. Mechanical forces emergent from coupling of internal electronic and external motional degrees of freedom are exploited for a variety of applications, Ref. [26, 33, 34]. In particular, vibrational signatures, similar to those found in molecular systems, are accessible on exaggerated length and time scales in the order of μm and μs , Ref. [35–37].

In this thesis, we aim to explore the idea of constructing a quantum simulator for molecular phenomena, utilizing trapped atom arrays. We present a general Hamiltonian for our simulation framework, capable of simulating vibrational dynamics for virtually any conceivable molecular shape. To do so, the position dependence of strong dipole-dipole interactions, emergent between Rydberg atoms, Ref. [38], is used to introduce coupling between electronic and motional degrees of freedom. From our Hamiltonian we deduce that only small arrays may be treated on a classical computer. Our findings underline the relevance for such a quantum simulator as well as its experimental feasibility and excellent capacity for control, with state-of-the-art technologies. Subsequently, we focus on three simple array shapes – a dumbbell, a triangle, and a tetrahedron. These, generally, are still complex settings. We simplify them with constraints imposed via Rydberg facilitation, Ref. [39]. These constraints allow to reduce the dimension of the electronic Hilbert space. Applying our Hamiltonian description, which is closely related to Ref. [37], we report interesting vibration signatures and structural instabilities in our artificial molecule, introduced by electron-phonon coupling. We calculate critical coupling strengths using a multimode Bogoliubov transformation and give analytical expressions, illustrating the critical behaviour, for the ground state energy in certain parameter lim-

its. Moreover, we discuss experimental implications and conduct a numerical analysis, unveiling how the unstable regime manifests itself in the time evolution of vibrations in the trapped atom arrays. Our discoveries extend the work Ref. [37], in that we consider additional contributions to the dipole-dipole interaction Hamiltonian. These specific interactions are responsible for the instabilities and interesting vibration patterns. Our considerations illustrate the importance of these additional contributions. In the broader context of our work, these results demonstrate that the simulator is capable of producing non-trivial phenomena. Hence, we conclude that the system employed as the simulation platform is not constructed in a too simplified (compared to molecules in nature) manner.

In an attempt to understand the instabilities origin, we perform a Born-Oppenheimer approximation (BOA), Ref. [1] and examine the Born-Oppenheimer ground state potential energy surface (GSBOES). This approach is motivated by Ref. [37], where the GSBOES was employed to gain insight on a regime in which the Jahn-Teller effect, Ref. [40, 41], is evident. In essence, this effect from molecular physics states that polyatomic (non-linear) molecules with degenerate electronic ground state energy distort the molecules spatial shape in order to lower the ground state energy. By this process, spontaneous symmetry breaking is introduced and the degeneracy is lifted. Given that our simulator hosts additional interactions compared to Ref. [37], we are interested in the implications instabilities, caused by these interactions, impose on the Jahn-Teller regime. With the help of the GSBOES, we provide an intuitive picture for the connection of these phenomena. Subsequently, attention is directed to analytical expressions for the ground state energy in the BOA, in special parameter regimes. We analytically identify quantum corrections to the ground state energy obtained within the BOA through comparison to the full quantum results. These corrections we trace back to squeezed phonon wave functions in the motional ground state. To illustrate their practical impact, we investigate a structural transition into and out of the Jahn-Teller regime employing exact numerical methods. In the BOA we identify a critical point for the transition, opposed to the quantum result where the transition is not abrupt. Our studies reveal that quantum corrections, similar to the ones found analytically, are responsible for the ‘washed-out’ nature of the full result. Our findings show that the simulator hosts practically relevant quantum effects and demonstrate their physical implications. Moreover, comparison to the quantum results offers a neat way to observe breakdown of the BOA for too slow electronic dynamics. In the wider scope of this project, we conclude that our simulation framework is able to capture qualitatively important quantum phenomena and is well described by a BOA in the parameter regimes where it should be. The latter point is of course important to verify, for a simulator intended to mimic molecular phenomena.

Finally, we propose a protocol consisting of laser pulses and adiabatic ramps, enabling preparation of practically relevant molecular states. Exemplary, we focus on a triangular array. We show how the protocol could be used to enable measurement of a purely electronic order parameter for the Jahn-Teller regime. We close from this result that the simulator permits extraction of physically meaningful observables.

This thesis is structured into four chapters. In the first chapter, we develop the general framework for our simulator, taking advantage of ideas from Ref. [27, 37, 42, 43]. Goal of this chapter is to motivate and derive the Hamiltonian which is simulated. In the second chapter we focus on arrays resembling a dumbbell, a triangle, and a tetrahedron. First, we constrain the electronic state manifold using Rydberg facilitation for all three arrays. Further, we rewrite the Hamiltonian in vibrational modes tailored to the intrinsic symmetries of the arrays. The eigenmodes of the arrays are employed to gain insight on the possible vibration patterns. Following this discussion, we find the multimode Bogoliubov transformation and make use of it to discover the instabilities, extract critical parameters and calculate analytical expressions for the ground state energy. In the third chapter the BOA is

introduced; here we lay out where approximations are made exemplarily for the dumbbell. We calculate the GSBOES, identify analytical expressions for quantum corrections to the ground state energy in the BOA and clarify what we refer to with this notion. Following that, we move on to the triangular array. Here we use the GSBOES to give an intuitive explanation for the unstable regime and its connection to the Jahn-Teller effect. Finally, we address the significance of the quantum corrections through comparison of the ground state energy in the BOA and full quantum picture.

In the final chapter, a step-by-step derivation for the ground state preparation protocol is laid out. The protocol is split into an entirely electronic and vibronic part. Lastly, the protocol is tested by simulating the laser pulse sequence and adiabatic ramps.

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I. General Setting: Hamiltonian of an Artificial Molecule

This chapter's purpose is to set the stage for the remainder of the thesis. We aim at making use of atoms trapped in optical tweezers to model a system exhibiting dynamics similar to vibrations in molecular physics. The goal is to arrive at a general Hamiltonian for our quantum simulation framework, which is feasible to work with experimentally, but still implements the desired dynamics. To this end, we discuss a variety of phenomena for motivating why these types of composite quantum systems are a versatile choice for our intentions. However, it should be made clear already at this point that the following derivation is not focused on giving an exhaustive discussion on all possible aspects in full generality, as this is not the goal of this thesis and a rather challenging (maybe impossible) endeavour. Before getting started, it should be noted that Ref. [27, 37, 43] and the documentation of the *Python* package Ref. [42], were used as inspiration for this chapter.

We structure this chapter into three parts: First, we elaborate on the electronic part of the Hamiltonian. Followed by this is a section on dipole-dipole interactions emergent for spatially close ($\sim 1\mu\text{m}$) Rydberg atoms. These interactions then lead to the desired vibrational dynamics, reminiscent of molecular motion. Finally, we combine the electronic and vibrational dynamics in a composite quantum system which we then use to pinpoint our simulation framework.

I.1. Atomic Hamiltonian

The goal of this section is to construct the purely electronic part of our simulator. We start with a brief introduction of Rydberg atoms. Subsequently, we focus on a model Hamiltonian for an array of trapped atoms in the limit of vanishing atom-atom interactions, i.e. where all atom pairs are far separated.

I.1.1. Rydberg Atoms

Under the term *Rydberg atom* we understand an atom excited to an electronic state of large principal quantum number, referred to as *Rydberg state*, Ref. [38, 42] (both references serve as inspiration for this short introduction). We will focus on single valence electron Rydberg atoms, as this type will be the object of interest. Thus, we are primarily interested in alkaline Rydberg atoms, i.e. Li, Na, K, Rb, Cs. Rydberg atoms inherit many properties of the Hydrogen atom since states of large principal quantum number come with large radial electron-nucleus separation. The valence electron far from the electron cloud near the nucleus, effectively experiences a single positive charge, as the Z positive charged nucleons are screened by the $Z - 1$ electrons in the vicinity of the core. Therefore, states of the Rydberg electron are naturally similar to Hydrogen eigenstates. Thus, we shall consider the Hamiltonian of the Hydrogen atom, written as

$$H_H = \frac{\mathbf{p}_p^2}{2m_p} + \frac{\mathbf{p}_e^2}{2m_e} + \frac{1}{k_C} \frac{(-e)e}{\|\mathbf{r}_e - \mathbf{r}_p\|}, \quad (\text{I.1})$$

where we have introduced Coulomb's constant $k_C = 4\pi\epsilon_0$, the electron's and proton's momentum $\mathbf{p}_e$ and $\mathbf{p}_p$, the corresponding position vectors $\mathbf{r}_e$ and $\mathbf{r}_p$ and the elementary charge $e > 0$. It is

beneficial to transform to relative and centre-of-mass (CMS) co-ordinates via

$$\mathbf{r}_{\text{rel}} = -\mathbf{r}_p + \mathbf{r}_e, \quad \mathbf{r}_{\text{cms}} = \frac{m_p}{m_p + m_e} \mathbf{r}_p + \frac{m_e}{m_p + m_e} \mathbf{r}_e, \quad (\text{I.2})$$

$$\mathbf{p}_{\text{rel}} = -\frac{m_p}{m_p + m_e} \mathbf{p}_p + \frac{m_e}{m_p + m_e} \mathbf{p}_e, \quad \mathbf{p}_{\text{cms}} = \mathbf{p}_p + \mathbf{p}_e, \quad (\text{I.3})$$

yielding the well known form of the Hydrogen Hamiltonian,

$$H_{\text{H}} = \frac{\mathbf{p}_{\text{cms}}^2}{2(m_p + m_e)} + \frac{\mathbf{p}_{\text{rel}}^2}{2m} - \frac{1}{k_{\text{C}}} \frac{e^2}{\|\mathbf{r}_{\text{rel}}\|}, \quad (\text{I.4})$$

with the reduced mass $m = m_p m_e / (m_p + m_e) \approx m_e \approx 500 \text{ keV}$ (natural units). As $\mathbf{p}_{\text{cms}}$ is decoupled from the relative atomic degrees of freedom, we transform into the CMS frame such that only the relative dynamics remain. For simplicity, we drop the subscript ‘rel’ from this point onward. Upon canonical quantization of the remaining contributions,

$$\mathbf{r} \mapsto \hat{\mathbf{r}}, \quad \hat{\mathbf{r}}(\psi)(\mathbf{r}) = \psi(\mathbf{r})\mathbf{r}; \quad \mathbf{p} \mapsto \hat{\mathbf{p}}, \quad \hat{\mathbf{p}}(\psi)(\mathbf{r}) = -i\hbar(\nabla\psi)(\mathbf{r}), \quad (\text{I.5})$$

we obtain the Schrödinger equation for the Hydrogen atom in position space:

$$\left[i\hbar\partial_t + \frac{\hbar^2}{2m}\nabla^2 + \frac{e^2}{k_{\text{C}}} \frac{1}{r} \right] \psi(\mathbf{r}, t) = 0, \quad r \equiv \|\mathbf{r}\| \quad (\text{I.6})$$

Because this is a linear differential equation in time, we may solve it using the eigenfunctions $\psi_{nlm}(\mathbf{r})$ of the Hamiltonian, defined by

$$\left[\frac{\hbar^2}{2m}\nabla^2 + \frac{e^2}{k_{\text{C}}} \frac{1}{r} \right] \psi_{nlm}(\mathbf{r}) = E_n \psi_{nlm}(\mathbf{r}), \quad (\text{I.7})$$

famously obtained first by Schrödinger himself, Ref. [5, 44]:

$$\psi_{nlm}(\theta, \varphi) = R_{nl}(r) Y_{lm}(\mathbf{r}) \quad (\text{I.8})$$

$$R_{nl}(r) = \sqrt{\left(\frac{2}{n\bar{a}_0} \right)^3 \frac{(n-l-1)!}{2n(n+l)!}} e^{-r/(n\bar{a}_0)} \left(\frac{2r}{n\bar{a}_0} \right)^l L_{n-l-1}^{2l+1}(2r/(n\bar{a}_0)) Y_{lm}(\theta, \varphi) \quad (\text{I.9})$$

$$Y_{lm}(\theta, \varphi) = (-1)^m \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} e^{im\varphi} P_{lj}(\cos(\theta)) \quad (\text{I.10})$$

$$n = 1, 2, 3, 4, 5, \dots \quad (\text{I.11})$$

$$l = 1, 2, \dots, n-1 \quad (\text{I.12})$$

$$m = -l, -(l-1), \dots, (l-1), l \quad (\text{I.13})$$

In the above expression, n denotes the principal quantum number, l the total orbital angular momentum, m angular momentum with respect to the quantization axis (z-axis in this case), L_k^α (Normalization: $L_0^{(\alpha)}(x) = 1, L_1^{(\alpha)}(x) = 1 + \alpha - x$) *generalized Laguerre polynomials*, Y_{lm} *spherical harmonics*, P_{ij} (Normalization: $P_{00} = 1$) *associated Legendre polynomials*, θ, φ the azimuthal and polar angle of $\mathbf{r}$ and $\bar{a}_0 = \hbar^2 k_{\text{C}} / (me^2)$ the *reduced Bohr radius*.

Making use of these eigenstates, we can compute the energies and radial matrix elements,

$$E_n = -\frac{R_{\text{H}}}{n^2}, \quad \langle r \rangle_{nl} = \frac{\bar{a}_0}{2} (3n^2 - l(l+1)). \quad (\text{I.14})$$

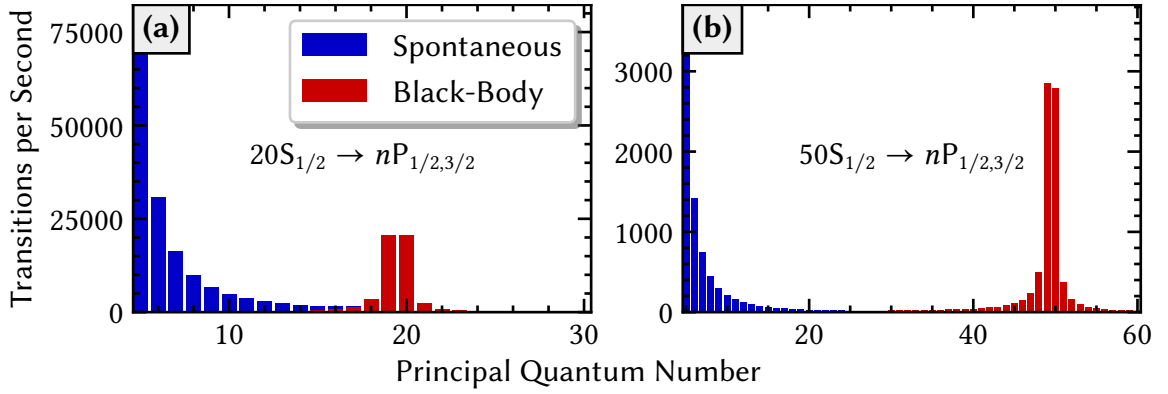


Figure I.1.: Radiative Decay. Radiative decay of ^{85}Rb fine structure Rydberg states to $nP_{1/2}$ and $nP_{3/2}$, for a variety of principal quantum numbers, starting from the ground state, $n = 5$. Initial states are (a) $20S_{1/2}$ and (b) $50S_{1/2}$. In blue, spontaneous emission rates are shown and rates resulting from black-body radiation in red. For data generation the *Alkaline Rydberg Calculator* (ARC) was employed, Ref. [42].

with the Rydberg constant for Hydrogen $R_H = 109.678 \text{ nm}^{-1}$. Note the typical quadratic scaling of the radial expectation values in the principal quantum number n , implying large dipole moments associated to Rydberg states.

If we want to describe an atom different from Hydrogen, e.g. one of the Alkaline atoms for our purposes, we must take electron-electron interactions, the finite core size and spin-orbit coupling into account. Solutions then, generally, become feasible only numerically, Ref. [42]. A different approach opposed to a full numerical treatment is *quantum defect theory*, which introduces a quantum defect δ_{nlj} to the principal quantum number n , where an effective principal quantum number $n^* = n - \delta_{nlj}$ is defined and the quantum number j describes the total angular momentum (electron spin and orbital angular momentum), Ref. [38, 42, 43, 45, 46]. Reference [47] provides an expression for the Rydberg energy wave functions within quantum defect theory, reading

$$R_{n^*l}(r) = \left(\frac{1}{\tilde{a}_0}\right)^{3/2} \frac{1}{\sqrt{n^{*2}\Gamma(n^* + l + 1)\Gamma(n^* - 1)}} W_{n^*, l + \frac{1}{2}}\left(\frac{2r}{n^* \tilde{a}_0}\right), \quad (\text{I.15})$$

where Γ is the Gamma-functions and W the *Whittaker-function* (introduced by Sir Edmund Taylor Whittaker). Angular dependence is still given by the unchanged spherical harmonics, because the effective core potential is assumed to be rotationally symmetric. With this wave function at hand, expectation values of observables for the Rydberg atom are computable, where δ_{nlj} are either obtained numerically or by spectroscopy experiments, Ref. [48–50]. The latter method, generally, yields more precise results, due to not introducing additional sources of errors, e.g. through the model description of the finite sized ionic core.

Another key aspect of Rydberg atoms, responsible for their application in quantum simulators, Ref. [51], are long radiative lifetimes of Rydberg states, allowing unitary evolution. Long radiative lifetimes of alkaline Rydberg atoms are a result of the weak Rydberg to ground state coupling, Ref. [38]. Radiative decay of the Rydberg state is mainly rooted in two mechanisms, namely zero-temperature spontaneous decay and black-body radiation, with average lifetimes τ_0 and τ_{bb} respectively. Combining both contributions yields the Rydberg to ground state transition rate

$$\Gamma_{\text{r} \rightarrow \text{g}} = \frac{1}{\tau_0} + \frac{1}{\tau_{\text{bb}}}. \quad (\text{I.16})$$

Analytical expression for these rates are stated and obtained for example in Ref. [52]. For illustration of the μs -lifetimes and the importance of both contributions, we plot the transition rates for a p-state in the spectrum of ^{85}Rb in Fig. I.1. We clearly observe the Rydberg state with higher principal quantum number to be longer lived. Moreover, it is clear that black-body radiation carries an important contribution. Summation over all rates yields the approximate inverse lifetime, such that we find $\tau \approx 4.6 \mu\text{s}$ for $20\text{S}_{1/2}$ and $\tau \approx 63 \mu\text{s}$ for $50\text{S}_{1/2}$.

With this brief discussion on Rydberg atom lifetimes we conclude this section and turn our attention to an array of trapped atoms.

I.1.2. Light-Matter Interaction of Trapped Atom Array

Focus shall now be directed to an array of $N \in \mathbb{N}$ optically trapped atoms. We start by modelling the atoms as two-level systems with a ground state $|g\rangle$ and an excited (Rydberg) state $|r\rangle$, where transitions are driven by a laser (*not* equal to the far off-resonant laser of the trap) with frequency ω_L . We justify this drastic simplification by assuming the driving laser to be far off-resonant with respect to all dipole-allowed atomic transitions except $|g\rangle \leftrightarrow |r\rangle$. Note that in practice it is beneficial to implement 2 or 3 photon-excitation schemes to drive the ground to Rydberg state transition, see e.g. Ref. [53, 54]. For the laser drive we use a classical description of the light field, i.e. a non-depletable photon bath. In this semiclassical picture and in the dipole approximation, atom-light interactions are captured by the electric field coupling to the electric dipole moment of the atom ($\hbar \equiv 1$):

$$H_{\text{array}} = H_{\text{atoms}} + H_{\text{atoms-light}} + H_{\text{trap}} \quad (\text{I.17})$$

$$H_{\text{atoms}} = \omega_A \sum_{k=1}^N |r\rangle\langle r|^{(k)} \quad (\text{I.18})$$

$$H_{\text{atoms-light}} = \sum_{k=1}^N eE_0 \cdot \hat{\mathbf{r}}_e^{(k)} \cos(\omega_L t) \quad (\text{I.19})$$

$$= \sum_{k=1}^N \cos(\omega_L t) (|g\rangle\langle r|^{(k)} \langle g|eE_0 \cdot \hat{\mathbf{r}}_e^{(k)}|r\rangle^{(k)} + |r\rangle\langle g|^{(k)} \langle r|eE_0 \cdot \hat{\mathbf{r}}_e^{(k)}|g\rangle^{(k)}) \quad (\text{I.20})$$

Here the laser coupling to the electric dipole moment of the Rydberg atom is modelled by an electric field oscillating in time. H_{trap} describes the trap and its interaction with the atoms. E_0 is the electric field amplitude of the drive and ω_A the level spacing of each atom. Superscripts in parentheses denote the atomic subspace which the operator is associated with, e.g. $|r\rangle\langle r|^{(2)} = 1_2 \otimes |r\rangle\langle r| \otimes 1_2 \otimes \dots \otimes 1_2$, with the two-by-two identity 1_2 . In total, there are $N - 1$ identities in this string of operators. The relative (to the nucleus) electron position of atom k is described by the operator $\hat{\mathbf{r}}_e^{(k)}$ and $e > 0$ is the elementary charge. We assume that the spatial extent of the electron's wave packet in the ground state is much smaller than optical wave lengths, such that omitting the spatial dependence of the driving laser field is justified. For the spatially large Rydberg state ($\sim 100 \text{ nm}$) it is not obvious why this description is valid, and generally is not. Nevertheless, this model description is typically employed for purposes similar to ours, as it yields a good effective description. As examples we give Ref. [37, 55]. In the basis of the atomic states the atom-light interaction Hamiltonian has only off-diagonal elements and the well-known dipole selection rules¹ apply, dictating which states may be

¹ For obtaining the usual selection rules, the parity operator's action, $\pi^\dagger \hat{\mathbf{r}} \pi = -\hat{\mathbf{r}}$, $\pi |g\rangle = \pm |g\rangle$ and $\pi |r\rangle = \pm |r\rangle$ is used to compare matrix elements of $\mathbf{r}$ and $-\mathbf{r}$ in the electronic states of the atom.

coupled. In the following, the Rabi frequency emergent from the laser coupling to the atomic dipole moment is defined as $\Omega \equiv 2 \langle g | e \mathbf{E}_0 \cdot \hat{\mathbf{r}}_e^{(k)} | r \rangle^{(k)}$. The factor 2 is added just for convenience to cancel the $1/2$ stemming from $2 \cos(x) = \exp(ix) + \exp(-ix)$. For simplicity, we choose the laser coupling between $|g\rangle$ and $|r\rangle$ such that $\Omega \geq 0$. Next, we will change the frame of reference to one oscillating with the laser frequency, ω_L , in order to identify fast and slow time scales to ultimately perform a rotating wave approximation (RWA). Under the unitary

$$U_L(t) = \exp \left(-i\omega_L t \sum_{k=1}^N |r\rangle\langle r|^{(k)} \right), \quad (I.21)$$

the transformed Hamiltonian of the array acquires the form

$$H_{\text{array,L}}(t) = \underbrace{-iU_L^\dagger(t)\dot{U}_L(t)}_{=-\omega_L H_{\text{atoms}}/\omega_A} + U_L^\dagger(t)H_{\text{atoms}}U_L(t) + U_L^\dagger(t)H_{\text{atoms-light}}U_L(t) + U_L^\dagger(t)H_{\text{trap}}U_L(t) \quad (I.22)$$

$$= \Delta \sum_{k=1}^N |r\rangle\langle r|^{(k)} + \Omega \sum_{k=1}^N (|g\rangle\langle r|^{(k)} (1 + e^{i2\omega_L t}) + \text{h.c.}) + U_L^\dagger(t)H_{\text{trap}}U_L(t) \quad (I.23)$$

$$\approx \Delta \sum_{k=1}^N |r\rangle\langle r|^{(k)} + \Omega \sum_{k=1}^N (|g\rangle\langle r|^{(k)} + \text{h.c.}) + U_L^\dagger(t)H_{\text{trap}}U_L(t) \quad (I.24)$$

where the laser detuning $\Delta \equiv \omega_L - \omega_A$ is defined, and the approximation sign signals that the rapidly oscillating terms with frequency ω_L are omitted (RWA). To get intuition about the processes associated with the neglected terms, the quantum description of the light field is employed – first discussed by Jaynes and Cummings in 1963 and experimentally validated by Rempe, Walther and Klein in 1987, Ref. [56, 57]. It can be seen that the fast oscillating terms correspond to processes where the atom becomes excited and *emits* a photon and vice versa. The high energy cost of these processes renders their probability negligible in comparison to the more intuitive processes where atom excitation results in photon mode de-excitation.

In the following we assume state independent trapping of the atoms. Generally, the trapping potential each atom experiences depends on the electronic state it is in, and in particular its associated polarizability. A trap with so called ‘magic wavelength’ cancels the state dependence by making sure the ground and Rydberg state accumulate equal light shifts upon excitation, keeping the energy distance fixed. Mathematically, we implement this condition by setting $[U_L(t), H_{\text{trap}}] = 0$, directly showing that trapping is not effected by atomic degrees of freedom. Mechanisms yielding the magic trapping condition (for atoms and ions) and experimental evidence is given in Ref. [58–60].

Represented in the basis $(|g\rangle, |r\rangle)$ on each atomic subspace, the now time-independent Hamiltonian of the array reads

$$H_{\text{array,L}} = \Delta \sum_{k=1}^N n^{(k)} + \Omega \sum_{k=1}^N \sigma_x^{(k)} + H_{\text{trap}}, \quad n^{(k)} \equiv \frac{1_2 - \sigma_z^{(k)}}{2}, \quad (I.25)$$

with the Pauli matrices on each subspace, e.g. $\sigma_{(x,y,z)}^{(2)} = 1_2 \otimes \sigma_{(x,y,z)} \otimes 1_2 \otimes \dots \otimes 1_2$. We refrained from introducing this basis representation from the start because it is less apparent which states couple to which. It should be emphasized that all steps taken up until now did not depend on Δ and Ω being stationary, such that explicit time-dependence is simply implemented by substituting $\Delta \mapsto \Delta(t)$ and $\Omega \mapsto \Omega(t)$. Note that this Hamiltonian is merely a sum of driven two-level systems, with drive Ω and energy difference Δ *plus* the trapping Hamiltonian.

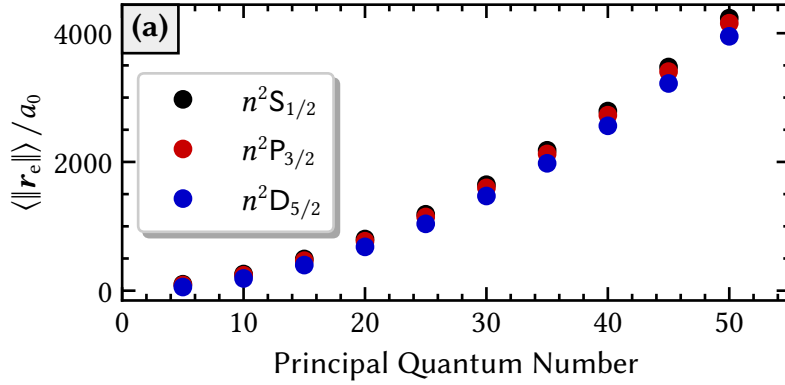


Figure I.2.: Radial Expectation Values. (a) Expectation values of electron-nucleus distance $\| \mathbf{r}_e \|$ for the ^{85}Rb atom depending on the principal quantum number n . To numerically obtain the quantum defects δ_{nlj} , the Python package *Alkaline Rydberg Calculator* (ARC) was employed, Ref. [42].

I.2. Dipole-Dipole Interactions

Previously, an expression for the Hamiltonian of a trapped, non-interacting atom array was derived. The included light-matter interactions are restricted to a single atom. In order to simulate molecular dynamics in the array, atom-atom interactions are obviously desired. Rydberg atom arrays naturally come equipped with these types of interactions, because of strong electric dipole moments associated with states of large principal quantum number. Therefore, non-negligible dipole-dipole interactions emerge if two Rydberg atoms reside sufficiently close in space. For the purpose of illustration, expectation values of $\| \mathbf{r}_e \|$ are plotted for ^{85}Rb in Fig. I.2:

$$\langle \| \mathbf{r}_e \| \rangle = \frac{a_0}{2} (3(n - \delta_{nlj})^2 - l(l + 1)) \quad (\text{I.26})$$

Here, $a_0 = \hbar/(m_e c \alpha) \approx 52.9 \text{ pm}$ is Bohr's radius and δ_{nlj} the quantum defect. l and j are the orbital and total (l and electron spin) angular momentum quantum number, respectively. At $n = 50$ the atomic radius turns out to be approximately $4000a_0 \approx 200 \text{ nm}$, resulting in an electric dipole moment of roughly 9.6 kD .

In the following we will discuss phenomena rooted in these strong interactions. Of particular interest are the so called *Rydberg blockade* and *Rydberg facilitation*, initially called *antiblockade* in the literature, Ref. [39, 61–64]. Ultimately, we will make use of these effects to constrain our system to a suitable Hilbert subspace. However, before finding a model Hamiltonian for the Rydberg blockade and facilitation, we will focus on how the dipole-dipole interaction Hamiltonian emerges. On the way, we will also discuss a method for engineering the interaction potentials' shape using *microwave (MW) dressing*, Ref. [43, 62, 65]. For simplicity, we focus on two atoms, as this is sufficient to find all relevant effects for the scope of this work. Final goal is to obtain an effective description of atom-atom interactions which we can base our Hamiltonian on.

I.2.1. Dipole-Dipole Interactions and Microwave Dressing

Starting point of the discussion are two atoms with nuclei positioned at $\mathbf{r}_n^{(k)}$ and relative electronic position vectors $\mathbf{r}_e^{(k)}$, where $k = 1, 2$ labels the atom. For convenience we define the distance vector $\mathbf{R} = \mathbf{r}_n^{(2)} - \mathbf{r}_n^{(1)}$ and its norm $R = \|\mathbf{R}\|$. Electric dipole moments associated to the atoms are defined as $\boldsymbol{\mu}^{(k)} = -e\mathbf{r}_e^{(k)}$, where $e > 0$ is the elementary charge. A sketch of the situation is presented

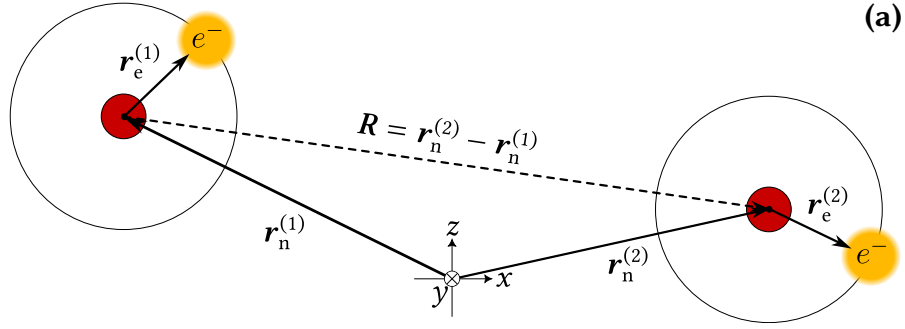


Figure I.3.: Interacting Rydberg Atoms. (a) Sketch of interacting atomic dipole moments $\mu^{(1,2)} = -er_e^{(1,2)}$, ($e > 0$). $r_n^{(1,2)}$ are the nuclei and $r_e^{(1,2)}$ the relative electron position vectors.

in Fig. I.3. There are four terms contributing to the interaction energy, V_C , of this configuration, because the two charges of the second atom feel the Coulomb potential of the core and the electron of atom one. Explicitly,

$$V_C(r_e^{(1)}, r_e^{(2)}, R) = \frac{1}{k_C} \left(\frac{ee}{\|r_n^{(2)} - r_n^{(1)}\|} + \frac{e(-e)}{\|r_n^{(2)} - (r_n^{(1)} + r_e^{(1)})\|} + \frac{(-e)e}{\|r_n^{(2)} + r_e^{(2)} - r_n^{(1)}\|} + \frac{(-e)(-e)}{\|r_n^{(2)} + r_e^{(2)} - (r_n^{(1)} + r_e^{(1)})\|} \right), \quad (I.27)$$

where $k_C = 4\pi$ is Coulomb's constant in units $\hbar = \epsilon_0 = 1$. If we assume the atoms to be sufficiently far separated, more precisely $\|r_e^{(k)}\|/R \ll 1$, we may expand V_C to find the leading order contribution of the multipole expansion:

$$V_C(\mu^{(1)}, \mu^{(2)}; R) = \frac{1}{k_C} \frac{\mu^{(1)} \cdot \mu^{(2)} - 3(\mu^{(1)} \cdot \nu_R)(\mu^{(2)} \cdot \nu_R)}{R^3} + \mathcal{O}\left(\frac{\|r_e^{(1,2)}\|^3}{R^3}\right), \quad \nu_R = \frac{R}{R} \quad (I.28)$$

By promoting dipole moments to operators, $\mu^{(k)} \mapsto \hat{\mu}^{(k)}$, we obtain the interaction Hamiltonian

$$H_{\text{int}} = V_C(\hat{\mu}^{(1)}, \hat{\mu}^{(2)}; R). \quad (I.29)$$

Because the dipole operators have non-zero transition matrix elements with more than just the ground state, our two-level approximation of the atoms is not valid anymore. However, since the transition matrix elements $\langle r | \hat{\mu}^{(k)} | g \rangle^{(k)}$ scale as $(n - \delta_{nlj})^{-3/2}$, Ref. [43, Table 1], we find

$$\langle r, r | V_C(\hat{\mu}^{(1)}, \hat{\mu}^{(2)}, R) | g, g \rangle, \langle r, g | V_C(\hat{\mu}^{(1)}, \hat{\mu}^{(2)}, R) | g, r \rangle \sim (n - \delta_{nlj})^{-3}. \quad (I.30)$$

All other matrix elements vanish as a consequence of selection rules. Because of the scaling, we can ignore the coupling of the ground and Rydberg state introduced by V_C . To be concrete, some typical values for the dipole matrix elements in the fine structure spectrum of ^{85}Rb are provided in Tab. I.1. For the transition $5^2S_{1/2} \rightarrow 50^2P_{3/2}$, we roughly get a dipole moment of $0.001ea_0$, such that even for an unphysical value $R = 0.1 \mu\text{m}$ we find the magnitude $V_C(|g\rangle \leftrightarrow |r\rangle) \sim 10 \text{ feV}$. In conclusion, we are only interested in coupling to neighbouring Rydberg states. Often, the dipole-dipole interaction is dominated by only a few pair states $|r_i, r_j\rangle = |r_i\rangle^{(1)} \otimes |r_j\rangle^{(2)}$, energetically close to $|r, r\rangle$, as outlined in Ref. [27, 43]. To second order perturbation theory, we find the energy correction from populating other pair states:

$$E^{(2)} = \sum_{|r', r''\rangle \neq |r, r\rangle} \frac{|\langle r', r'' | V_C(\hat{\mu}^{(1)}, \hat{\mu}^{(2)}, R) | r, r \rangle|^2}{E_{r, r} - E_{r', r''}} \equiv \frac{C_6}{R^6} \quad (I.31)$$

Table I.1.: Typical Dipole Moments. Typical values for dipole matrix elements of σ^+ , π and σ^- transitions in the fine-structure spectrum of ^{85}Rb . Exemplarily, three different transitions are considered. Values are given in units of ea_0 and rounded to 4 significant digits. Data is generated using Ref. [42].

Transition	σ^+	π	σ^-
$5^2S_{1/2} \rightarrow 50^2P_{3/2}$	0.0008339	-0.001179	0.0008339
$49^2S_{1/2} \rightarrow 50^2P_{3/2}$	149.3	-211.2	149.3
$50^2S_{1/2} \rightarrow 50^2P_{1/2}$	1764	-1248	-1764

We observe that there is only a correction to second order perturbation theory, because of dipole selection rules. In case there are *Förster resonances*, i.e. there are at least two pair states with equal energy, one must apply degenerate perturbation theory, which can result in an additional contribution C_3/R^3 , Ref. [66].

We now want to take a look at the simplest case in order to get intuition on the spatial shape of the interaction energy. To do so, we assume there is only one other pair state contributing significantly and the remaining ones are suppressed by the *Förster defect* $E_{r,r} - E_{r',r''}$ and the weak dipole coupling, Tab. I.1. Of course, in a practical application, one would select a state $|r\rangle$ such that only a few pair states lay energetically close to it in the first place. Before computing the energy correction in this simplified picture, we note that there is no way to tailor the energy correction's shape to our needs. We therefore introduce another field by applying a Maser in order to couple the Rydberg pair states externally. In particular, we couple the two Rydberg states by a classical Maser such that we again obtain a two-level system description analogous to Eq. (I.25). We thus write for the Hamiltonian

$$H_{\text{MW}}^{(k)} = \begin{bmatrix} 0 & \Omega_{\text{MW}} \\ \Omega_{\text{MW}} & \Delta_{\text{MW}} \end{bmatrix}, \quad (\text{I.32})$$

with the detuning of the MW field Δ_{MW} and the arising Rabi frequency Ω_{MW} . Remark: Generally, this form of the Hamiltonian can impose inconsistency with the previously conducted RWA employed to arrive at Eq. (I.25). This is because we have neglected fast oscillating terms with twice the frequency of the maser, $2\omega_{\text{M}}$. However, for a Rydberg transition frequency in the THz range, the detuning Δ may be of similar order as ω_{M} . To make a point, we assume Δ not to be comparable to the omitted time-dependent terms. Nonetheless, this is something which must be considered in general, meaning one must start with the full Hamiltonian in the laboratory frame and go beyond the RWA, e.g. using so called Floquet theory (named after mathematician *Gaston Floquet* (1847-1920)), Ref. [67, 68]. As only Rydberg pair states interact significantly anyway, we assume now both atoms to be in their Rydberg state. We also assume the excited state not to be an atomic eigenstate but a MW dressed state $\psi_{\pm}$, i.e. an eigenstate of Eq. (I.32):

$$E_{\pm} = \frac{\Delta_{\text{MW}}}{2} \pm \frac{\sqrt{\Delta_{\text{MW}}^2 + 4\Omega_{\text{MW}}^2}}{2}, \quad \psi_{\pm} = \frac{1}{\sqrt{1 + \left(\frac{E_{\mp}}{\Omega_{\text{MW}}}\right)^2}} \begin{bmatrix} -\frac{E_{\mp}}{\Omega_{\text{MW}}} \\ 1 \end{bmatrix} \quad (\text{I.33})$$

The mixing of the Rydberg states $|r\rangle$ and $|r'\rangle$ enforces the experimentally controllable parameters Δ_{MW} and Ω_{MW} onto the coefficient C_6 and also enforces a coefficient C_3 , naturally. We may represent the full Hamiltonian of the excited atoms

$$H = H_{\text{MW}} \otimes 1 + 1 \otimes H_{\text{MW}} \quad (\text{I.34})$$

in the basis $(|r, r\rangle, |r, r'\rangle, |r', r\rangle, |r', r'\rangle)$,

$$H_{\text{MW}} = \begin{bmatrix} 0 & \Omega_{\text{MW}} & \Omega_{\text{MW}} & 0 \\ \Omega_{\text{MW}} & \Delta_{\text{MW}} & 0 & \Omega_{\text{MW}} \\ \Omega_{\text{MW}} & 0 & \Delta_{\text{MW}} & \Omega_{\text{MW}} \\ 0 & \Omega_{\text{MW}} & \Omega_{\text{MW}} & 2\Delta_{\text{MW}} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & v_{rr;r'r'} \\ 0 & 0 & v_{rr';r'r} & 0 \\ 0 & v_{rr';r'r}^* & 0 & 0 \\ v_{rr;r'r'}^* & 0 & 0 & 0 \end{bmatrix}, \quad (\text{I.35})$$

where the matrix on the right, from now on called $\tilde{H}_{\text{int}}$, describes the dipole-dipole interaction with the coupling being the transition matrix elements of V_{C} in the pair states. If we simplify our picture even further, by assuming all $v_{rr,ij}$ real and equal, we can apply second order perturbation theory to compute the energy correction to our Rydberg state ψ_+ :

$$2E_+ + \psi_{++}^\dagger \tilde{H}_{\text{int}} \psi_{++} + \sum_{j,k=\pm,\pm} \frac{|\psi_{++}^\dagger \tilde{H}_{\text{int}} \psi_{jk}|^2}{\sqrt{\Omega_{\text{MW}}^2 + \Delta_{\text{MW}}^2}} \equiv 2E_+ - \frac{C_3(\Omega_{\text{MW}}, \Delta_{\text{MW}})}{R^3} + \frac{C_6(\Omega_{\text{MW}}, \Delta_{\text{MW}})}{R^6}, \quad (\text{I.36})$$

with $\psi_{\pm\pm} = \psi_{\pm} \otimes \psi_{\pm}$. Combining both terms gives rise to a (6, 3)-Lennard-Jones potential, Ref. [69], clearly reminiscent of potentials studied in molecular physics. The coefficients C_3 and C_6 take a more simple form in the dressing regime $|\Omega_{\text{MW}}|/|\Delta_{\text{MW}}| \ll 1$, Ref. [43]:

$$C_3(\Omega_{\text{MW}}, \Delta_{\text{MW}}) \approx \frac{\mu^2 \Omega_{\text{MW}}^2}{\pi \Delta_{\text{MW}}^2}, \quad C_6(\Omega_{\text{MW}}, \Delta_{\text{MW}}) \approx \frac{\mu^4}{16\pi^2} \frac{\Delta_{\text{MW}}^2 + 6\Omega_{\text{MW}}^2}{2\Delta_{\text{MW}}^3} \quad (\text{I.37})$$

In the limit of $\Omega_{\text{MW}} \rightarrow 0$ the C_3 term vanishes, as expected from selection rules. Note that we have used non-degenerate perturbation theory although the fine-structure states are degenerate in the magnetic quantum number m_j , as we do not have Zeeman splitting², Ref. [70]. With the help of the expressions for the C_3 and C_6 coefficient we can visualize the potential for different ratios of $\Omega_{\text{MW}}/\Delta_{\text{MW}}$, see Fig. I.4. For simplicity we assume the angle to be such that the angel dependent part of V_{C} vanishes.

As a final remark, we note that the coefficients C_3 and C_6 generally depend on the angle between the relative electron position vector and the atom separation vector R , i.e. on the angular dependence of the electron wave function, Ref. [71]. To eliminate the angular dependence in an experiment one could fix the quantization axis using a magnetic field, such that the angle is constant or use Rydberg states without angular dependence in the first place. Of course, the angular dependence gives rise to yet another way of tailoring the interaction potentials shape. For illustration, Fig. I.5 presents the angular dependence of the C_6 dispersion coefficient for different states in the fine-structure spectrum of ^{85}Rb .

Before diving into the next section, we want to conclude what we have found by the previous considerations. First, we have observed that we only get an energy correction from the atom-atom interactions if both atoms are in their respective Rydberg states. This circumstance was clear from the start, at least to some extent – essentially it means that only energy shifts appear if *both* atoms are excited to a state with large associated electric dipole moment. Secondly, we motivated that the interaction energy is approximately R -dependent in a (6, 3)-Lennard-Jones like fashion, where we can control the potentials shape by tuning MW field parameters. Again, it should be emphasized that this derivation is not aimed at giving a full theoretical description for an experiment in the most general way. Its purpose is to motivate that dipole-dipole interactions indeed are capable of

2 The discussion here is simplified because our aim is to motivate that potentials of Lennard-Jones type arise. For a detailed discussion on the subject the interested reader is referred to Ref. [27] and references therein.

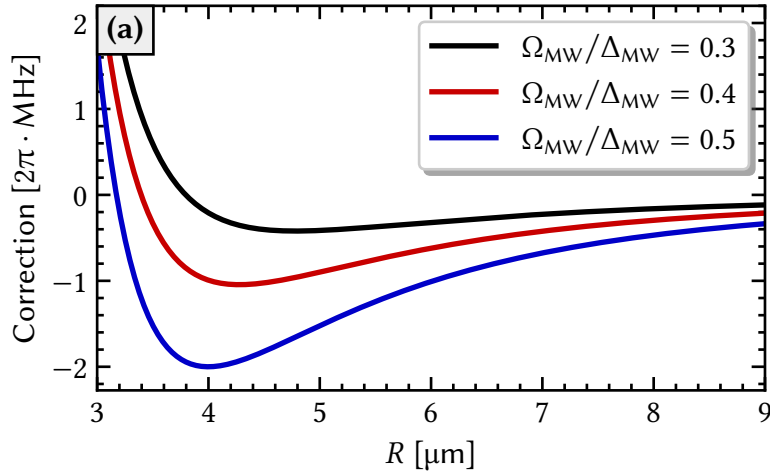


Figure I.4.: Energy Correction from Dipole-Dipole Interaction. (a) Second order perturbation theory energy correction to the pair state energy $2E_+$ due to dipole-dipole interactions. $R = \|\mathbf{R}\|$ is the atomic distance, for simplicity assumed to be orthogonal to the dipole moments in Eq. (I.36). We choose ^{85}Rb , $\Omega_{\text{MW}}/\Delta_{\text{MW}} = 0.1, 0.3, 0.5$, fine structure states $50^2S_{1/2}$ and $50^2P_{1/2}$ with energy difference of roughly 30 GHz, Ref. [42], and finally, the detuning $\Delta_{\text{MW}} = 2\pi \cdot 10$ MHz. From Tab. I.1 we take the dipole moment $\mu = \langle 50^2S_{1/2} | \hat{\mu}^{(k)} | 50^2P_{1/2} \rangle \approx -1284ea_0$ for the π -transition.

generating potentials reminiscent of interactions known from molecular physics. There are many more things that can be taken into account. For instance, it is clearly a drastic simplification that only two pair states interact. In practice, multiple masers could be used to increase control over the coefficients C_3 and C_6 , leading to more complex potentials and more control.

In conclusion we take from this section that under carefully configured experimental conditions we may model atom-atom interactions between two Rydberg atoms in good approximation by an effective Hamiltonian of the form

$$H_{\text{int}} = V(R)n^{(1)}n^{(2)}. \quad (\text{I.38})$$

This Hamiltonian implements our results, because if atom 1 is excited, the projected Hamiltonian takes the form $V(R)n^{(2)}$, leading to a level shift $V(R)$ of the second atom's Rydberg state. At this point we have achieved our goal of finding an effective Hamiltonian for atom-atom interactions.

I.2.2. Rydberg Blockade and Facilitation

In the last section we have derived an expression for an effective Hamiltonian modelling atom-atom interaction. Goal of this paragraph is to introduce the for this work relevant effects, emergent from the previously derived two-atom interaction Hamiltonian. Therefore, we will stay within the two-atom picture and consider the two-atom Hamiltonian

$$H = \Delta(n^{(1)} + n^{(2)}) + \Omega(\sigma_x^{(1)} + \sigma_x^{(2)}) + V(R)n^{(1)}n^{(2)}, \quad (\text{I.39})$$

where $V(R)$ describes a potential depending on experimental parameters (such as Ω_{MW} and Δ_{MW}) and in particular the nuclei distance of the atoms R , i.e. the distance of the atoms in the traps. For now, we will keep R fixed and neglect atomic displacements from the trap centres. Later, we will reintroduce such contributions.

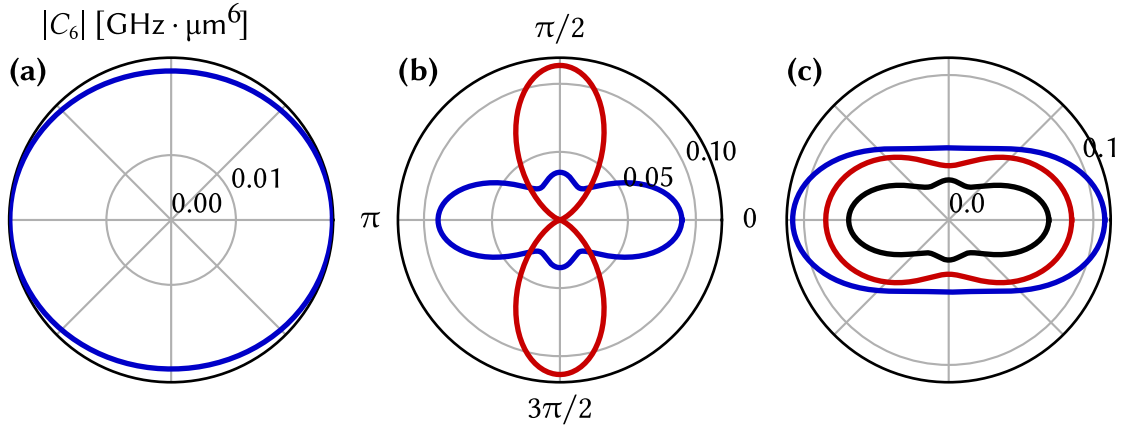


Figure I.5.: Angular Dependence of C_6 . Angular dependence of the dispersion coefficient C_6 , for the fine-structure states (a) $30^2S_{1/2}$ (weak angular dependence), (b) $30^2P_{3/2}$ and (c) $30^2D_{5/2}$ (strong dependence). States are taken from the spectrum of ^{85}Rb . In the perturbative series for C_6 , the principal quantum numbers $n \in [30 - 5, 30 + 5] \cap \mathbb{Z}$ are considered and Förster defects up to 25 GHz. Blue lines represent $m_j = 1/2$, red lines $m_j = 3/2$ and the black line $m_j = 5/2$. Data is generated using Ref. [42]. Note that the angle dependence of to the azimuthal angle, θ , with respect to the quantization axis is shown, such that all plots are mirrored on the horizontal axis, since $\theta \in [0, \pi)$.

Starting with this Hamiltonian, we can observe Rydberg blockade and facilitation by considering two scenarios, (a) and (b), both captured by Eq. (I.39):

- (a) Suppose atom 1 and atom 2 are in their respective ground states. With a resonant laser defined by the Rabi frequency Ω , a laser pulse is directed at atom 1, such that it is excited to its Rydberg state. Now the same pulse is directed at atom 2. During the second pulse the system is no longer described by Eq. (I.39), but rather the projected Hamiltonian

$$\langle \mathbf{r} | H | \mathbf{r} \rangle^{(1)} = V(R)n^{(2)} + \Omega\sigma_x^{(2)}. \quad (\text{I.40})$$

Clearly, the potential $V(R)$ can be tuned such that $|V(R)| \gg |\Omega|$, prohibiting excitation of atom 2, although the laser is resonant with respect to the undisturbed atomic transition. Hence, excitation of atom 1 is *blocking* excitation of atom 2. Note that this prohibition becomes apparent already from the well-known physics of driven two-level systems, but can also be formalized with effective Hamiltonians, Ref. [73]. A graphical description of the situation is given in Fig. I.6 (a).

- (b) For the counter effect we take two lasers, one resonant with respect to the undisturbed atomic transition and one detuned by $V(R)$. We again excite atom 1 to its Rydberg state and leave atom 2 in its ground state. The second laser shall be detuned by $|\Delta| \gg |\Omega|$, such that effectively the transition in atom 2 cannot be driven. Increasing $V(R)$ such that the so called ‘facilitation condition’, $\Delta = -V(R)$, is obeyed, the off-resonant laser now is resonant to the shifted transition, i.e. the interaction cancels the detuning, such that the laser can drive the atom to its Rydberg state after all, Ref. [37, 72]. Figure I.6 (b) shows a qualitative picture of the energy level structure.

For completeness it should be mentioned that one can instead discover these effects in a more rigorous manner, namely by starting the discussion before the RWA is applied and see which states

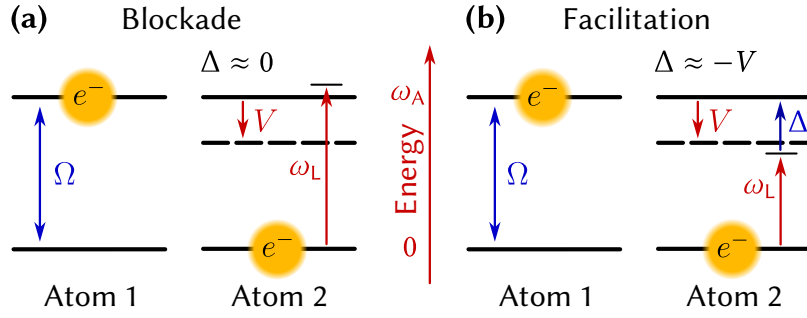


Figure I.6.: Rydberg Blockade and Facilitation. Shifts in the energy level structure caused by dipole-dipole interactions, V , between Rydberg states. ω_L is the laser frequency, ω_A the atomic energy level separation, Ω the Rabi frequency and Δ the detuning of the laser; $|\Omega| \ll |\Delta|$ is assumed. *Rydberg blockade* (a): A resonant laser cannot drive the transition in atom 2, because it is detuned by V . An excited Rydberg atom can therefore *block* excitation of atom 1. *Rydberg facilitation* (b): Although detuned by Δ , the laser is able to drive the transition since V cancels the detuning. The relation $\Delta = -V$ defines the so called ‘facilitation condition’, Ref. [37, 72].

couple on which time scales. This dynamical approach is used in Ref. [43, 64], but omitted here as this is somewhat outside of scope. Noteworthy is that because $V = V(R)$ with long range algebraic scaling R^{-3} , Eq. (I.36), there need exist some radius beyond which blockade and facilitation fail, as the atoms effectively do not see each other. In the literature this radius is called *blockade radius*, Ref. [43] and is defined by the condition $V(R_b) = \Omega$. This definition grasps the fact that blockade and facilitation really are phenomena in the regime $|V(R)|, |\Delta| \gg |\Omega|$. Else, the drive Ω is so strong that none of the other parameters makes a difference and the laser is practically resonant. Example: In a spin chain realized with Rydberg atoms where two neighbouring atoms shall not be excited simultaneously, the lattice spacing must be smaller than the blockade radius. This must be kept in mind when building kinetically constrained systems, as in Ref. [37, 74, 75], where blockade and/or facilitation are imposed. Finally, to give a typical value for R_b , we may take the coefficients from Eq. (I.37) and use the definition to find

$$R_b = \sqrt[3]{\frac{\sqrt{C_3^2 + 4\Omega_{MW}C_6} - C_3}{2\Omega_{MW}}} \approx 2.9 \mu\text{m}, \quad (\text{I.41})$$

with values taken from Fig. I.4 and the ratio $\Omega_{MW}/\Delta_{MW} = 0.4$. Of course, if a specific region of the potential is desired as well as blockade/facilitation, e.g. the minimum, one must ensure that it has a large enough value to still lay within the ball defined by R_b .

In conclusion we have found Rydberg blockade and facilitation in our effective description of the system. Moreover, we have discussed that one has to be careful when building kinetically constrained systems to take the finite blockade radius into account. This is relevant for our simulator, if one wants to consider molecular shapes where some atom pairs are further separated than the remaining ones. A simple example would be a AX_2E_0 type linear ‘chain’ of three atoms.

I.3. Artificial Molecular Hamiltonian

It is now time to make use of the preceding sections to formulate the class of Hamiltonians this thesis will be concerned with, i.e. our simulator will simulate. Therefore, we continue the discussion

of Sec. I.1 by introducing dipole-dipole interactions to Eq. (I.25). Equation I.38 allows a neat way of implementing them. With atomic positions $\mathbf{r}^{(k)}$, we write

$$H = \Delta \sum_{k=1}^N n^{(k)} + \Omega \sum_{k=1}^N \sigma_x^{(k)} + H_{\text{trap}} + \sum_{k=1}^N \sum_{k' < k} V(\|\mathbf{r}^{(k)} - \mathbf{r}^{(k')}\|) n^{(k)} n^{(k')}. \quad (\text{I.42})$$

Opposed to the initial assumption of constant nucleus separations, we now allow variance in $R_{kk'} = \|\mathbf{r}^{(k)} - \mathbf{r}^{(k')}\|$, in a regime where the tweezers can still be modelled as harmonic potentials³ defined by the beam intensity profile. This assumption manifests itself in an effective trap frequency ω in the range of several 10-100kHz and even MHz, Ref. [27, 37]. Excitations of motional degrees of freedom in the trap will thus be described by the sum of free harmonic oscillators:

$$H_{\text{trap}} = \omega \sum_{k=1}^N \sum_{j=x,y,z} a_j^{(k)\dagger} a_j^{(k)} \quad (\text{I.43})$$

In principal, one could use non-isotropic traps with frequencies $\omega_x, \omega_y, \omega_z$, but we will stick with a single ω for the sake of simplicity. Excitations of the mode $a_x^{(k)}$ are phonons describing the displacement of atom k in x -direction and analogously for the remaining modes. An excitation in one of the modes therefore induces an energy change in V as a result of $R_{kk'}^0 \mapsto R_{kk'}^0 + \delta R_{kk'}$, where $R_{kk'}^0$ are the atomic equilibrium distances defined by the equilibrium position vectors $\mathbf{r}_0^{(k)}$. For the absolute atomic position we write $\mathbf{r}^{(k)} = \mathbf{r}_0^{(k)} + \delta \mathbf{r}^{(k)}$. Note that the assumption of small displacements compared to atomic equilibrium distances is in line with the multipole expansion in Eq. (I.28), since it prevents cases where $\|\mathbf{r}_n^{(k)} - \mathbf{r}_n^{(k')}\| \approx \|\mathbf{r}_e^{(k)}\|$. For this reason, it is natural to expand V and the atomic norm distance with respect to the trap equilibrium positions, yielding

$$\begin{aligned} V(\|\mathbf{r}^{(k)} - \mathbf{r}^{(k')}\|) &\approx V(R_{kk'}^0) + (\nabla V)(\mathbf{r}_0^{(k)}, \mathbf{r}_0^{(k')}) \begin{bmatrix} \delta \mathbf{r}^{(k)} \\ \delta \mathbf{r}^{(k')} \end{bmatrix} \\ &+ \frac{1}{2} [\delta \mathbf{r}^{(k)\text{t}}, \delta \mathbf{r}^{(k')\text{t}}] (\nabla \otimes \nabla^t V)(\mathbf{r}_0^{(k)}, \mathbf{r}_0^{(k')}) \begin{bmatrix} \delta \mathbf{r}^{(k)} \\ \delta \mathbf{r}^{(k')} \end{bmatrix} + \mathcal{O}\left(\left[\frac{\|\delta \mathbf{r}^{(k,k')}\|}{R_{kk'}^0}\right]^3\right), \end{aligned} \quad (\text{I.44})$$

where we define the gradient,

$$(\nabla V)(\|\mathbf{r}_0^{(k)} - \mathbf{r}_0^{(k')}\|) = \frac{V'(R_{kk'}^0)}{R_{kk'}^0} [1, -1] \otimes (\mathbf{r}_0^{(k)} - \mathbf{r}_0^{(k')})^t, \quad (\text{I.45})$$

and the Hessian,

$$\begin{aligned} (\nabla \otimes \nabla^t V)(\mathbf{r}_0^{(k)}, \mathbf{r}_0^{(k')}) &= V''(R_{kk'}^0) \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \otimes \frac{\mathbf{r}_0^{(k)} - \mathbf{r}_0^{(k')}}{R_{kk'}^0} \otimes \frac{(\mathbf{r}_0^{(k)} - \mathbf{r}_0^{(k')})^t}{R_{kk'}^0} \\ &+ \frac{V'(R_{kk'}^0)}{R_{kk'}^0} \left(\begin{bmatrix} 1_3 & -1_3 \\ -1_3 & 1_3 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \otimes \frac{\mathbf{r}_0^{(k)} - \mathbf{r}_0^{(k')}}{R_{kk'}^0} \otimes \frac{(\mathbf{r}_0^{(k)} - \mathbf{r}_0^{(k')})^t}{R_{kk'}^0} \right), \end{aligned} \quad (\text{I.46})$$

where the gradient operator is defined as $\nabla = (\partial_{r_{0,1}^{(k)}}, \partial_{r_{0,2}^{(k)}}, \partial_{r_{0,1}^{(k')}}, \partial_{r_{0,2}^{(k')}})$ and, with slight abuse of notation, we write $V(\mathbf{r}^{(k)}, \mathbf{r}^{(k')}) = V(\|\mathbf{r}^{(k)}, \mathbf{r}^{(k')}\|)$. Note that there are, by construction, only polynomial

³ Atoms experience a spatially resetting potential defined by the beam intensity profile. Therefore, a Taylor expansion with respect to the equilibrium position yields a harmonic oscillator.

terms in the displacements – linear and quadratic ones in case of our truncated series. Clearly, the symmetry introduced by the solely distance dependent potential is evident in the matrix structure. Essentially, the Hessian is the Kronecker product of the gradient with itself plus the additional term, independent of the trap zero positions' vector structure. We quantize the atomic motion in the oscillator modes of the trap:

$$\delta r_j^{(k)} \mapsto \frac{x_0}{\sqrt{2}}(a_j^{(k)\dagger} + a_j^{(k)}), \quad p_j^{(k)} \mapsto \frac{i}{\sqrt{2}x_0}(a_j^{(k)\dagger} - a_j^{(k)}), \quad j = x, y, z \quad (\text{I.47})$$

Here, also conjugated momenta, $\mathbf{p}^{(k)}$, are defined, since we will need them later on. The naturally appearing length scale is the harmonic oscillator length, $x_0 = 1/\sqrt{m_A\omega}$, defined by the atom mass m_A and the trap frequency. Looking at the gradient and Hessian, it can be seen that full information on the array's spatial structure (atom positions) is already encoded into the gradient contribution. To give a typical value for the oscillator length, consider ^7Li and a trapping frequency of $\omega = 2\pi \cdot 20$ kHz, which results in $x_0 \approx 270$ nm, such that our expansion parameter for an atomic distance of roughly $3 \mu\text{m}$ is of order $x_0/R_{kk'}^0 \sim 0.1$. Substituting the expansion and quantization of the potential into Eq. (I.44), leads to the full Hamiltonian of the atom array in the tweezers:

$$H = \Delta \sum_{k=1}^N n^{(k)} + \Omega \sum_{k=1}^N \sigma_x^{(k)} + \omega \sum_{k=1}^N \sum_{j=x,y,z} a_j^{(k)\dagger} a_j^{(k)} + \sum_{k=1}^N \sum_{k' < k} \left[\mu_{kk'} h_{\text{ep},\mu}^{(0);k,k'} + \kappa_{kk'} h_{\text{ep},\kappa}^{(1);k,k'} + \nu_{kk'} \kappa_{kk'} h_{\text{ep},\nu\kappa}^{(2);k,k'} + \xi_{kk'} h_{\text{ep},\xi}^{(2);k,k'} \right] n^{(k)} n^{(k')} \quad (\text{I.48})$$

In the above expression we have introduced the following coupling constants, all with units of energy: $\mu_{kk'} = V(R_{kk'}^0)$, $\kappa_{kk'} = x_0 V'(R_{kk'})/\sqrt{2}$, $\xi_{kk'} = x_0^2 V''(R_{kk'})/2$. Further, the ratios $\nu_{kk'} = x_0/R_{kk'}^0$ are defined. Local electron-phonon⁴ interaction Hamiltonians $h_{\text{ep},\alpha}^{(p);k,k'}$ are defined via comparison to Eq. (I.44); α labels the coupling, k, k' the interacting atom pair and p the polynomial degree in the phonon modes, e.g. $p = 1$ for linear terms. We can clearly see that all electron-phonon interaction Hamiltonians are direct consequences of the dipole-dipole terms. As an overview for now and later considerations, we shall list correspondences between appearing parameters and physical quantities, Tab. I.2, to aid interpretation of results and also keep track of restrictions introduced by the assumptions leading to this Hamiltonian.

At this point, we have a Hamiltonian which we can in principle work with, and should be able to simulate in an experiment. This was the main goal of this chapter, which we report to be achieved. Hence, this is a good point in the discussion to take a closer look at our result before proceeding and quickly conclude our findings. Additionally we note that the necessary technicalities in this chapter might have somewhat convoluted the top view on our system. Therefore, we want to give a summarizing picture of our quantum simulator, emphasizing the important aspects, Fig. I.7.

In our Hamiltonian, we have terms modelling vibrational dynamics (phonons), as desired. These depend on the electronic state of each atom. For instance, in the extreme case of zero excited atoms, no interactions show, in contrast to all atoms being excited. On first sight it may seem that one can just take this Hamiltonian and theoretically examine all sorts of atomic arrays (at least numerically), as it is a set of harmonic oscillators coupled to spin degrees of freedom. However, the problematic part is that each atom comes with three bosonic modes. For bosonic modes truncated to dimension

⁴ We choose the terminology 'electron phonon' coupling to emphasize that the internal atomic degrees of freedom couple to the external vibrational ones.

Table I.2.: Typical Experimental Values. Overview of parameters appearing in Hamiltonian Eq. (I.48). In the first column typical magnitudes for the parameters are listed and in the second column their dependencies on physical quantities. I is the intensity of the laser responsible for Rabi oscillations, $\mathbf{r}_e$ the electronic position with respect to the nucleus, V' the potential's slope, V'' its curvature, $R_{kk'}^0$ the equilibrium atomic distances and m_A the atomic mass. Finally, Δ_{MW} and Ω_{MW} are the parameters of the maser.

Parameter	Magnitude	Dependence
Δ	MHz-GHz	ω_L, ω_A
Ω	kHz-MHz	$\sqrt{I}, \langle \mathbf{r} \mathbf{r}_e g \rangle$
$\kappa_{kk'}$	kHz	$V', \Delta_{MW}, \Omega_{MW}$
$\xi_{kk'}$	kHz	$V'', \Delta_{MW}, \Omega_{MW}$
$\nu_{kk'}$	~ 0.1	$R_{kk'}^0, m_A$

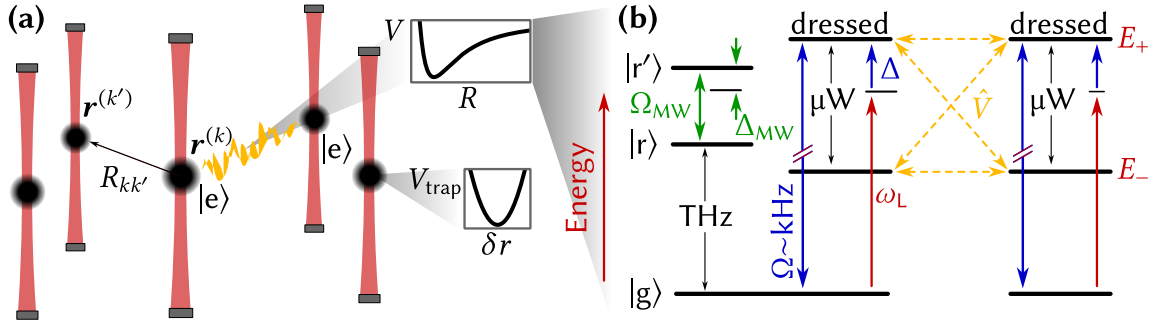


Figure I.7.: Overview of Simulation Framework. Graphical (simplified) summary of model described by the Hamiltonian in Eq. (I.48). (a) Atomic array in optical tweezers with atoms positioned at $\mathbf{r}^{(k)}$. $|e\rangle$ emphasizes that the atom is excited. Each trapped atom experiences the harmonic trap potential V_{trap} . Interactions are governed by the (6, 3)-Lennard-Jones potential V . (b) Relevant energy level structure of the two atoms labelled $|e\rangle$ in (a). Ω, Δ define the laser drive (blue), Ω_{MW}, Δ_{MW} describe the maser interaction (green) and ω_L is the laser frequency in the laboratory frame. The energies $E_{\pm}$ of the states resulting from MW dressing are defined in Eq. (I.33). Interactions stemming from the dipole-dipole operator $\hat{V}$ between the dressed states of the two atoms are described by dashed double arrows (yellow).

N_t , i.e. up to $N_t - 1$ bosons are allowed in each mode, the Hilbert space dimension is $2^N N_t^{3N}$. For 3 atoms and N_t as small as 5, this expression already evaluates to $2^3 \cdot 5^{3 \cdot 3} \approx 16 \cdot 10^6$. It complies to non-local operators needing roughly 4000 TB of storage when using 128-bit complex numbers (in a dense matrix representation). Therefore, to get results of non-trivial arrays it is not only beneficial but crucial to have a platform to simulate such systems – nature has no problem with many bosonic modes. We clearly see the relevance of building a quantum simulator for molecular systems by this rough estimate of resources needed for numerical approaches. Note also that we have followed the spirit of Manin and Feynman here, who famously introduced the notion of a quantum simulator. First, molecules in nature are much more complicated than our derived Hamiltonian. Secondly, we have control over the Hamiltonian we wish to simulate. All fine so far, but we are yet to proof the simulator not be too simple, in the sense that we indeed are able to reproduce the physics we wish to observe, that is phenomena linked to molecules in nature. At the end of this thesis, we will answer

this question with a clear: ‘Yes, the simulator works as intended!’, but this has to be demonstrated.

II. Analysis of Electron-Phonon Interactions

Previously, we have obtained a description for a system of N trapped atoms, which allows for tailored interatomic interactions. Our aim is to make use of this system to simulate molecular phenomena. At the end of the last chapter, the question if this kind of system inhabits the desired dynamics, was left unanswered. More generally, it is not clear what phenomena are observable and if they are of interest. In order to give an answer, we are motivated to investigate the implemented electron-phonon interactions. To this end, we consider the most simple possible arrays. Again, as already discussed, we cannot treat very large ($\gtrsim 8$ atoms) arrays numerically, because of hardware limitations, but the idea is that if simple, numerically still treatable, arrays host intriguing physics, we are confident that the simulator is working as intended. To be concrete, we will look at a dumbbell (2 atoms), an equilateral triangle (3 atoms) and tetrahedron (4 atoms). These three arrays will be the working platform for the remainder of this thesis.

As an outline for this chapter: In the first section we start with deriving Hamiltonians for the three arrays, where we make use of the Rydberg blockade and Rydberg facilitation, introduced in the previous chapter, to constrain the electronic Hilbert space to a smaller one. On the way, we want to look at what kind of vibration signatures we may observe in our simulator, for the different array shapes. In the second section, we discuss instabilities emerging in all three arrays and give (simple) analytical expressions for the critical values. Finally, we want to link this thesis to Ref. [37], where the same triangular array is considered; this is done in the last section. In Ref. [37] a Jahn-Teller regime was found and a concomitant structural phase transition. For this thesis a result like this is of great importance, because it is a phenomenon from molecular physics and our simulator can reproduce it, i.e. exactly what we are searching for. Moreover, up until now there is no work that has investigated this transition in this framework *with* the Hessian contribution and the emergent instabilities included. Therefore, we also briefly discuss how the Hessian contributions affect this transition by employing exact numerical methods.

II.1. Hamiltonians for Particular Molecular Shapes and their Vibration Signatures

In this section we want to derive expressions for the Hamiltonians of the dumbbell, triangle and tetrahedron. Starting point for all three arrays is the general Hamiltonian for our simulator, given in Eq. (I.48). We start the discussion with the simplest array in which electron-phonon interactions appear, i.e. the dumbbell. Afterwards, we consider an equilateral triangle and its three dimensional completion – an equilateral tetrahedron. However, focus shall mostly lay on the diatomic and triangular array, because most properties of interest for this thesis are already encoded in these two shapes. As will be seen, the diatomic array is described by a single vibrational mode, such that we may expect additional contributions to the Hamiltonian to appear, in case the vector structure, present only for more than one mode, comes into play (we will make this statement precise during the course of this section). The triangle and tetrahedron allow to examine these contributions in the most simple way, compared to other array shapes. This circumstance shall further motivate why these three array shapes are selected. Finally, we remark the dumbbell's relation to Rydberg macrodimers, Ref. [76–79].

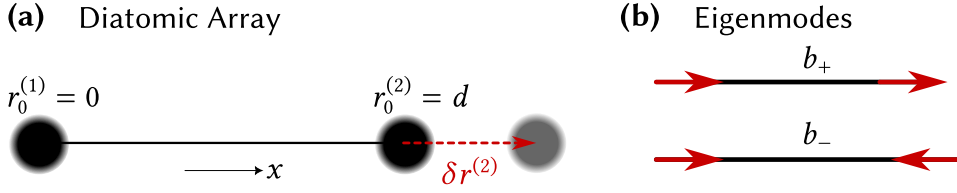


Figure II.1.: Sketch of Dumbbell. Sketch of (a) diatomic array (dumbbell) and (b) eigenmodes $b_{\pm}$. The trap zero positions are denoted $r_0^{(1)}$ for the first and $r_0^{(2)}$ for the second atom. Exemplarily, the second atom is displaced by $\delta r^{(2)}$.

II.1.1. Diatomic Array

In order to obtain a Hamiltonian for the diatomic array, we start by defining a co-ordinate system where we place the first atom at position 0 on the x -axis and the second one at position d . The equilibrium atom positions are denoted $r_0^{(1)}$ and $r_0^{(2)}$, both real numbers, Fig. II.1 (a). The corresponding bosonic modes are labelled $a^{(1)}$ and $a^{(2)}$. If we apply the expressions for the gradient and the Hessian of the potential $V(|r^{(2)} - r^{(1)}|)$, Eq. (I.45) and Eq. (I.46), we find

$$(\nabla V)(r_0^{(2)}, r_0^{(1)}) = V'(d)[1, -1], \quad (\nabla^t \otimes \nabla V)(r_0^{(2)}, r_0^{(1)}) = V''(d) \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \quad (\text{II.1})$$

We observe the second contribution to the Hessian is missing as a consequence of considering a one-dimensional problem. With the gradient and Hessian at hand, we find the Hamiltonian

$$H_2 = \Delta(n^{(1)} + n^{(2)}) + \Omega(\sigma_x^{(1)} + \sigma_x^{(2)}) + \omega(a^{(1)\dagger}a^{(1)} + a^{(2)\dagger}a^{(2)}) + \left[\mu_{21} + \kappa_{21}(a^{(1)} - a^{(2)} + \text{h.c.}) + \frac{\xi_{21}}{2}(a^{(1)} - a^{(2)} + \text{h.c.})^2 \right] n^{(2)}n^{(1)}. \quad (\text{II.2})$$

The subscript 2 emphasizes that this is the Hamiltonian for the diatomic array. We transform to the new bosonic modes $b_{\pm} = (a^{(2)} \pm a^{(1)})/\sqrt{2}$ by rotation, such that the Hamiltonian may be written (with the simplifying replacements $\mu_{21} \mapsto \mu$, $\kappa_{21} \mapsto \kappa$ and $\xi_{21} \mapsto \xi$) as:

$$H_2 = \Delta(n^{(1)} + n^{(2)}) + \Omega(\sigma_x^{(1)} + \sigma_x^{(2)}) + \omega b^{\dagger}b + [\mu + \sqrt{2}\kappa(b^{\dagger} + b) + \xi(b^{\dagger} + b)^2] n^{(2)}n^{(1)} \quad (\text{II.3})$$

Note that we have omitted the free-theory of mode b_+ and replaced b_- with b for brevity. Physically, the modes $b_{\pm}$ are clearly interpretable as the mode describing relative oscillation, b_- , and the mode responsible for collective shifts, b_+ , Fig. II.1 (b). Therefore, it is not surprising that b_+ decouples from the dynamics completely, since this would violate conservation of momentum guaranteed by Noether's theorem for a translation (with respect to the centre of mass) invariant system. Next, we represent the Hamiltonian in the partially rotated basis ($|\downarrow\downarrow\rangle$, $|\pm\rangle = 1/\sqrt{2}(|\downarrow\uparrow\rangle \pm |\uparrow\downarrow\rangle)$, $|\uparrow\uparrow\rangle$), where $|\downarrow\rangle$, $|\uparrow\rangle$ is the ground and excited state of the atom, respectively. For the Hamiltonian we thus find the matrix

$$H_2 = \begin{bmatrix} \omega b^{\dagger}b & \sqrt{2}\Omega & 0 \\ \sqrt{2}\Omega & \omega b^{\dagger}b + \Delta & \sqrt{2}\Omega \\ 0 & \sqrt{2}\Omega & \omega b^{\dagger}b + \sqrt{2}\kappa(b^{\dagger} + b) + \xi(b^{\dagger} + b)^2 + 2\Delta + \mu \end{bmatrix}. \quad (\text{II.4})$$

Note that we have omitted the row and column corresponding to $|\rightarrow\rangle$, as it decouples from the dynamics. This is merely a consequence of the driving laser, i.e. the laser responsible for Ω , irradiating

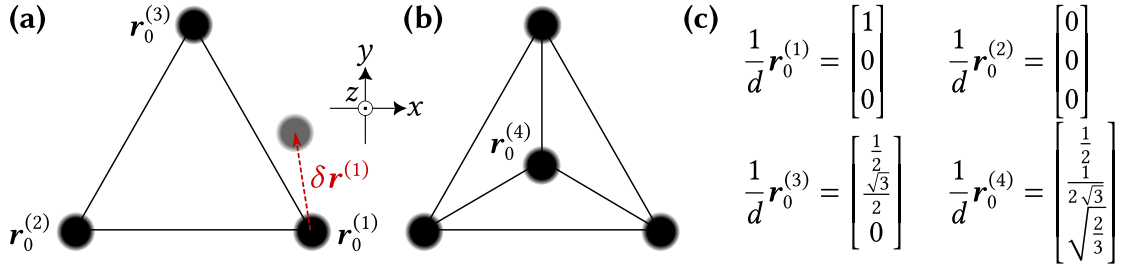


Figure II.2.: Sketch of Triangle and Tetrahedron. Sketch of (a) triangular array and (b) tetrahedral array. The trap zero positions are denoted $\mathbf{r}_0^{(k)}$ for atom k and given explicitly in (c). Note that the co-ordinate frame is chosen compatible with the triangle and tetrahedron position vectors.

the array homogeneously. Finally, the Rabi frequency Ω is increased by a factor of $\sqrt{2}$ as a result of the rotated basis. Introducing the facilitation condition $\mu + \Delta = 0$, we observe that the row and column associated to $|\downarrow\downarrow\rangle$ is decoupled to order $|\Omega/\Delta| \ll 1$, Ref. [73], and the remaining electronic dynamics ($\kappa, \xi = 0$) is resonant. As a result, the effective Hamiltonian is given by

$$H_2 = \begin{bmatrix} \omega b^\dagger b & \sqrt{2}\Omega \\ \sqrt{2}\Omega & \omega b^\dagger b + \sqrt{2}\kappa(b^\dagger + b) + \xi(b^\dagger + b)^2 \end{bmatrix}, \quad (\text{II.5})$$

for an initial state in the manifold $|+\rangle, |\uparrow\uparrow\rangle$. Note that we have shifted the baseline of the energy by adding $-\Delta$. This Hamiltonian provides a clear picture on the situation: If either atom is in its ground state, we describe the vibrational degrees of freedom via a harmonic oscillator of frequency ω . In case both atoms are in their excited states, the phonon dynamics is governed by a *shifted* and *squeezed* harmonic oscillator, because of the gradient and Hessian of the dipole-dipole interaction potential, respectively.

II.1.2. Equilateral Triangle and Tetrahedron

After considering a diatomic array, we have seen the last contribution of the Hessian in Eq. (I.46) not to play a role. This will change in higher dimensions. Note that for any form of atom arrangement which is one dimensional, this contribution does not come into play.

In order to obtain a Hamiltonian for the triangular and tetrahedral array, we again start by defining atom equilibrium positions $\mathbf{r}_0^{(1)}, \mathbf{r}_0^{(2)}, \mathbf{r}_0^{(3)}$ and $\mathbf{r}_0^{(4)}$, where the first three vectors describe the triangular array and the fourth one extends it to a tetrahedron. A sketch of both configurations is presented in Fig. II.2. For the triangular array the gradients result in the following expressions:

$$(\nabla V)(\mathbf{r}_0^{(2)}, \mathbf{r}_0^{(1)}) = \frac{\sqrt{2}}{x_0} \kappa [-1, 1] \otimes [-1, 0] \quad (\text{II.6})$$

$$(\nabla V)(\mathbf{r}_0^{(3)}, \mathbf{r}_0^{(1)}) = \frac{\sqrt{2}}{x_0} \kappa [-1, 1] \otimes [-\frac{1}{2}, \frac{3}{2}] \quad (\text{II.7})$$

$$(\nabla V)(\mathbf{r}_0^{(3)}, \mathbf{r}_0^{(2)}) = \frac{\sqrt{2}}{x_0} \kappa [-1, 1] \otimes [\frac{1}{2}, \frac{\sqrt{3}}{2}] \quad (\text{II.8})$$

Symmetry of the array allows to define the collective electron-phonon coupling $\kappa = \kappa_{21} = \kappa_{31} = \kappa_{32}$. Analogously, the Hessian is obtained and the corresponding coupling constants $\xi = \xi_{21} = \xi_{31} = \xi_{32}$ and $\nu\kappa, \nu = \nu_{21} = \nu_{31} = \nu_{32}$. These simplifications are the result of choosing arrays

with edges of equal length; this is why we call the dumbbell, (equilateral) triangle and tetrahedron the most ‘simple’ arrays. Note that the coupling $\nu\kappa$ appears solely because the array is not one-dimensional, opposed to the diatomic array, where no such coupling occurred and the Hamiltonian was not explicitly dependent on the ratio $\nu = x_0/d$. For the Hamiltonian in the electronic basis ($|\downarrow\downarrow\downarrow\rangle, |\uparrow\uparrow\downarrow\rangle, |\downarrow\uparrow\downarrow\rangle, |\downarrow\downarrow\uparrow\rangle, |\downarrow\uparrow\uparrow\rangle, |\uparrow\downarrow\uparrow\rangle, |\uparrow\uparrow\downarrow\rangle, |\uparrow\uparrow\uparrow\rangle$), the following expression is obtained:

$$H_3 = 1_8 \otimes \omega \mathbf{a}^\dagger \mathbf{a} + \begin{bmatrix} 0 & 0 & \Omega & 0 & \Omega & 0 & \Omega & 0 \\ 0 & 2\Delta + \mu + h_1 & \Omega & 0 & 0 & 0 & \Omega & \Omega \\ \Omega & \Omega & \Delta & \Omega & 0 & 0 & 0 & 0 \\ 0 & 0 & \Omega & 2\Delta + \mu + h_3 & \Omega & 0 & 0 & \Omega \\ \Omega & 0 & 0 & \Omega & \Delta & \Omega & 0 & 0 \\ 0 & 0 & 0 & 0 & \Omega & 2\Delta + \mu + h_5 & \Omega & \Omega \\ \Omega & \Omega & 0 & 0 & 0 & \Omega & \Delta & 0 \\ 0 & \Omega & 0 & \Omega & 0 & \Omega & 0 & 3\Delta + 2\mu \end{bmatrix} \quad (\text{II.9})$$

$$h_m = \kappa \mathbf{u}_m \cdot (\mathbf{a}^{\dagger t} + \mathbf{a}) + \frac{1}{2}(\mathbf{a}^{\dagger t} + \mathbf{a})^\dagger (\xi v_m + \kappa \nu w_m) (\mathbf{a}^{\dagger t} + \mathbf{a}), \quad m = 1, 3, 5 \quad (\text{II.10})$$

Note that this choice of basis renders the centre (6×6) -block circulant in the Rabi frequency Ω . Moreover, transposition shall only act on $\mathbb{C}^n$, not on the bosonic Hilbert spaces, and the Hamiltonians h_m represent electron-phonon interactions, where the vectors $\mathbf{u}_m$ and the matrices v_m and w_m result from the expressions in Eq. (I.45) and Eq. (I.46). Next, we demand the facilitation constraint $\Delta + \mu = 0$, in order to decouple the first and last basis vector. The remaining states in the basis we rename to $|1\rangle, \dots, |6\rangle$, such that, up to magnitudes of $|\Omega/\Delta|$, cf. Tab. I.2, the Hamiltonian takes the form

$$H_3 = \omega \mathbf{a}^\dagger \mathbf{a} + \sum_{m=1}^6 [|m\rangle\langle m+1| + \text{h.c.}] + \sum_{m=1,3,5} h_m P_m, \quad |7\rangle \equiv |1\rangle, \quad (\text{II.11})$$

after shifting the energy base line by $-\Delta$. In the above expression the projectors $P_m = |m\rangle\langle m|$ are defined and the Rabi Hamiltonian rewritten in terms of the 6 remaining basis states $|m\rangle$. This Hamiltonian is exactly the one considered in Ref. [37], however, without the two contributions from the Hessian. For physical intuition it useful to keep the *tight-binding* picture in mind, suggested in the same work. Hopping between the atomic states defines a ring structure, where the Rabi frequency describes motion along the ring-lattice, Fig. II.3 (a). It is clear that the modes $\mathbf{a}$ are not particularly suited for the problem. However, there is no set of modes that can be used to eliminate mixing of modes on all atomic subspaces simultaneously, which can be seen by the fact that, generally, $[\xi v_m + \nu \kappa w_m, \xi v_{m'} + \nu \kappa w_{m'}] \neq 0$, i.e. not all operators h_m ($m = 1, 3, 5$) can be diagonalized simultaneously with the same unitary. Therefore, we have to take another approach in finding modes reflecting the problem’s symmetry. To do so, we observe that the structure of Eq. (II.9) suggests that the electronic ground state, $|\text{GS}\rangle$, must asymptotically approach the ground state of the hopping Hamiltonian for $\Omega \rightarrow \infty$. This is because for large enough Ω , compared to the energy scale of the phonons, the system will not be influenced significantly by the diagonal entries. As the ground state must reflect the symmetry of the problem, we define the eigenmodes of the system to be eigenvectors of the matrix obtained by projection to the atomic ground state:

$$\langle \text{GS} | \sum_{m=1,3,5} h_m P_m | \text{GS} \rangle = (\mathbf{a}^{\dagger t} + \mathbf{a})^\dagger M (\mathbf{a}^{\dagger t} + \mathbf{a}) + \text{linear in } \mathbf{a}, \quad M = \frac{1}{6} \sum_{m=1,3,5} (\xi v_m + \nu \kappa w_m) \quad (\text{II.12})$$

In this particular case, the atomic ground state can be directly calculated since the matrix is circulant, with the result $|\text{GS}\rangle = \sum_{m=1,3,5} [|m\rangle - |m+1\rangle]$. Diagonalizing the matrix M with an orthogonal

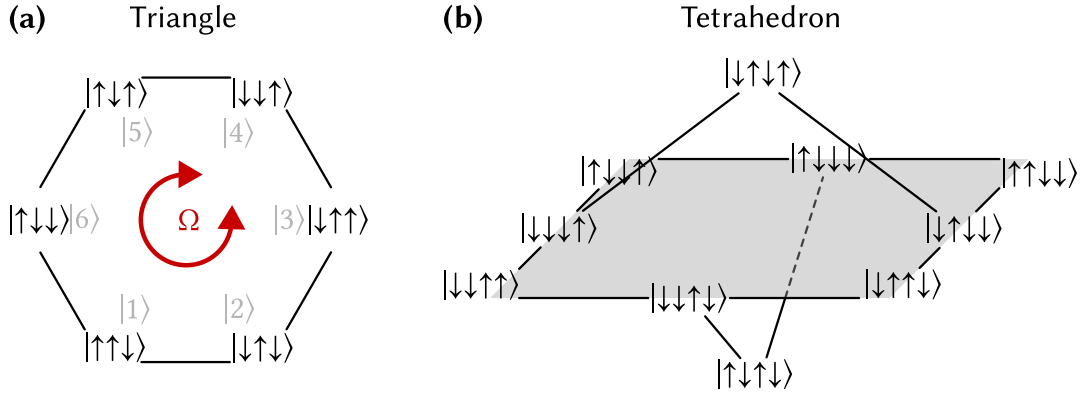


Figure II.3.: Tight-Binding Models for Laser Drive. Visualization of Rabi Hamiltonians as hopping in a tight-binding model lattice. (a) Ring lattice from Ref. [37] for triangle. (b) Lattice structure for tetrahedron.

transformation (rotation),

$$T = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2\sqrt{3}} & -\frac{1}{2\sqrt{3}} & 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{2\sqrt{3}} & -\frac{1}{2\sqrt{3}} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{3}} & 0 \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2\sqrt{3}} & -\frac{1}{2\sqrt{3}} & 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{2\sqrt{3}} & -\frac{1}{2\sqrt{3}} & \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{3}} & 0 \\ 0 & 0 & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & 0 & 0 & \frac{1}{\sqrt{3}} & 0 \end{bmatrix}, \quad (\text{II.13})$$

indeed, yields the same normal modes $\mathbf{b} = T^\dagger \mathbf{a}$, found in the supplemental material of Ref. [37]. Two eigenvalues of the resulting diagonal matrix are zero, such that only four modes remain. From T we close that the last two modes, b_5 and b_6 must decouple as they describe collective shifts of the triangle, again forbidden by conservation of momentum. It might be surprising to see that four modes remain and not three, as one could expect from the triangle's edges similarity to the diatomic array. For the purpose of visualization Fig. II.4 shows the displacement characteristics of the (non-decoupling) modes $\mathbf{b}$. Note that the canonical displacements, defined in Fig. II.4 (a), are *not* orthogonal, such that we continue with the modes b_1 , b_2 and b_3 , instead. b_4 is renamed to b_t for 'twist'. For the Hamiltonian one arrives at the expression in Eq. (II.11), with the transformed free-theory of the phonons and the following bosonic electron-phonon interaction Hamiltonians ($m = 1, 3, 5$):

$$h_m = \sqrt{2}\kappa(\ell_m^\parallel + \text{h.c.}) + \xi(\ell_m^\parallel + \text{h.c.})^2 + \frac{\nu\kappa}{\sqrt{2}}(\ell_m^\perp + \text{h.c.})^2 \quad (\text{II.14})$$

$$\ell_1^\parallel = \frac{1}{\sqrt{2}}(b_1 - b_2), \quad \ell_1^\perp = \frac{1}{\sqrt{2}}(b_3 + b_t) \quad (\text{II.15})$$

$$\ell_3^\parallel = \frac{1}{\sqrt{2}}\left(b_1 + \frac{1}{2}b_2 - \frac{\sqrt{3}}{2}b_3\right), \quad \ell_3^\perp = \frac{1}{\sqrt{2}}\left(\frac{\sqrt{3}}{2}b_2 + \frac{1}{2}b_3 - b_t\right) \quad (\text{II.16})$$

$$\ell_5^\parallel = \frac{1}{\sqrt{2}}\left(b_1 + \frac{1}{2}b_2 + \frac{\sqrt{3}}{2}b_3\right), \quad \ell_5^\perp = \frac{1}{\sqrt{2}}\left(-\frac{\sqrt{3}}{2}b_2 + \frac{1}{2}b_3 - b_t\right) \quad (\text{II.17})$$

Note that the Hamiltonians h_m have the same structure on each subspace as a result of the eigenvalues of the matrices v_m and w_m not depending on m . This circumstance is a direct manifestation of the 120° rotational symmetry of the Hamiltonian. Only the linear combinations of the modes b_1 , b_2

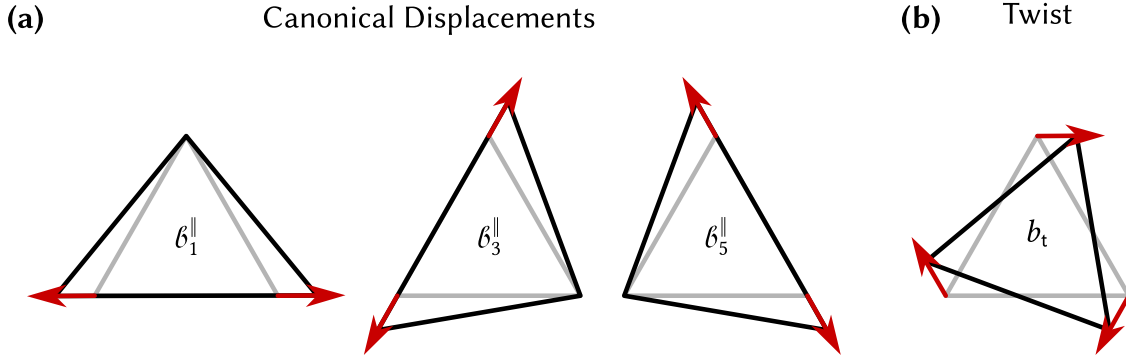


Figure II.4.: Characteristic Displacements for Triangle. Displacement characteristics of the eigenmodes $\mathbf{b}$ for a triangular atom array. In grey, the undisturbed triangle is drawn as a reference. In black, the displaced triangle is shown. The direction of the red arrows is dictated by the columns of T in Eq. (II.13). (a) Linear combinations of modes b_1 , b_2 and b_3 , yielding the *canonical displacements*, $\delta_m^\parallel$. (b) Mode ‘twisting’ the triangular shape, b_t , also *not* decoupling from dynamics.

and b_3 are different on each subspace. Further, we observe that the mode b_t only couples through $\nu\kappa$. This is expected, because no analogous mode was present for the diatomic array. Interestingly, without the Hessian the mode b_t decouples. The operators $\delta_m^\parallel$ and $\delta_m^\perp$ earn their names by describing displacements parallel and perpendicular to the atomic distance vector, defined by the two excited atoms on the subspace $|m\rangle$. It should again be emphasized that $\delta_m^{\parallel,\perp}$ are *not* bosonic modes, further emphasized by the different kind of letters used for notation. However, on each subspace separately, they fulfil bosonic algebra:

$$[\delta_m^{\parallel\dagger}, \delta_m^\perp] = 0 \quad (\text{II.18})$$

From this result we take that the free-theory for the bosons on subspace $m = 1, 3, 5$, *not* absorbed in the operators h_m , is simply rewritten as

$$\omega \mathbf{b}^\dagger \mathbf{b} = \omega \delta_m^{\parallel\dagger} \delta_m^\parallel + \omega \delta_m^{\perp\dagger} \delta_m^\perp + \omega c_m^{\dagger 1} c_m^1 + \omega c_m^{\dagger 2} c_m^2 \quad (\text{II.19})$$

on each subspace, where c_m^1 and c_m^2 are some modes acting on different subspaces than $\delta_m^{\parallel,\perp}$. In combination these results illustrate that the operators $\delta_m^{\parallel,\perp}$ are the ‘correct’ modes in case one considers only a single subspace m and diagonalizes the Hamiltonian h_m . In other words: We can diagonalize the Hamiltonian h_m , however, the modes in which this Hamiltonian is diagonal are not independent of the modes, i.e. do not act on different subspaces, needed to diagonalize $h_{m'}$, for $m \neq m'$. This incompatibility is a direct manifestation of the fact that displacements on one edge of the triangle affect the displacements of the remaining two edges. Each bosonic Hamiltonian for the triangular array is now completely expressed in the intuitive displacement operators $\delta_m^{\parallel,\perp}$ and the, on each subspace individually, decoupled modes $c_m^{1,2}$, i.e. $[c_m^{1\dagger}, \delta_{m'}^{\parallel,\perp}] \neq 0$, for $m \neq m'$.

For the tetrahedral array we select an analogous manifold of atomic states, i.e. where either one or two atoms are excited. This assures comparability to the result for the triangular array. Moreover, this decision is again implemented via the facilitation constraint. We may expect more electron-phonon contributions coupling through $\nu\kappa$, as more spatial directions allow for more complex vibrational motion. We number all possible electronic states in the following way: $|1; 1\rangle = |\downarrow\downarrow\downarrow\rangle, \dots, |1; 4\rangle = |\uparrow\downarrow\downarrow\rangle$ and $|2; 1\rangle = |\uparrow\uparrow\downarrow\rangle, |2; 2\rangle = |\downarrow\downarrow\uparrow\rangle, |2; 3\rangle = |\uparrow\downarrow\uparrow\rangle, |2; 4\rangle = |\downarrow\uparrow\uparrow\rangle, |2; 5\rangle = |\uparrow\uparrow\uparrow\rangle, |2; 6\rangle = |\downarrow\uparrow\uparrow\rangle$, where the first number denotes the number of excited atoms on the subspace and the second number the particular state, e.g. $|2; 3\rangle$ would be state (randomly) numbered 3 on the

subspace where two atoms are excited. There is no sense in numbering the states in a ‘linear’ fashion as it was done for the triangle, because hopping, defined by the Rabi Hamiltonian, is not governed by motion on a simple ring lattice structure, but rather on the object depicted in Fig. II.3 (b), not allowing closed boundary conditions. Although there are differences to the triangular array, we can also write the Hamiltonian for the tetrahedron in the same fashion as for the triangle in Eq. (II.11),

$$H_4 = \omega \mathbf{a}^\dagger \mathbf{a} + \Omega \sum_{(m,n) \in C} [|2; m\rangle \langle 1; n| + \text{h.c.}] + \sum_{m=1}^6 h_m P_{2;m}, \quad (\text{II.20})$$

with the set of all connections between states, C , shown in Fig. II.3 (b) and the projectors $P_{2;m} = |2; m\rangle \langle 2; m|$. To express the bosonic Hamiltonians h_m in modes tailored to the problem, we again make use of the ground state of the electronic Hamiltonian:

$$|\text{GS}\rangle = -\frac{1}{2\sqrt{2}} \sum_{n=1}^4 |1; n\rangle + \frac{1}{\sqrt{12}} \sum_{m=1}^6 |2; m\rangle \quad (\text{II.21})$$

An interesting side effect of the more complex tight-binding lattice structure is that ‘flat bands’ emerge. Namely, the two eigenstates

$$|\text{FB}, 1\rangle = \frac{1}{2}(-|2; 1\rangle - |2; 2\rangle + |2; 3\rangle + |2; 4\rangle), \quad |\text{FB}, 2\rangle = \frac{1}{2}(-|2; 1\rangle - |2; 2\rangle + |2; 5\rangle + |2; 6\rangle) \quad (\text{II.22})$$

span a basis of the eigenspace associated to the eigenvalue 0. For the triangular array, such states were not possible. However, they might be of interest for practical applications, because if the system is in one of these states, a quench in Ω does not change the electronic state (to order $|\Omega/\Delta|$). This mechanism may aid in preparing states of interest in the simulator. As an example: Suppose the system is in state $|a\rangle$ and $\Omega = 0$. If this state is desired at $\Omega \neq 0$, an adiabatic ramp in Ω could be employed to increase Ω without changing the state, Ref. [80]. However, it is generally favourable to avoid adiabatic ramps when preparing experimentally relevant states. The reason for this is rooted in the fact that coherent time evolution is limited, Ref. [42]. Therefore, if the adiabatic ramp for preparing the state of interest is long, there is less time left to do physics. Coming back to the problem of finding suitable modes, we project the Hamiltonian to the atomic ground state, analogously to the procedure for the triangle before. For the matrix that mixes the modes we obtain $\sum_{m=1}^6 (\xi v_m + \nu \kappa w_m)$. This matrix has only real-valued components, yielding again a rotation T which is used to define the normal modes $\mathbf{b}$ via $\mathbf{b} = T^\dagger \mathbf{a}$. For completeness, the rotation T is given in Supp. S.2. This time, three modes decouple, as a result of the three spatial directions for collective shifts. In total, 9 modes $b_1, \dots, b_9$ remain. Analogous to the triangle we identify canonical displacements. For each edge, there is a parallel displacement operator $b_m^\parallel$, where $m = 1, \dots, 6$ denotes the subspace associated to the projector $P_{2;m}$. Exemplarily, the displacement operator for the subspace where atom 1 and atom 2 are excited is given in terms of the normal modes $\mathbf{b}$:

$$b_1^\parallel = \frac{1}{\sqrt{3}} b_1 - \frac{1}{\sqrt{6}} b_5 + \frac{1}{\sqrt{3}} b_6 + \frac{1}{\sqrt{6}} b_8 \quad (\text{II.23})$$

The operators $b_{m=2,\dots,6}^\parallel$ are obtained by appropriate 60° and 120° rotations of the coefficient vector used to define $b_1^\parallel$. If the Hessian of the dipole-dipole interaction potential would not be considered, we could express the Hamiltonian fully in the operators $b_m^\parallel$. However, as we have seen already in case of the triangular array, the Hessian introduces contributions with displacements perpendicular to the atomic distance vector. To each edge of the tetrahedron there are associated two faces, such that

we expect two perpendicular displacement operators for each subspace $P_{2,m}$ – one for displacements within each surface. Indeed, this is what is found by direct calculation. In particular we find the perpendicular displacement operators to be given by appropriate 60° and 120° rotations of

$$\mathcal{b}_{12,3}^\perp = \frac{1}{3\sqrt{2}}b_3 + \frac{2}{3}b_4 - \frac{1}{\sqrt{3}}b_7 + \frac{1}{\sqrt{6}}b_9, \quad (\text{II.24})$$

$$\mathcal{b}_{12,4}^\perp = \left(\frac{1}{6\sqrt{2}} + \frac{1}{\sqrt{3}}\right)b_3 + \left(\frac{1}{3} - \frac{1}{2\sqrt{6}}\right)b_4 + \left(-\frac{1}{2\sqrt{2}} - \frac{1}{2\sqrt{3}}\right)b_7 + \left(-\frac{1}{2} + \frac{1}{2\sqrt{6}}\right)b_9, \quad (\text{II.25})$$

where $\mathcal{b}_{12,3}$ describes displacements perpendicular to the distance vector of atoms 1 and 2 in the plane of the triangle with corner atoms 1, 2, and 3. This set of canonical displacements operators suffices to express the Hamiltonian in a clear fashion. For ease of notation, we collect the perpendicular operators in a vector $\mathcal{b}_m^\perp$. Note that the operators $\mathcal{b}_m^\parallel, \mathcal{b}_{12,3}^\perp$ and $\mathcal{b}_{12,3}^\perp$ obey bosonic algebra *only* on each individual subspace:

$$[\mathcal{b}_m^\parallel, \mathcal{b}_m^\perp] = 0, \quad m = 1, \dots, 6 \quad (\text{II.26})$$

Rewriting the Hamiltonian in Eq. (II.20) in the $\mathcal{b}$ -modes, results in the expression

$$h_m = \sqrt{2}\kappa(\mathcal{b}_m^\parallel + \text{h.c.}) + \xi(\mathcal{b}_m^\parallel + \text{h.c.})^2 + \frac{\nu\kappa}{\sqrt{2}}(\mathcal{b}_m^\perp + \text{h.c.})^2, \quad (\text{II.27})$$

for the electron-phonon interaction Hamiltonians.

Intuitively, this form of the Hamiltonian is clear: The perpendicular displacements are necessary to describe the ‘mixing’ process, where with the word ‘mixing’ we mean the change of electronic subspace by hopping along the tight-binding lattices in Fig. II.3 for $\Omega \neq 0$. For instance: Let the array be displaced along the distance vector of atom 1 and atom 2 and reside in the subspace $|2; 1\rangle$. If the laser drive mixes in the subspace $|2; 2\rangle$ by hopping to the corresponding site, the already displaced atoms must adjust to the new displacement signature by performing perpendicular motion. This observation also explains why no such contributions appear for the diatomic array: There is no other displacement signature to mix in. Further, it is clear why we cannot eliminate mixing of modes for the triangle and tetrahedron Hamiltonian collectively for all electronic subspaces: If atom 1 and atom 2 are displaced along their distance vector, it is a simple geometric argument which implies that also the perpendicular displacements of the remaining subspaces $|2; m\rangle$ must play a role in that they become displaced.

Before turning to the next section, we want to underline numerically that all modes are important parts of the dynamics and will become excited during time-evolution. As an example we take the triangular array, more precisely, we take a closer look at the associated canonical displacements as well as b_t . Displacements in the mode b_t may seem somewhat unintuitive, because it breaks the problems symmetry. We take initial states of the form $|\psi_0\rangle \otimes |0\rangle_{b_t}$. Note that initially the mode b_t is in the vacuum. For an initial state respecting the Hamiltonians symmetry, there are no displacements in b_t (at least up to machine precision 10^{-16}), as can be seen in Fig. II.5 (e). In contrast, Fig. II.5 (f) shows that finite displacements are found for an initial state, $|\psi_0\rangle$, breaking the Hamiltonians symmetry. In panels Fig. II.5 (a-d) the canonical modes are shown, where even for the symmetry conserving initial state in panel (a) at time $t = 0$, the observables $\mathcal{b}_m^\perp$ are displaced, as a direct consequence of the problems geometry. Finally, against intuition, Fig. II.5 (e) shows that even for the symmetric initial state the mode b_t cannot be omitted, since it hosts excitations for $t > 0$, evident from $\langle b_t^\dagger b_t \rangle > 0$.

In summary, we have found in this section that our ‘simple’ arrays described by the Hamiltonian of our simulator, exhibit interesting motional dynamics. Additionally, we have rewritten the Hamiltonians in a more transparent form, facilitating further examination as well as giving us intuition on the physics side of the problem.

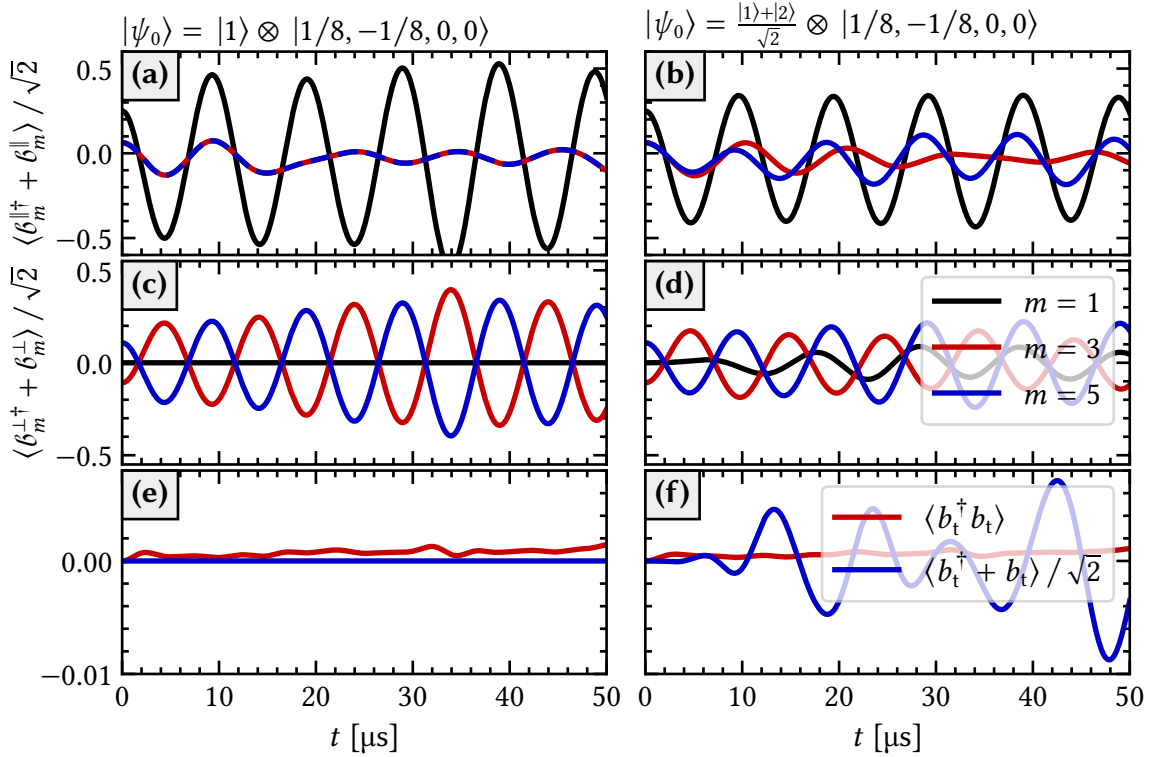


Figure II.5: Time Evolution. Time evolution of the canonical displacement operators, $\hat{b}_m^{\parallel,\perp}$, (a-d) and the ‘twist’ mode b_t (e, f) for the triangular array, Fig. II.4. Panels in the left column show time evolution for the symmetry conserving initial state $|\psi_0\rangle \otimes |0\rangle_{b_t} = |1\rangle \otimes |1/8, -1/8, 0, 0\rangle$. In the right column the symmetry breaking initial state $|\psi_0\rangle \otimes |0\rangle_{b_t} = (|1\rangle + |2\rangle)/\sqrt{2} \otimes |1/8, -1/8, 0, 0\rangle$ is evolved. Note that all panels have the same time axis. The legend in panel (d) is the same as for panels (a-c) and the legend in panel (f) is the same as in panel (e). Parameters are set to $\kappa = 0.1\omega$, $\xi = 0.05\omega$ and $\nu = 0.3$. The Rabi frequency is set to $\Omega = 0.3\omega$ and the trap frequency is assumed to be $\omega = 2\pi \cdot 100$ kHz. In each mode maximally 30 excitations were allowed for the simulation.

II.2. Structural Instabilities from Electron-Phonon Coupling

In the previous subsection we have derived Hamiltonians for a diatomic array and an equilateral triangle and tetrahedron. Also we have gained intuition on the vibration signatures encoded through electron-phonon coupling. Now we want to go further in exploring the phenomena rooted in the gradient and Hessian of the dipole-dipole interaction potential. Therefore, we wish to take a look at the spectrum of the aforementioned Hamiltonians, and in particular the ground state energy. This path is motivated by Ref. [37], where the ground state energy dependence on Ω is also used to demonstrate non-trivial behaviour. Particularly, a Jahn-Teller regime, i.e. a regime in which the triangular shape is distorted to minimize the ground state energy (this will briefly be discussed later in Sec. II.3), is unveiled. We want to make use of this approach to prove the emergence of instabilities. To do so, we employ a Bogoliubov transformation to determine the critical coupling strengths. We approach this by calculating an analytical expression for the ground state energy in the limits $\Omega = 0$ and $\Omega \rightarrow \infty$. With the limit $\Omega \rightarrow \infty$ we mean that we consider $\Omega \gg \omega, \kappa, \xi$. For the sake of simple notation we summarize these conditions with ‘ $\Omega \rightarrow \infty$ ’. Finally, we conduct a numerical analysis to

look at the instabilities also from a dynamical perspective, in case of finite laser drives Ω .

We structure the proceedings of this section into two parts. In the first subsection we discuss the technicalities needed to obtain the instabilities. In the second subsection, we derive and discuss the instabilities for the three array shapes.

II.2.1. 1-D. Bogoliubov Transformation

In this subsection we derive a Bogoliubov transformation (BT) which will aid us in finding instabilities of the three arrays. Further, this section helps to get some technicalities cleared up before we turn attention to the physics in the next subsection. Starting point of the discussion is a Hamiltonian which is general enough to include all cases of the bosonic Hamiltonians, h_m , emergent for the considered arrays. For real valued coupling constants $\alpha, \beta, \gamma, \delta$ and the assumption $\alpha > 0$, we define the quadratic (bosonic) family of Hamiltonians

$$h(\alpha, \beta, \gamma, \delta) = \alpha b^\dagger b + \beta(b + b^\dagger) + \gamma(b + b^\dagger)^2 + \delta, \quad (\text{II.28})$$

with the annihilation and creation operators b and $b^\dagger$, respectively. Of course, the constant δ is arbitrary, however, it is introduced in this case to make notation more consistent and later facilitate comparison to our specific cases. In order to diagonalize h , we need to eliminate contributions of the form b^2 and $b^{\dagger 2}$. For the new modes c , resulting from the BT of b , we write

$$c = ub + vb^\dagger, \quad c^\dagger = u^*b^\dagger + v^*b, \quad |u|^2 + |v|^2 = 1, \quad (\text{II.29})$$

where the complex coefficients u and v are introduced. The last equation ensures that the transformation is symplectic, i.e. leaves bosonic algebra invariant, and the first two equations ensure the transformation to be in the subset of Bogoliubov transformations, within the larger group of symplectic maps. Finally, the asterisk shall denote complex conjugation. If we substitute the backwards transformation

$$b = u^*c - vc^\dagger, \quad b^\dagger = uc^\dagger - v^*c, \quad (\text{II.30})$$

into the Hamiltonian in Eq. (II.28), we arrive at the expression

$$\begin{aligned} h(\alpha, \beta, \gamma, \delta) = & ((\alpha + 2\gamma)(|u|^2 + |v|^2) - 2\gamma(uv^* + u^*v))c^\dagger c \\ & + [(\gamma(u^2 + v^2) - (\alpha + 2\gamma)uv)c^{\dagger 2} + \text{h.c.}] + \beta((u - v)c^\dagger + \text{h.c.}) \\ & + (\alpha + 2\gamma)|v|^2 + \gamma + \delta - \gamma(uv^* + u^*v). \end{aligned} \quad (\text{II.31})$$

We clearly eliminate contributions of the form c^2 and $c^{\dagger 2}$ by demanding u and v to obey the equation

$$\gamma(u^2 + v^2) - (\alpha + 2\gamma)uv = 0. \quad (\text{II.32})$$

Note that two complex numbers u and v constitute 4 degrees of freedom. One is a global phase to the operators c , not fixed by the bosonic algebra. Additionally, we are allowed to put the relative phase between u and v completely into v and thus demand $u \geq 0$ without loss of generality, since we do not lose any physical solutions by this particular choice. Next, we observe that in case $u = 0$, $|u|^2 - |v|^2 = 1$ cannot be satisfied. Thus, no solutions are discarded by assuming a strictly positive u , such that we may divide Eq. (II.32) by u^2 , yielding

$$\gamma(1 + x^2) - (\alpha + 2\gamma)x = \gamma x^2 - (\alpha + 2\gamma)x + \gamma = 0, \quad (\text{II.33})$$

with the ratio $x = v/u$. Physical solutions¹, x_{ph} , are unique for $u > 0$ and can be expressed in the following way:

$$x_{\text{ph}}(\alpha, \gamma) = \begin{cases} 1 + \frac{\alpha}{2\gamma} - \text{sign}(\frac{\alpha}{2\gamma})\sqrt{(1 + \frac{\alpha}{2\gamma})^2 - 1} & ; \quad \frac{\gamma}{\alpha} > -\frac{1}{4}, \gamma \neq 0 \\ 0 & ; \quad \gamma = 0 \\ \text{no solution} & ; \quad \text{else} \end{cases} \quad (\text{II.34})$$

This result may be somewhat surprising, because it tells us that we cannot write the Hamiltonian in terms of a creation and annihilation operator with bosonic algebra. For now we will take this result as it is and will later give an intuitive explanation for this behaviour. Moreover, to arrive at this expression one must be careful to not loose solutions at any point, such that one can be sure that indeed no physical solutions exist in the regime $2\gamma/\alpha \in (-\infty, -1/2]$. For brevity, steps to arrive at Eq. (II.34) are placed in the supplemental material Supp. S.1. Finally, we note that because the bosonic Hamiltonians in Eq. (I.48) all have the same structure as in Eq. (II.10), for all possible array shapes on a single electronic subspace, this result is applicable to all artificial molecular systems, in the sense of Eq. (I.48). This is important because it tells us that, at least mathematically, the Hamiltonian we can simulate always comes with the interesting property of a regime where the BT is not well-defined.

With the physical solutions x_{ph} for x at hand, we find that $u^2 - |v|^2 = 1$ yields the (unique for the choice $u > 0$) solutions

$$u(\alpha, \gamma) = \frac{1}{\sqrt{1 - x_{\text{ph}}^2(\alpha, \gamma)}}, \quad v(\alpha, \gamma) = \frac{x_{\text{ph}}(\alpha, \gamma)}{\sqrt{1 - x_{\text{ph}}^2(\alpha, \gamma)}}. \quad (\text{II.35})$$

Both solutions are real valued, such that we can omit complex conjugation in Eq. (II.31). We find the Hamiltonian, after substituting the solutions for u and v , in the form

$$h(\alpha, \beta, \gamma, \delta) = \alpha' c^\dagger c + \beta' (c^\dagger + c) + \gamma' (c^{\dagger 2} + c^2) + \delta', \quad (\text{II.36})$$

with the following constants:

$$\alpha' = (\alpha + 2\gamma) \frac{1 + x_{\text{ph}}^2(\alpha, \gamma)}{1 - x_{\text{ph}}^2(\alpha, \gamma)} - 4\gamma \frac{x_{\text{ph}}(\alpha, \gamma)}{1 - x_{\text{ph}}^2(\alpha, \gamma)} = \alpha \left(1 + 4\frac{\gamma}{\alpha}\right)^{1/2} \quad (\text{II.37})$$

$$\beta' = \beta \sqrt{\frac{1 - x_{\text{ph}}(\alpha, \gamma)}{1 + x_{\text{ph}}(\alpha, \gamma)}} = \beta \left(1 + 4\frac{\gamma}{\alpha}\right)^{-1/4} \propto (\gamma - \gamma_c)^{-a_c} \quad (\text{II.38})$$

$$\gamma' = 0 \quad (\text{II.39})$$

$$\delta' = \delta + \gamma + (\alpha + 2\gamma) \frac{x_{\text{ph}}^2(\alpha, \gamma)}{1 - x_{\text{ph}}^2(\alpha, \gamma)} - 2\gamma \frac{x_{\text{ph}}(\alpha, \gamma)}{1 - x_{\text{ph}}^2(\alpha, \gamma)} = \delta + \frac{\alpha}{2} \left(1 + 4\frac{\gamma}{\alpha}\right)^{1/2} - \frac{\alpha}{2} \quad (\text{II.40})$$

The second equal signs are obtained by somewhat tedious, but straight forward algebraic steps, such that they are omitted here. Of course, γ' is equal to zero by construction. Observe that β' diverges for $\gamma \searrow \gamma_c = -\alpha/4$ with ‘critical exponent’ $a_c = 1/4$. As a final step in diagonalizing the Hamiltonian, we change the frame of reference with the unitary transformation $U(\alpha', \beta') = \exp(-\beta'/\alpha'(c^\dagger - c))$. In particular, we find the expression

$$U^\dagger(\alpha', \beta') h(\alpha, \gamma) U^\dagger(\alpha', \beta') = \alpha' c^\dagger c - \frac{\beta'^2}{\alpha'} + \delta', \quad (\text{II.41})$$

¹ With ‘physical solutions’ we mean solutions where $|x| < 1$; otherwise $|u|^2 - |v|^2 = |u|^2(1 - |x|^2) \neq 1$.

yielding the energy spectrum (for $n = 0, 1, 2, \dots$):

$$E_n(\alpha, \beta, \gamma, \delta) = \alpha \sqrt{1 - \gamma/\gamma_c} n - \frac{\beta^2}{\alpha} \frac{1}{1 - \gamma/\gamma_c} + \delta + \frac{\alpha}{2} \sqrt{1 - \gamma/\gamma_c} - \frac{\alpha}{2} \quad (\text{II.42})$$

Clearly we see diverging energies in the limit $\gamma \searrow \gamma_c$, diverging as $(\gamma - \gamma_c)^{-b_c}$ with exponent $b_c = 1$. As a final remark, the additional terms $\delta' - \delta$ are a direct consequence of the non-zero canonical commutator of the mode b , such that they are a direct manifestation of the ‘quantum nature’ of the problem. The significance of these commutator contributions (CC) will be discussed later in this work. They will not show up in a semi-classical description, applied later in this thesis. Therefore, the CC turn out to be a neat way of directly accessing ‘quantum corrections’ in an analytical fashion.

II.2.2. Instabilities of Particular Molecular Shapes

Attention shall now be directed at the particular shapes, dumbbell, triangle and tetrahedron. This subsection is the second part of this section, where we apply the technical discussion from before to the arrays, in order to demonstrate the emergence of instabilities and compute the critical values. Starting point is the dumbbell; more precisely the Hamiltonian for it, given in Eq. (II.5). Note that the facilitation constraint is already applied in this expression. Comparing the Hamiltonian on the subspace $|\uparrow\uparrow\rangle$ (22-entry in the matrix of Eq. (II.5)) to the expression for h in Eq. (II.28) allows to identify the constants $\alpha = \omega, \beta = \sqrt{2}\kappa, \gamma = \xi$ and $\delta = 0$. In the Bogoliubov transformed mode c , the Hamiltonian of the diatomic array takes the form, Eq. (II.36):

$$H_2 = \begin{bmatrix} \omega b^\dagger b & \sqrt{2}\Omega \\ \sqrt{2}\Omega & \omega \sqrt{1 + 4\xi/\omega} c^\dagger c + \frac{\sqrt{2}\kappa}{(1+4\xi/\omega)^{1/4}} (c^\dagger + c) + \frac{\omega}{2} \sqrt{1 + 4\xi/\omega} - \frac{\omega}{2} \end{bmatrix}. \quad (\text{II.43})$$

This form of the Hamiltonian gives rise to a clear interpretation of the system, constrained by the facilitation condition: If the system resides in the atomic state $|+\rangle = (|\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle)/\sqrt{2}$, the free theory of mode b describes the vibrational dynamics. On the subspace $|\uparrow\uparrow\rangle$ the mode c describes spatial distortions, because on this subspace both atoms reside in the Rydberg state, i.e. the vibrational mode describes a different kind of distortions due to the contributions of the dipole-dipole interaction potential. The homogeneously applied laser generating the Rabi frequency $\sqrt{2}\Omega$ mixes the different vibration signatures. We find the instability is defined solely by the coupling constant $\xi \sim V''(d)$, because undefined terms in the Hamiltonian only occur as a consequence of $1 + 4\xi/\omega \leq 0$. This implies the emergence of this specific instability to strictly be a result of the potential’s curvature, not its gradient, $\kappa \sim V'(d)$. For the remainder of this thesis, the critical coupling constant introduced by the curvature is denoted $\xi_c = -\omega/4$.

Next, we want to understand the origin of the instability. For $\xi \searrow \xi_c$ the excitation energy of the oscillator with mode c vanishes, i.e. exciting a phonon does not cost energy. If we move deeper into the unstable regime, α' is expected to become negative, such that the system can even gain energy upon excitation of a phonon, such that also the modes b can be excited arbitrarily high, after giving the dynamics enough time to explore the far regions of phase space. We conclude that the potential introduced by the electron-phonon interaction Hamiltonian is able to cancel the parabolic potential of the trap, until it becomes flat at $\xi = \xi_c$ and even flips sign for $\xi < \xi_c$. By considering time evolution in a numerical description of the system, this behaviour is evident in the fact that it becomes more challenging to obtain a converged result in the Fock space cut-off dimension for the bosonic modes. In Fig. II.6 results for system dynamics are shown for different values of ξ and different Fock space cut-off dimensions in the case of a triangular array (which will turn out to have

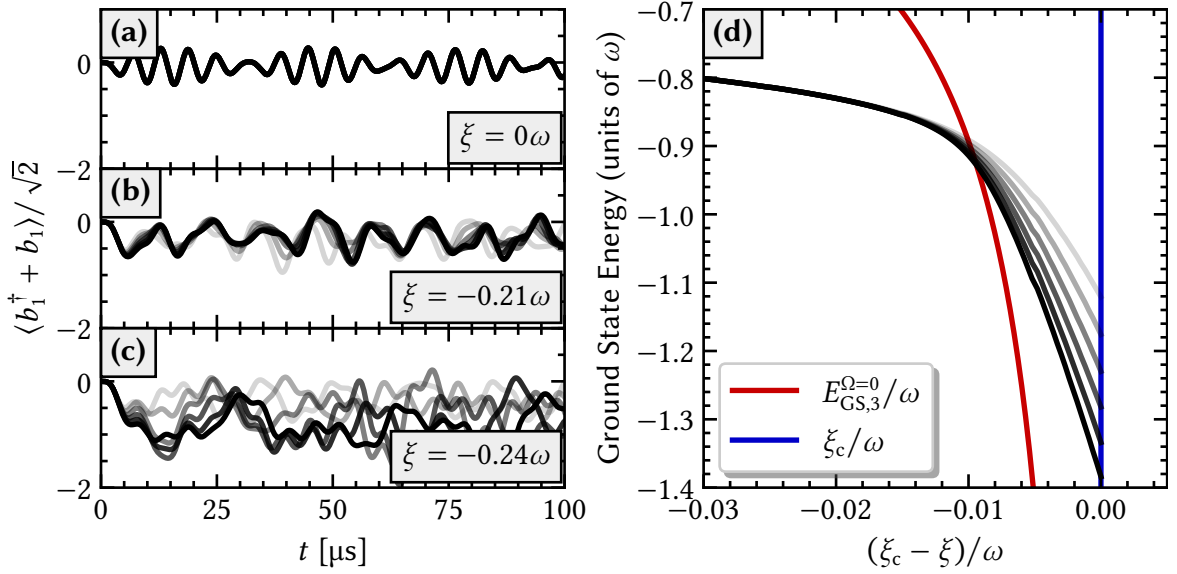


Figure II.6.: Instability in Time Evolution and Energy for the Triangle. (a-c) Time evolution of symmetric displacement (breathing), $\langle b_1^\dagger + b_1 \rangle / \sqrt{2}$, in units of the harmonic oscillator length, x_0 , for a triangular atom array. Parameters for the simulation are set to $\Omega = 0.3\omega$, $\kappa = 0.1\omega$, $\nu = 0.1$ and the initial state is chosen as $|\psi_0\rangle = (|2\rangle + |4\rangle + |6\rangle)/\sqrt{3} \otimes |0, 0, 0, 0\rangle$, with the vacuum of all four b -modes, $|0, 0, 0, 0\rangle$. More opaque lines represent larger allowed phonon numbers per mode, ranging from maximally 5 to 10 excitations per mode. Near the critical ξ the lines show larger discrepancy, i.e. are not yet converged. (d) In red, the ground state energy at $\Omega = 0$, $E_{\text{GS},3}^{\Omega=0}$, is plotted. Black lines show the ground state energy, $E_{\text{GS},3}$ at $\Omega = 0.1\omega$, depending on ξ . Here, the maximally allowed excitations per mode vary from 10 to 15 (most opaque line), indicating the instability to lay at $\xi_c = -\omega/4$, also for $\Omega \neq 0$ (note small values on x-axis of panel (d)). The remaining parameters are set to the same ones as for panels (a-c).

this instability as well, later). Finally, we want to understand the explicit value of ξ_c . The definition of ξ allows to rewrite the inequality characterising the *stable* regime:

$$\xi > -\frac{\omega}{4} \iff -x_0^2 V''(d) < T_0 = U_0, \quad T_0 = \frac{1}{2} m(x_0 \omega)^2 \quad (\text{II.44})$$

Here, m denotes the atomic mass and x_0 the harmonic oscillator length of the trap. Therefore, T_0 is the characteristic kinetic energy of the trapping potential and, because of the (quantum version) of the virial theorem, U_0 the potential energy scale. Hence, the system becomes unstable if the characteristic potential energy of the trap is ‘cancelled’ by electron-phonon coupling at $\xi = \xi_c$. In practice, there is no ‘hard’ transition into instability. This can be seen by looking at the expectation value of the displacement operator in the b -mode, which is related to the c -mode through the BT in Eq. (II.30):

$$\langle b^\dagger + b \rangle = \frac{1}{(1 - \xi/\xi_c)^{1/4}} \langle c^\dagger + c \rangle \quad (\text{II.45})$$

This quantity, generally, will become large over the course of time evolution, because near the instability, excitations of the c -mode come with almost no energy cost and, additionally, the coefficient diverges with exponent $a_c = 1/4$. However, if this expectation value becomes large, the derivation of the system’s Hamiltonian breaks down in the first place, as for large enough atomic displacements

the dipole-dipole approximation, magic-wavelength trapping and the expansion of the potential are not justified; cf. Chap. I. In summary we state the following:

If the characteristic potential energy $-x_0^2 V''(d)$ (positive near the instability), introduced by quadratic electron-phonon coupling, is comparable to the potential energy scale of the trapping potential, U_0 , trapping becomes more challenging and the Hamiltonian must be extended to describe the system correctly. Physically, the condition for a *stable* system, described by our Hamiltonian, is summarized with

$$|U_0 + x_0^2 V''(d)| \sim \omega. \quad (\text{II.46})$$

This statement also shows that in case the potentials curvature V'' is chosen large and positive, i.e. $|U_0 + x_0^2 V''(d)| \gg \omega$, the array's spatial shape is stabilized, keeping Eq. (II.45) in mind. Namely, the denominator becomes large and, as a consequence, atomic displacements small and costly (in terms of energy) to excite.

In order to achieve our goal of finding an analytical expression for the ground state energy for $\Omega = 0$, we transform into a displaced frame by applying the unitary transformation

$$U = |+\rangle\langle +| + \exp\left(-\frac{\sqrt{2}\kappa}{\omega(1-\xi/\xi_c)^{3/4}}(c^\dagger - c)\right) |\uparrow\uparrow\rangle\langle\uparrow\uparrow| \quad (\text{II.47})$$

to the Hamiltonian in Eq. (II.43). Explicitly, we find

$$U^\dagger H_2 U = \begin{bmatrix} \omega b^\dagger b & \sqrt{2}\Omega \exp\left(-\frac{\sqrt{2}\kappa}{\omega(1-\xi/\xi_c)^{3/4}}(c^\dagger - c)\right) \\ \sqrt{2}\Omega \exp\left(\frac{\sqrt{2}\kappa}{\omega(1-\xi/\xi_c)^{3/4}}(c^\dagger - c)\right) & \omega\sqrt{1-\xi/\xi_c}c^\dagger c - \frac{2\kappa^2}{1-\xi/\xi_c} + \frac{\omega}{2}\sqrt{1-\xi/\xi_c} - \frac{\omega}{2} \end{bmatrix}. \quad (\text{II.48})$$

When $\Omega = 0$, we can just read off the spectrum, and in particular the ground state energy:

$$E_{\text{GS},2}^{\Omega=0} = \omega \cdot \min(0, \mathcal{E}), \quad \mathcal{E} = -\frac{2\kappa^2}{\omega^2} \frac{1}{1-\xi/\xi_c} + \frac{1}{2}\sqrt{1-\xi/\xi_c} - \frac{1}{2} \quad (\text{II.49})$$

Evidently, the instability in ξ is linked to an effective increase in κ and additional terms from the non-zero commutator of the mode c becoming ill-defined. If we go to very large ξ , the constant κ is effectively reduced and the ground state is given by the symmetric sum of the single excited electronic states, $|+\rangle$, hosting no spatial distortion. This is in accordance with the statement that large ξ stabilize the array's shape. For the counter case $\Omega \rightarrow \infty$, the displaced Hamiltonian does not help in finding the ground state energy. However, from the form of Eq. (II.5) we close that in this limit the ground state must approach a ground state of product form² $(|+\rangle - |\uparrow\uparrow\rangle)/\sqrt{2} \otimes |\text{phonons}\rangle$. Therefore, to obtain the ground state energy, we need to diagonalize the projected Hamiltonian

$$\frac{1}{2}(\langle +| - \langle \uparrow\uparrow|)H_2(|+\rangle - |\uparrow\uparrow\rangle) = -\sqrt{2}\Omega + \omega b^\dagger b + \frac{\kappa}{\sqrt{2}}(b^\dagger + b) + \frac{\xi}{2}(b^\dagger + b)^2. \quad (\text{II.50})$$

From identification with $h(\omega, \kappa/\sqrt{2}, \xi/2, -\sqrt{2}\Omega)$, we arrive at the ground state energy. Eq. (II.42),

$$E_{\text{GS},2}^{\Omega \rightarrow \infty} \approx -\sqrt{2}\Omega - \frac{\kappa^2}{2\omega} \frac{1}{1-\xi/(2\xi_c)} + \frac{\omega}{2}\sqrt{1-\xi/(2\xi_c)} - \frac{\omega}{2}. \quad (\text{II.51})$$

² There are two electronic eigenstates, namely $(|+\rangle \pm |\uparrow\uparrow\rangle)/\sqrt{2}$, with associated energies $\pm\sqrt{2}\Omega$. For large Ω the state with energy $-\sqrt{2}\Omega$ becomes more and more favourable as the ground state. Therefore, we close that for $\Omega \rightarrow \infty$, we obtain the product form $(|+\rangle - |\uparrow\uparrow\rangle)/\sqrt{2} \otimes |\text{phonons}\rangle$ for the ground state.

The approximation sign emphasizes the assumption of a factorized ground state. Formally, we cannot take the limit $\Omega \rightarrow \infty$, but the approximation is justified if $\Omega \gg \omega, |\kappa|, |\xi|$ and $|\kappa/(1 - \xi/\xi_c)| \ll \Omega$, i.e. the parameters are such that the ‘near criticality’ is overshadowed by the ‘largeness’ of Ω . We again summarize these conditions by writing $\Omega \rightarrow \infty$, because in this limit all of these conditions are fulfilled. From these conditions we also note that the ground state energy must asymptotically approach the slope in Ω , Eq. (II.51), as Ω becomes larger. On first sight, the expression for the energy may seem wrong, because the instability is not at ξ_c , but rather it is located below ξ_c , deeper inside the unstable regime. Note that ξ_c is *not* dependent on Ω , since Eq. (II.43), hosts a diverging coupling constant in front of the $c^\dagger + c$ contribution regardless of Ω . Mathematically, this seemingly present contradiction is explained by the fact that the limits $\Omega \rightarrow 0$ and $\xi \searrow \xi_c$ do not commute. A physical interpretation will be given later, as the Born-Oppenheimer energy surfaces will provide a clear picture, see Sec. III.2, of the situation. However, this is included already in the conditions for a factorizing ground state, in particular in the assumption $|\kappa/(1 - \xi/\xi_c)| \ll \Omega$. If we are close to the instability, the left hand side becomes large and one must increase Ω even further to achieve factorization.

Now that we have unveiled that electron-phonon coupling can be responsible for instabilities in the system, we want to examine the triangular and tetrahedral array. Motivating this path is the fact that the instability defined by ξ_c is not depending on the artificial molecules spatial structure (other than the atomic distance), such that it is anticipated to find this instability in any atomic array. In Sec. II.1 it was observed that the triangular and tetrahedral array gain additional electron-phonon interactions from the vector structure of the Hessian³. Interestingly, these contributions host their own instability, as can be seen directly by writing the Hamiltonian for the triangular array in the Bogoliubov transformed modes. To arrive at this expression we proceed in the following way: First, we look at the subspaces corresponding to the projectors P_m ($m = 1, 3, 5$) individually. On these subspaces we have the bosonic Hamiltonians defined by the free-theory in Eq. (II.19) and the interactions h_m , Eq. (II.14):

$$\omega b_m^{\parallel\dagger} b_m^{\parallel} + \omega b_m^{\perp\dagger} b_m^{\perp} + \omega c_m^{\dagger 1} c_m^1 + \omega c_m^{\dagger 2} c_m^2 + \sqrt{2}\kappa(b_m^{\parallel} + \text{h.c.}) + \xi(b_m^{\parallel} + \text{h.c.})^2 + \frac{\nu\kappa}{\sqrt{2}}(b_m^{\perp} + \text{h.c.})^2 \quad (\text{II.52})$$

Next, analogous to the procedure for the triangle, we make use of the BT to rewrite each of these Hamiltonians ($m = 1, 3, 5$). To do so, we identify the expression in Eq. (II.52) with

$$\omega c_m^{\dagger 1} c_m^1 + \omega c_m^{\dagger 2} c_m^2 + h_{b_m^{\parallel}}(\omega, \sqrt{2}\kappa, \xi, 0) + h_{b_m^{\perp}}(\omega, 0, \nu\kappa/\sqrt{2}, 0). \quad (\text{II.53})$$

The Hamiltonian h_a is the Hamiltonian given in Eq. (II.28) with respect to the bosonic mode a . To the modes resulting from the BT we assign the names $c_m^{\parallel}$ and $c_m^{\perp}$. The names c_m^1 and c_m^2 are kept, because the BT for these modes is trivial. Before we write down the Hamiltonian we perform a final transformation into a displaced frame of reference, defined by the unitary

$$U = (P_2 + P_4 + P_6) + \sum_{m=1,3,5} U_m P_m, \quad U_m = \exp\left(-\frac{\sqrt{2}\kappa}{\omega(1 - \xi/\xi_c)^{3/4}}(c_m^{\parallel\dagger} - c_m^{\parallel})\right). \quad (\text{II.54})$$

Note the similarity to Eq. (II.47) in the discussion for the dumbbell. With this transformation we cancel linear terms in the modes $c_m^{\parallel}$ on each subspace. Finally, we arrive at the Hamiltonian in the

3 With ‘vector structure’ we relate to the fact that $(\nabla^i \otimes \nabla^j)\|\mathbf{x}\| = 1 - \hat{\mathbf{x}} \otimes \hat{\mathbf{x}}^i$, $\hat{\mathbf{x}} = \mathbf{x}/\|\mathbf{x}\|$ is non-zero for dimensions larger than 1.

displaced frame:

$$\begin{aligned}
 U^\dagger H_3 U &= \Omega h_\Omega + \omega \mathbf{b}^\dagger \mathbf{b} (P_2 + P_4 + P_6) \\
 &+ \omega \sum_{m=1,3,5} \left[\sqrt{1 - \xi/\xi_c} c_m^\parallel c_m^\parallel + \sqrt{1 + 2\sqrt{2}\nu\kappa/\omega} c_m^{\perp\dagger} c_m^\perp + c_m^{\perp\dagger} c_m^\perp + c_m^{2\dagger} c_m^2 \right] P_m \\
 &+ \omega \left[-\frac{2\kappa^2}{\omega^2} \frac{1}{1 - \xi/\xi_c} + \frac{1}{2} \sqrt{1 - \xi/\xi_c} + \frac{1}{2} \sqrt{1 + 2\sqrt{2}\nu\kappa/\omega} - 1 \right] (P_1 + P_3 + P_5) \quad (\text{II.55})
 \end{aligned}$$

In this expression the Hamiltonian $h_\Omega = U^\dagger \sum_{m=1}^6 (|m\rangle\langle m+1| + \text{h.c.}) U$ is the transformed Rabi Hamiltonian. It is of a similar form as the off diagonal entries in Eq. (II.48) for the dumbbell. We do not give the explicit expression here because it is not relevant in the case $\Omega = 0$, i.e. the case of interest for us, since $\Omega h_\Omega \rightarrow 0$, for $\Omega \rightarrow 0$. Note that we are only interested in the case $\Omega = 0$, because for $\Omega \neq 0$ we cannot give analytical results and for numerical discussions we do not need h_Ω either, because we directly use Eq. (II.11) in the b -modes. As a reminder, the b -modes were defined by the rotation given in Eq. (II.13). As a side remark, note that for $\kappa, \xi \rightarrow 0$ the modes $c_m^\parallel$ are equal to the operators $b_m^{\parallel,\perp}$, obeying bosonic algebra on each individual subspace $m = 1, 3, 5$. From Eq. (II.55) we immediately identify the ground state energy at $\Omega = 0$ to be

$$E_{\text{GS},3}^{\Omega=0} = \omega \cdot \min(0, \mathcal{E}), \quad \mathcal{E} = -\frac{2\kappa^2}{\omega^2} \frac{1}{1 - \xi/\xi_c} + \frac{1}{2} \sqrt{1 - \xi/\xi_c} + \frac{1}{2} \sqrt{1 - \kappa/\kappa_c} - 1. \quad (\text{II.56})$$

Evidently, we obtain a new instability, this time in the coupling constant κ rooted in the gradient of the dipole-dipole potential, with critical coupling strength $\kappa_c = -\omega/(2\sqrt{2}\nu)$. The instability caused by the coupling constant $\nu\kappa$ is more subtle, since it does not manifest in increasingly large atomic displacements near the critical value κ_c . In fact, the ground state energy at $\Omega = 0$ does not even diverge continuously, but is of finite value for all $\kappa > \kappa_c$ and abruptly becomes negative infinity for $\kappa < \kappa_c$. It also depends explicitly on the ratio of the harmonic oscillator length to the equilibrium atom distance, $\nu = x_0/d \sim 0.1$. As a consequence, κ_c will typically be an order of magnitude larger than ξ_c , at least roughly. It means that this instability can only be implemented experimentally for large ξ in the sense that $\kappa/(1 - \xi/\xi_c)$ must be kept small enough to yield physically sensible displacements, such that the dipole approximation, magic-trapping and the expansion of the dipole-dipole potential is still justified, cf. Chap. I. Finally, notice that this instability is explicitly dependent on the arrays spatial shape, not only the atomic distance. The structure is contained in the pre-factor $1/(2\sqrt{2})$. A clear picture is obtained by plotting the ground state energy at $\Omega = 0$ for different values of κ , ξ and ν , Fig. II.7 (a, b).

For completeness, we note that an analogous procedure can be employed to obtain the ground state energy of the tetrahedral array, Fig. II.7 (c, d),

$$E_{\text{GS},4}^{\Omega=0} = \omega \cdot \min(0, \mathcal{E}), \quad \mathcal{E} = -\frac{2\kappa^2}{\omega^2} \frac{1}{1 - \xi/\xi_c} + \frac{1}{2} \sqrt{1 - \xi/\xi_c} + \sqrt{1 - \kappa/\kappa_c} - \frac{3}{2}. \quad (\text{II.57})$$

Evidently, a factor of 2 is gained in front of the energy contributions from the perpendicular modes, compared to the result for the triangle. Clearly, this is a consequence of the two perpendicular modes arising from the additional spatial degree of freedom in 3 dimensions and the fact that each tetrahedron edge is part of *two* triangular arrays.

From the plots of $E_{3,\text{GS}}^{\Omega=0}$ and $E_{4,\text{GS}}^{\Omega=0}$ in Fig. II.7, it becomes clear that the ‘correct linear combinations’, i.e. the state to 0th-order perturbation theory, defining the ground state, are parameter dependent, opposed to the result without the Hessian from Ref. [37]. This is a manifestation of the trapping

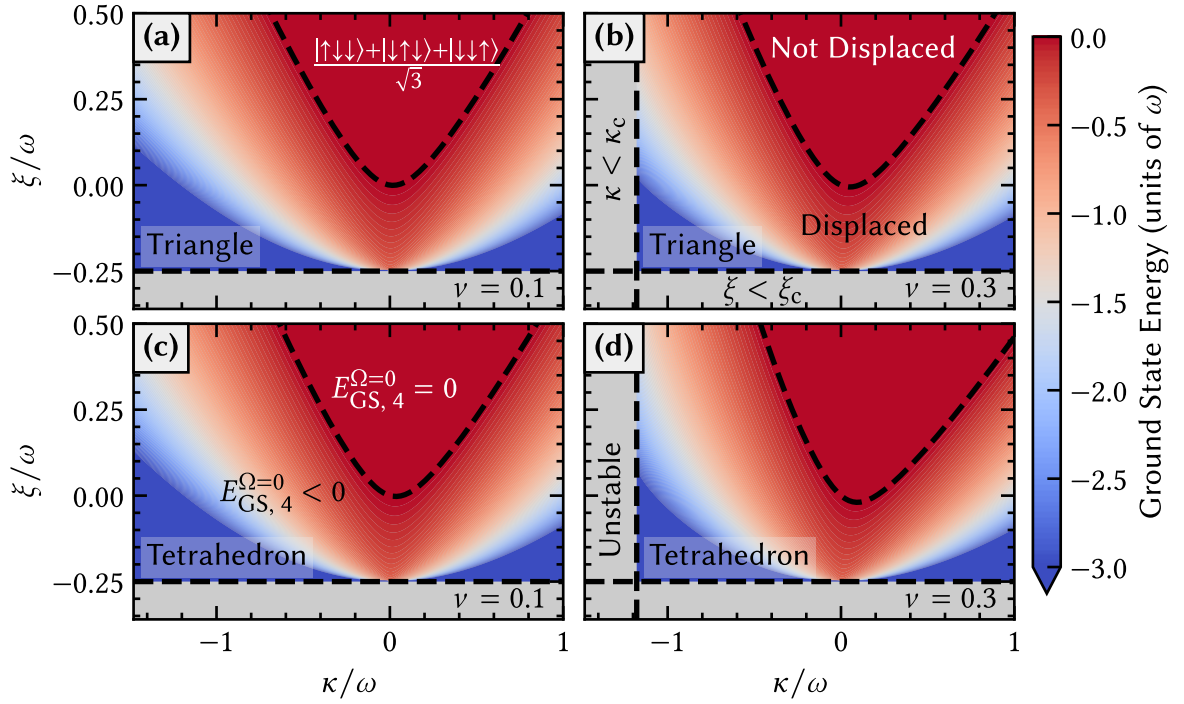


Figure II.7.: Ground State Energy for Triangle and Tetrahedron. Ground state energy, $E_{\text{GS},3/4}^{\Omega=0}$, of the triangular (a, b) and tetrahedral (c, d) array for $\Omega = 0$ and two different values of ν . Energy is given in units of the trapping frequency ω . Axis are equal in all four panels. In grey, the unstable regimes are marked and the colour-axis is clipped to -3ω for visibility. For the triangular array, the electronic ground state in the regime $E_{\text{GS},3}^{\Omega=0} = 0$ is shown in the corresponding area of the plot. Because only single excited electronic states are present, all displacements vanish. Dashed black lines mark the critical line at which (and beyond) the displacements vanish, because $\xi \sim V''(d)$ stabilizes the array in the tweezers. For $\nu = 0.1$, κ_c lays outside the plotted area. The visibility of this regime is compromised to keep the subplots comparable and the colour structure well distinguishable. There are no further instabilities to the top and right of each subplot.

potential being increased by electron-phonon coupling, i.e. stabilizing the array structure.

Finally, we can also give analytical results for the ground state energy in the limit of $\Omega \rightarrow \infty$ for the triangle, analogous to the result for the diatomic array. The reason for this is that for large Ω the system, again, must approach a ground state of product form, consisting of the electronic ground state and some bosonic state, yet to be determined. The explicit expression for the Hamiltonian and identifications with $\hbar$ are moved to Supp. S.3, for brevity. For the ground state energy we find the expression

$$E_{\text{GS},3}^{\Omega \rightarrow \infty} = -2\Omega - \frac{\kappa^2}{4\omega} \frac{1}{1 - \xi/(4\xi_c)} + \frac{\omega}{2} \left[\sqrt{1 - \xi/(4\xi_c)} + 2\sqrt{1 - \xi/(8\xi_c) - \kappa/(8\kappa_c)} + \sqrt{1 - \kappa/(4\kappa_c)} - 4 \right]. \quad (\text{II.58})$$

We can see that, just as it was the case for the diatomic array, the coupling constant ξ_c is shifted deeper into the unstable regime. In fact, the same holds true for κ_c . This implies that both instabilities are not seen by the system anymore, such that it is clear that the limit $\Omega \rightarrow \infty$ does not commute

with $\kappa \searrow \kappa_c$ either. Note how there are square roots containing $\nu\kappa$ and ξ as a linear combination, which was not the case for $\Omega = 0$. This stems from the fact that for $\Omega > 0$ there is mixing of the parallel, $b_m^{\parallel}$, and perpendicular operators, $b_m^{\perp}$, with the ones on the remaining subspaces m' , with $m' \neq m$. In the limit of $\Omega = 0$, we cannot see this, because the mixing of subspaces is ‘turned off’.

Of course, the same process can also be applied for the tetrahedral array, however, there does not come new physical insight with this result, such that it is not considered. Comparison of the analytical results to exact numerics is pushed to the next section, because it fits well into Fig. II.8, which cannot be discussed at this point, since there was no proper introduction of the Jahn-Teller regime.

II.3. Jahn-Teller Regime and Structural Transition

In this last section we want to review a result from Ref. [37], motivated already in the introduction of this chapter. In that work, the triangular array is discussed *without* contributions from the Hessian of the dipole-dipole interaction potential. It is reported that a transition between two regimes, hosting fundamentally different displacement signatures, can be observed. Namely, a Jahn-Teller (JT) regime is found for small Ω with respect to the electron-phonon coupling constants and the frequency of the trap. To identify the JT regime, the ground state energy and the ground state is considered, which shows that for small Ω the triangular array is (to good approximation) in a symmetric linear combination of symmetry broken triangular shapes. For large Ω , we have already seen that the ground state approximately factorizes and only the mode b_1 , describing symmetric breathing, is displaced. For instance, this is evident from the form of the Hamiltonian given in Supp. S.3, where linear contributions in bosonic modes, leading to finite displacements, only appear for b_1 . Clearly, this regime is fundamentally different from the Jahn-Teller regime. A pictorial version of this (rather simplified) argument is provided in the bottom right of panel (d) in Fig. II.8, for clarity.

In the ground state energy, the JT regime is manifest in the downwards ‘bend’ of the energy band. With ‘bending’ we refer to the fact that the ground state energy curves do not follow a simple slope for small Rabi frequencies Ω , as is evident in Fig. II.8. This ‘bending’ is a result of the system distorting the triangular shape, in order to reduce the energy. In addition to that, the degeneracy of the ground state is lifted. Further, upon measurement of the electronic state, this mechanism introduces spontaneous symmetry breaking of the spatial array shape. The combination of these phenomena is generally known under the name ‘Jahn-Teller effect’, Ref. [40, 41, 81], from which this regime inherits its name. The reason we take a look at this result in this thesis is twofold: First, it shows that in our simulator phenomena from molecular physics, indeed, are observable, which is exactly what we are searching for. On the other hand, we are interested in how this transition changes form in our simulator. Note that our simulator also hosts the contributions from the Hessian, not solely the gradient. Up to this point, it is unclear what the implications of the instabilities, introduced by the Hessian, are, regarding this transition. Motivated by this open question, we want to generate the spectrum for the dumbbell and the triangle. To this end, attention shall be directed at Fig. II.8. In panel (c) and (d), the JT regime is indicated in grey by virtue of an order parameter⁴ which takes the value 1 at $\Omega = 0$, where the JT regime is most prominent. Such a parameter allows us to quantify how prominent the JT regime still is for different Rabi frequencies Ω . In the limit of $\Omega \rightarrow \infty$ this order

⁴ The specific choice of this order-parameter becomes clear in Chap. IV. Details are omitted here because they are not essential at this point and would only lead to a more convoluted discussion. The notion of a Jahn-Teller regime will also become more clear in the next chapter.

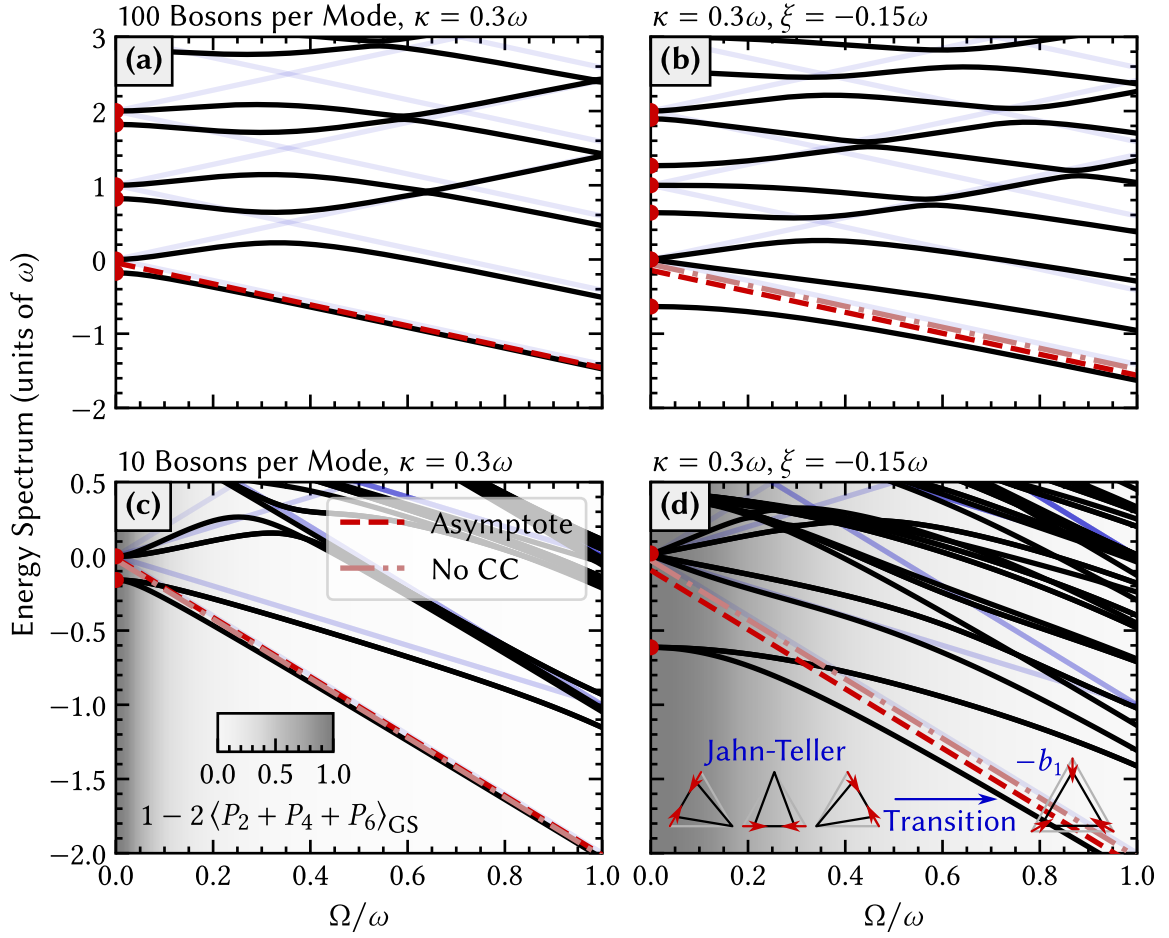


Figure II.8.: Energy Spectrum for Dumbbell and Triangle. Energy spectrum for diatomic (a, b) and triangular (c, d) array. Dashed red lines represent the asymptotes for the ground state energy in the limit of $\Omega \rightarrow \infty$, $E_{\text{GS},3/4}^{\Omega \rightarrow \infty}$. Dash-dotted light red lines show the results for $E_{\text{GS},3/4}^{\Omega \rightarrow \infty}$ without the commutator contributions (CC), rooted in Eq. (II.42), in order to understand their significance. The CC are given in Eq. (II.51) and Eq. (II.58). Note that in panel (a) no such contributions exist, such that the light red line is missing; see Eq. (II.51) for $\xi = 0$. Red dots represent the analytical results for $E_{\text{GS},3/4}^{\Omega=0}$. This also serves as a test if the spectrum is (numerically) converged in the Fock-space cut-off dimensions. In light blue, the energy spectrum for vanishing electron-phonon coupling constants $\kappa, \xi = 0$ is provided as a reference. Opaqueness of the blue lines shows their degeneracy. With this information, it can be closed which energies become non-degenerate because of finite κ, ξ . The Jahn-Teller regime is indicated in grey by plotting the order parameter's, $1 - 2\langle P_2 + P_4 + P_6 \rangle_{\text{GS}}$, dependence on Ω . Sketches of displaced triangles show the (dominant) contributions to the triangular ground state in the different regimes. Settings to reproduce the spectra are given on top of each panel.

parameter vanishes. We see that the JT regime is drastically widened for $\xi \searrow \xi_c$. A first intuitive explanation for this behaviour is that near the critical coupling strength ξ_c , the coupling constant κ^2 is effectively increased by the small denominator $1 - \xi/\xi_c$ in Eq. (II.56). Because the regime where the ground state energy is not a simple slope in Ω becomes wider for larger κ and more critical ξ ,

Fig. II.8 (c, d), one must apply even larger Rabi frequencies Ω , to render the symmetrically displaced triangular shape energetically favourable, i.e. bring the ground state energy dependence in Ω back to a slope. One may also see this the other way around: The curvature (chosen positive for this purpose) of the potential can be increased by tuning the MW field appropriately; cf. Sec. I.2.1. Therefore, the coupling constant κ^2 is not increased by the denominator $1 - \xi/\xi_c$, but rather decreased. As a result, the JT regime can be effectively removed for the array. Something like this is not possible without the interactions governed by the Hessian.

Perhaps, it would be interesting to see what kind of properties the JT regime can introduce in a whole lattice of triangles. This is of course not possible in an exact numerical fashion, because of the many atoms needed. However, this is exactly what the simulator would be useful for, because nature can handle any number of bosonic modes. Another example would be to consider arrays resembling octahedral complexes, such as SF_6 , in which the JT effect is a particularly prominent phenomenon, Ref. [81]. Here one could imagine that observables in the simulator are linked to macroscopic properties of such chemical compounds.

As a final remark, we observe also the dumbbell to show a ‘bend’ in the energy spectrum, although there is not a direct way to make sense of a JT regime; see panels (a, b) of Fig. II.8. The essence of this difference, compared to the triangular array, will become clear in the next chapter, where we inspect the Born-Oppenheimer energy surfaces with this question in mind. This shall already serve as a first motivation for the study of Born-Oppenheimer energy surfaces.

II.4. Conclusion

At this point we should quickly conclude what we have achieved in this chapter. This shall serve as an overview for the reader where we stand with our goals declared in the introduction. It has become clear that our simulation framework is able to implement non-trivial phenomena like instabilities and interesting vibrational signatures. Additionally, we have witnessed the possibility to simulate the Jahn-Teller effect and gave two side remarks on why the simulator could be useful with this effect in mind. Because the instabilities, Jahn-Teller effect and intriguing vibrational signatures are already encoded in the very simple, still numerically tractable, atomic arrays we have considered in this thesis, we are confident that our simulator is capable of producing non-trivial phenomena of interest and the simulation frameworks complexity (compared to molecules in nature) is sufficient⁵.

⁵ Generally, a quantum simulator is built in a more simple way than the system of interest. For instance, our arrays are much simpler quantum systems than molecules in nature. Therefore, one task in building a simulation framework is to convince ourselves that our simulator is not built in such a reduced form that most results become trivial.

III. Quantum Corrections and Jahn-Teller Regime

In the previous chapter we have considered three simple arrays; a dumbbell and an equilateral triangle and tetrahedron, all of which in the full quantum mechanical framework. We could observe that our simulator hosts non-trivial phenomena such as instabilities, interesting vibration signatures and a Jahn-Teller (JT) regime, as well as a concomitant structural phase transition, upon variation of the Rabi frequency Ω . These results motivate us to carry on with the discussion, because it seems that our simulator is working as intended. From this point onward, we reduce the discussion to the dumbbell and the triangle. Reason for this is that already for an array as simple as a tetrahedron, the numerical and analytical methods applied in this chapter become challenging. As an example, one could see that finding appropriate eigenmodes turned out to be quite tedious analytically. Moreover, exact simulations with 9 bosonic modes, generally, start to become challenging on a classical computer. Nonetheless, the considering of the tetrahedron earlier was not unfruitful. It has convinced us of the necessity for a quantum simulator.

As already pointed out, we have treated the system in the full quantum mechanical setting. For this we have used analytical as well as numerical methods. The general approach in describing molecules in nature is to simplify the problem by applying a Born-Oppenheimer approximation (BOA). Therefore, it is only natural to also apply such a procedure to our arrays. After all, if we try to mimic molecular systems, we should make sure that the go-to method in this field produces reasonable results, in the regimes where it is expected to. Importantly, this is to be seen as yet another proof of principle for our simulation framework. Apart from this, we are interested in a BOA in order to answer questions that arose in the previous chapter. This approach has proven fruitful already in Ref. [37], where the structural phase transition into and out of the JT regime could be made sense of with three degenerate minima in the ground state Born-Oppenheimer energy surface (GSBOES). For large enough Ω , it was shown that the minima approach each other, such that only a single one is left and there is no more spontaneous symmetry breaking. Specifically, we want to understand the connection between the JT regime and the instabilities we have found, because this has not become completely clear in the previous chapter. In particular, we have argued that the limits $\Omega \rightarrow 0$ and $\xi \searrow \xi_c$ do not commute. It will turn out that the GSBOES provides an intuitive reason for this circumstance. With the bigger picture of this thesis in mind, our result demonstrates that the typical Born-Oppenheimer (BO) approach, indeed, is also working well in our framework.

Another important aspect to proof the simulator to be useful is to investigate the ‘quantum nature’ of it. More precisely, it would be interesting to find out what part contributions stemming from the canonical commutation relations of the bosons play, in the transition into and out of the JT regime. This is relevant because it allows to identify what role quantum effects play in our simulation framework. Molecular physics undoubtedly relies on quantum mechanical phenomena, such that we must convince ourselves that we are able to reproduce practically relevant ones also in our simulator. We approach this question here by considering the triangular array, for which we compute the ground state energy in the framework of the BOA. Comparison of the results to the full quantum version of the problem yields a neat way to identify analytical expressions for ‘quantum corrections to the BOA’. This is possible in the limits of $\Omega \rightarrow 0$ and $\Omega \rightarrow \infty$. Those corrections are directly linked to the commutator contributions (CC) from the Hessian, found in Eq. (II.42). Further, a numerical analysis unveils that the non-zero commutators ‘wash-out’ the transition into and out

of the JT regime. The importance of the non-zero commutators to this transition was *not* previously discussed in the literature. Our results allow to close that our simulator is able to capture practically relevant quantum phenomena.

This chapter is structured into two main sections. In the first section we consider the dumbbell. This example gives us the chance to properly introduce the BOA and define where exactly approximations are applied. We further give conditions when the BOA is justified in our case. In addition, we clarify what is meant by ‘quantum corrections to the BOA’. In the second section we move on to the triangle. We discuss how well the BOA works for arbitrary Rabi frequencies Ω and investigate the significance of the quantum corrections with respect to the JT regime.

III.1. Introduction to the BOA and Quantum Corrections

Starting point of the discussion is the Hamiltonian in Eq. (II.5), describing the facilitated dynamics of the diatomic array. To this Hamiltonian we want to apply the BOA, first introduced by Max Born and J. Robert Oppenheimer, Ref. [1]. To this end, we assume the electronic dynamics to take place on a much shorter time scale than the vibrational motion of the atoms. Translated to our specific case: We assume $1/\Omega$ to be of much smaller magnitude than $1/\omega$, $1/\kappa$ and $1/\xi$, i.e. the energy scales of electron-phonon coupling are assumed small compared to the Rabi frequency Ω . This implies that we expect the BOA to yield trustworthy results in the limit of large Rabi frequencies Ω . As a shorthand for Ω much larger than all phonon energy scales we simply write: $\Omega \rightarrow \infty$. This is equivalent to being in the symmetrically displaced, *not* the JT, regime. Before the BOA may be applied, the Hamiltonian in Eq. (II.5) is again expressed in the quadrature $\hat{q} = x_0(b^\dagger + b)/\sqrt{2}$, such that we arrive at

$$H_2 = \sqrt{2}\Omega(|+\rangle\langle++| + |++\rangle\langle+|) + \left(\frac{\hat{p}^2}{2m} + \frac{m}{2}\omega^2\hat{q}^2 - \frac{\omega}{2}\right)|+\rangle\langle+| + \left(\frac{\hat{p}^2}{2m} + \frac{m}{2}\omega^2\hat{q}^2 - \frac{\omega}{2} + 2\kappa\frac{\hat{q}}{x_0} + 2\xi\frac{\hat{q}^2}{x_0^2}\right)|++\rangle\langle++|. \quad (\text{III.1})$$

As a reminder, it was defined $|+\rangle = (|\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle)/\sqrt{2}$ and m denotes the mass of the trapped atom species. To denote the quadratures, a hat is introduced to emphasize that these quantities are operators, in order to avoid confusion later. The constant energy offset, $-\omega/2$, is the result of the canonical commutator for the bosonic mode b . Before we proceed, the general idea of the BOA should be outlined. If the electronic dynamics is fast, compared to the vibrational degrees of freedom, the electronic state must adiabatically follow the motion of the atoms. Reason for this is that the electronic cloud is assumed to reconfigure fast enough to the new nuclei positions as they move. This idea is best explained with a simple example: One could imagine holding two glasses of water, one in each hand. The water almost immediately reacts if the glass is tilted, by adjusting the angle of the water surface relative to the glass’ base. So we model the electronic state as the water surface’s angle and the two glasses shall represent the nuclei positions in space. If one moves both glasses towards each other slowly, the water surface in each glass approximately stays flat during motion. Suppose now the vibrational dynamics is fast, in comparison to the ‘resetting’ time scale of the water surface. Then the fast motion of the glasses, at some point, will lead to drastic changes of the angle of the water surface (if one is fast enough, the water is spilled). At this point, the BOA breaks down, because the electronic state (water surface) is affected by the virtual forces introduced by the external motion of the nuclei (glasses). In our example with the glasses of water, it is clear what the possible states are in the limit of static glasses (nuclei) – the single possible state is the one where the water surface is flat. For an actual quantum mechanical system, the form of the electronic states for static

nuclei positions is, generally, not apparent from the start. In fact, typically the electronic state is also dependent on the atomic distances. In our simple picture this would relate to the water surface looking different for different (static) distances of the glasses. In order to obtain the electronic states for static nuclei (glass positions), we look at the electronic Hamiltonian for fixed nuclei positions. Formally, we rewrite the Hamiltonian for the dumbbell in the limit of vanishing kinetic energy of the atoms in the trap, i.e. $\hat{p} = 0$. Using x_0 , given by $x_0^{-2} = m\omega$ ($\hbar \equiv 1$), the reduced Hamiltonian takes the form

$$H_2^{(\text{fast})}(q) = \sqrt{2}\Omega(|\chi\uparrow\uparrow\rangle + |\uparrow\uparrow\chi\rangle) + \left(\frac{\omega}{2} \frac{q^2}{x_0^2} - \frac{\omega}{2}\right)|\chi\uparrow\uparrow\rangle + \frac{\omega}{2}(1 + 4\xi/\omega) \left(\left(\frac{q}{x_0} + \frac{2\kappa}{\omega} \frac{1}{1 + 4\xi/\omega}\right)^2 - \frac{4\kappa^2}{\omega^2} \frac{1}{(1 + 4\xi/\omega)^2} - \frac{\omega}{2} \right) |\uparrow\uparrow\chi\rangle. \quad (\text{III.2})$$

The superscript ‘fast’ stresses that this is the Hamiltonian responsible for the fast electronic dynamics. Note that we do have a classical problem now in the slow vibrational motion, because the displacement q can be treated classically, since it commutes with all remaining parts. For this reason it suffices to treat q as a real number, emphasized by the missing hat. One neat property of the dumbbell is that the (fast-degrees-of-freedom) Hamiltonian is simple enough to allow a full analytical description. For the two eigenvalues $E_{\min}$ and $E_{\max}$ we find, for arbitrary q ,

$$E_{\min}(q) = \min\{E_+(q), E_-(q)\}, \quad E_{\max}(q) = \max\{E_+(q), E_-(q)\}, \quad (\text{III.3})$$

$$E_{\pm}(q) = \frac{\omega}{2} \left[\frac{1}{2} (Q^2 + \frac{\bar{\omega}}{\omega} \bar{Q}^2) + \bar{\delta} \pm \sqrt{8 \frac{\Omega^2}{\omega^2} + \left(\frac{1}{2} (Q^2 - \frac{\bar{\omega}}{\omega} \bar{Q}^2) - \bar{\delta} \right)^2 - 1} \right], \quad Q = \frac{q}{x_0}, \quad (\text{III.4})$$

where the expressions

$$\bar{\omega} = \omega(1 - \xi/\xi_c), \quad \bar{Q} = \frac{q}{x_0} + \frac{2\kappa}{\bar{\omega}} \quad \text{and} \quad \bar{\delta} = -\frac{2\kappa^2}{\omega\bar{\omega}} \quad (\text{III.5})$$

represent (left to right) the dressed trap frequency $\bar{\omega}$ on the $|\uparrow\uparrow\rangle$ subspace, the analogue to the BT $\bar{Q}$ and the energy shift $\bar{\delta}$, caused by electron-phonon coupling. As a reminder, $\xi_c = -\omega/4$ is the critical coupling strength, derived from the curvature of the dipole-dipole potential. The eigenstates are the quantity corresponding to the static water surface. Because the eigenvalues depend on the position q , they are referred to as Born-Oppenheimer energy surfaces (BOES) or potential energy surfaces, Ref. [37, 82] in the literature. By construction, the two BOES come with the property $E_{\min}(q) \leq E_{\max}(q)$, i.e. the ground state surface is given by $E_{\text{GS}}(q) = E_{\min}(q)$. Note that for large Ω , we have $E_{\min}(q) \ll E_{\max}(q)$. It is not surprising that this is the case. It grasps the idea of staying in the electronic state no matter how q changes in time, as long as it is not too fast: If the electronic dynamics is fast (Ω large), the BOES are so far separated that the small energies associated to the vibrations cannot drive transitions between them, such that the electronic system is forced to adiabatically follow the eigenstate of $H_2^{(\text{fast})}$ it was initially in. Since the BOES are obtained analytically, we may plot their dependence on q to gain insight on how these surfaces are effected by the instability in ξ . In Fig. III.1 the BOES are shown for different values of Ω and ξ . Panels (a-c) provide the dependence on Ω , which shows how the GSBOES, $E_{\text{GS}} = E_{\min}$, increasingly separates from $E_{\max}$. In panels (d-f), the dependence on ξ is indicated. It becomes clear that if ξ approaches ξ_c , the minimum of $E_{\min}$ is pushed downwards, where at ξ_c it would reach negative infinity. It can also be seen that the minimum is further driven to the left of the plot, explaining also the (theoretically appearing)

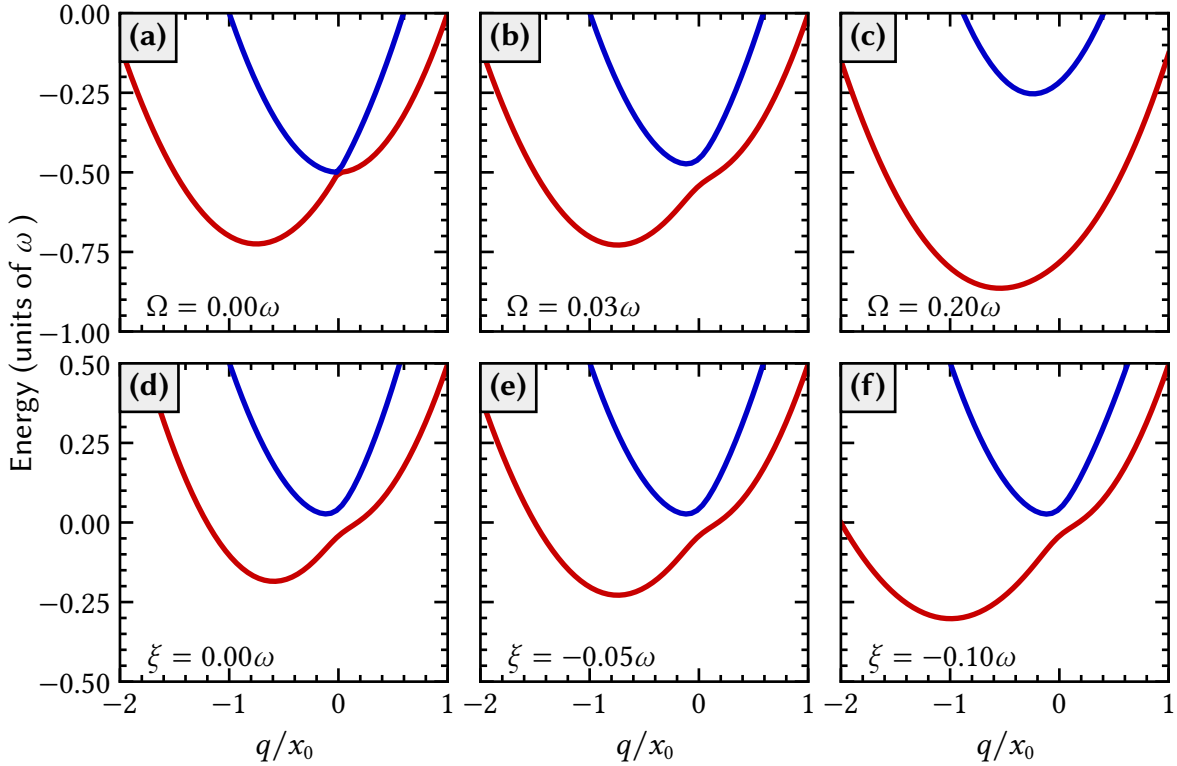


Figure III.1.: Born-Oppenheimer Energy Surfaces for Dumbbell. BOES for a diatomic array at different values for Ω and ξ . (a-c) BOES for fixed ξ and three different Ω ; values are placed in the respective panels. Ω defines the energy distance between $E_{\min}$ (red) and $E_{\max}$ (blue). Parameters are set to $\kappa = 0.3\omega$ and $\xi = -0.05\omega$. (d-f) Surfaces for fixed Ω and variable ξ . More critical ξ , closer to $\xi_c = -\omega/4$, yield smaller ground state energies and the displacement at the minimum, $q_{\min}$, is shifted to larger $|q_{\min}|$. Values for ξ are placed in each panel. Parameters are set to $\kappa = 0.3\omega$ and $\Omega = 0.03\omega$. Note that panel (b) and (e) are equal.

divergence of the displacement. In conclusion, the limit $\xi \searrow \xi_c$ shifts the minimum to the left and $\Omega \rightarrow \infty$ pushes it back to the right, towards zero. Moreover, we clearly see that the red curves, $E_{\min}$, are *not* strictly monotonic left and right of the minimum. We will later see the ‘dent’ in this curve turns into a true minimum for the triangular array, where the global minimum defines if the system resides in the JT regime or not. That for the dumbbell the parameters cannot be chosen such that a second minimum is present is a manifestation of the fact that the topology of a diatomic array is not rich enough to host spontaneous symmetry breaking.

To come back to our goal of introducing the BOA, we use the electronic eigenstates to define the unitary transformation $U(q)$, diagonalizing $H_2^{(\text{fast})}$. The dependence on q is a result of the eigenstates depending on it. The unitary transformation U is applied to the full quantum Hamiltonian. It means that we transform the Hamiltonian into a new frame of reference defined by $U(\hat{q})$, which is obtained by the substitution $q \mapsto \hat{q}$. We arrive at

$$U^\dagger(\hat{q})H_2U(\hat{q}) = \frac{\omega x_0^2}{2} \left(\hat{p}^2 - \{i\hat{p}, U^\dagger(\hat{q})U'(\hat{q})\} - (U^\dagger(\hat{q})U'(\hat{q}))^2 \right) + \begin{bmatrix} E_{\min}(\hat{q}) & \\ & E_{\max}(\hat{q}) \end{bmatrix}. \quad (\text{III.6})$$

In the above expression $\{A, B\} = AB + BA$ denotes the anti-commutator. Up to this point, all steps are exact and we describe the complete system just in a new basis, defined by $U(\hat{q})$. The derivative, U' , is explained by differentiation of U , understood as a real function in q , evaluated at the operator $\hat{q}$. Physically, the new frame provides a clear picture in that it separates the diagonal electronic Hamiltonian from terms which are potentially responsible for coupling the BOES. This is useful because in the BOA it is assumed that the electronic eigenstates do not couple. If they are not coupled, it simplifies the problem because only a 1-d. electronic Hilbert space remains. We can summarize the condition for decoupling electronic eigenstates by the following statement: Coupling between the electronic states may be neglected whenever the energy distance between $E_{\max} - E_{\min}$, scaling with Ω , is much larger than the largest energy scale associated to the first term in Eq. (III.6), proportional to ωx_0^2 . If this is the case, the off-diagonal terms, with respect to the basis defined by $U(\hat{q})$, in Eq. (III.6) are energetically not able to drive transitions between $E_{\min}$ and $E_{\max}$, such that it is sufficient to only consider a 1-d. electronic Hilbert space. Note that this is equivalent to the limit of large atomic masses: If the mass of the atom is approaching infinity, the first contribution to the Hamiltonian can be neglected. Physically, this is clear, because large atomic masses have more inertia. As a result, the vibrations become even slower, such that the BOA becomes more precise, because the ratio of electronic to vibrational time scales becomes smaller. So for large Ω and large atomic masses, we may entirely neglect the first contribution. In the following we will call this approximation $\text{BO}^{(0)}$. The superscript ‘(0)’ emphasizes that we neglect any contributions associated to the ratio of vibrational to electronic energy scales. One can also refine the approximation by taking the most relevant terms of first order in this ratio into account. To be concrete, terms containing derivatives of U are still neglected but the momentum contribution $\omega x_0^2 \hat{p}^2 / 2$ *not*. We justify this approximation by the fact that U' must become small for increasingly large Ω compared to Ω . Namely, if Ω is large, the diagonalization process yielding the eigenstates, i.e. the columns of U , does not ‘care’ about small entries on the diagonal, so the shape of the eigenvectors is mostly influenced by Ω and thus the derivative with respect to q small, compared to Ω . This was already evident in the fact that the ground state approximately factorizes in the limit of large Rabi frequencies; see footnote 2 in the previous chapter. We label this type of refined BOA with $\text{BO}^{(1)}$, suggestive of the typical notation that the precision is increased by one order – thus the superscript ‘(1)’ opposed to ‘(0)’.

Next, the unitary $U(\hat{q})$ should be inspected, because it gives insight on which contributions might actually vanish by construction. Generally speaking, $U(\hat{q})$ is a non-trivial operator on the full tensor product of Hilbert spaces $\mathcal{H}_{\text{atoms}} \otimes \mathcal{H}_{\text{phonons}} \cong \mathbb{C}^2 \otimes \mathcal{H}_{\text{phonons}}$, because the q -dependence of the electronic eigenstates can be rather complex. Still, there are statements to be made about $U^\dagger(q)U'(q)$. First, we should think of $U(q)$ as a 2-d. square matrix in the basis $(|+\rangle, |\uparrow\uparrow\rangle)$, with operator valued entries. Moreover, the operator $U^\dagger(\hat{q})U'(\hat{q})$ is anti-hermitian, because $(d/dq)^\dagger = -d/dq$. In our case $U(\hat{q})$ has only hermitian entries, implying $U^\dagger(\hat{q})U'(\hat{q})$ to solely host off-diagonal entries. Therefore, its square is a diagonal matrix. From this we conclude that the contribution $(U^\dagger(\hat{q})U'(\hat{q}))^2$ does not drive transitions between the BOES. On the other hand, contributions linear in $\hat{p}$ can drive such transitions. Considerations of this kind are important tools to estimate the precision of the BOA.

The Hamiltonian in Eq. (III.6) provides the basis to study the system in the BOA. For the remainder of this thesis, the term ‘BOA’ refers to the approximation where all coupling between the BOES $E_{\min}$ and $E_{\max}$ is neglected, i.e. the case we have labelled ‘ $\text{BO}^{(0)}$ ’, unless stated otherwise. Next, we want to finally make use of the BOA to compute the ground state energy for $\Omega = 0$ and $\Omega \rightarrow \infty$. These cases are chosen because they allow feasible analytical results and can be compared to the results in the full quantum picture, cf. Eq. (II.49) and Eq. (II.51). We start with $\Omega = 0$. The ground state energy

is obtained by minimizing E_{GS} in q . For $\Omega = 0$ and q close enough to the minimum, $E_{\text{GS}}(q) = E_{-}(q)$. Therefore, we minimize $E_{-}(q)$. This is a simple parabola, such that we immediately find

$$E_{\text{GS},2}^{(\text{BO}),\Omega=0} = -\frac{\omega}{2} - \frac{2\kappa^2}{\omega} \frac{1}{1 - \xi/\xi_c}. \quad (\text{III.7})$$

The superscript ‘BO’ emphasizes that the result is obtained in the BOA. Comparison to the correct result in Eq. (II.49) leads to the following observations: First, for vanishing κ and ξ we find the energy difference $-\omega/2$. This is just the result of having a classical description for the atomic position q , because we have neglected all terms containing the quantized momentum $\hat{p}$. It is an energy shift that contributes equally to E_{min} and E_{max} and has no effect on the minimums position, q_{min} . Therefore, there is no physical relevance to this shift, because it does not change the physics of the problem; after all, it is an additive constant to the Hamiltonian in Eq. (III.6). The shift is included in E_{min} and E_{max} in Eq. (III.3). Secondly, we notice that for non-zero ξ there are additional terms not captured by the BOA, namely

$$\frac{\omega}{2} (\sqrt{1 - \xi/\xi_c} - 1). \quad (\text{III.8})$$

The contributions that are missing are given by the commutator contributions (CC), where the term ‘commutator contribution’ is defined in the text right below Eq. (II.42) and the corresponding expressions are given in Eq. (II.42). Compared to the global energy offset $-\omega/2$, the CC are important for the physics. They introduce electronic state dependent energy shifts to the energy levels of the system, because such contributions only appear if the system is in the electronic state with two excited atoms. This is evident by the fact that the CC only depend on ξ , originating from interactions between *two* Rydberg atoms. Hence, the CC are *not* a global energy shift that can be accounted for by redefining the energy zero point. For the trivial offset $-\omega/2$ this is possible. In the following we call the CC ‘quantum corrections to the BOA’. The name ‘quantum corrections’ seems appropriate because the terms are the consequence of the non-zero canonical commutator $[\hat{q}, \hat{p}] \neq 0$. We justify the second part of the name by the fact that the only approximation we made is the BOA. Therefore the BOA must be the reason for the discrepancy between the result in the BOA and the full quantum picture. Later, in the next section, we establish one aspect of their physical implications. As a remark, it is quite interesting that the BOA reproduces the correct instability, even in the limit of (infinitely) slow electronic dynamics ($\Omega = 0$). One explanation could be that the approximation is saved by the fact that we compute the ground state energy, i.e. we are interested in the minimum of the red curve in Fig. III.1 (a). At the position of the energy minimum q_{min} , the blue curve is still sufficiently far separated. If one would consider long time evolution for small Ω , the approximation must break down because of the ergodic theorem. That is because at some point, the solution $\langle \hat{q}(t) \rangle$ of the equations of motion would come infinitely close to the point where the red and blue curves are nearest and the BOA is not well-justified.

Analogously to the limit $\Omega = 0$, we compute the ground state energy for $\Omega \rightarrow \infty$. Here, the BOES to be minimized is given by the asymptotic behaviour of $E_{\text{GS}}(q)$ for large Ω . The result is given by the asymptote

$$E_{\text{GS},2}^{(\text{BO}),\Omega \rightarrow \infty}(q) = -\sqrt{2}\Omega - \frac{\omega}{2} - \frac{\kappa^2}{2\omega} \frac{1}{1 - \xi/(2\xi_c)}. \quad (\text{III.9})$$

For a proof see Supp. S.4. Again, the CC from Eq. (II.51) are missing. Note that it is somewhat unexpected that the quantum corrections to the BOA are not evident, because in the limit of infinitely fast electronic dynamics the BOA should be exact. Seemingly, this is a contradiction, resolved by the

fact that the relative error

$$\left| \frac{E_{\text{GS},2}^{(\text{BO}),\Omega \rightarrow \infty}(q) - E_{\text{GS},2}^{\Omega \rightarrow \infty}(q)}{E_{\text{GS},2}^{\Omega \rightarrow \infty}(q)} \right| = \frac{\omega}{2} \frac{\sqrt{1 - \xi/(2\xi_c)} - 1}{\sqrt{2}\Omega} \xrightarrow{\Omega \rightarrow \infty} 0, \quad (\text{III.10})$$

for any q . It quantifies the imprecise statement that the results are ‘equal’ in the limit of infinite Rabi frequencies – where both expressions should be correct – because they both take the ‘value’ negative infinity. Of course, Eq. (III.9) also hosts the global energy shift $-\omega/2$.

III.2. Implications of Quantum Corrections and Jahn-Teller Regime

In this section we want to further illustrate that the simulator is well described by a BOA for large Ω , give an intuitive picture for the transition into and out of the Jahn-Teller (JT) regime, explain its connection to the instability in ξ , located at $\xi_c = -\omega/4$, and investigate the significance of quantum effects in our simulator. We approach these points by considering the triangular array, because it is the most simple array for which the JT regime may be studied.

Starting point is the Hamiltonian presented in Eq. (II.11) with the quadratic bosonic Hamiltonians h_m given in Eq. (II.14). In order to apply the BOA to this Hamiltonian, we proceed in the same manner as for the dumbbell in the previous section. First we rewrite the Hamiltonian again in the four quadratures $\hat{q}_i$ ($i = 1, 2, 3, \text{t}$), which we gather in the vector $\hat{\mathbf{q}}^{\text{t}} = (\hat{q}_1, \hat{q}_2, \hat{q}_3, \hat{q}_{\text{t}})$ (transposition acts only on $\mathbb{R}^4$). Hats for denoting operators are introduced to avoid confusion. Analogously as for the dumbbell, we extract the Hamiltonian for the fast electronic dynamics by formally setting $\hat{\mathbf{p}} = \mathbf{0}$. $\hat{\mathbf{p}}$ denotes the vector of conjugated momenta with respect to $\hat{\mathbf{q}}$. For the Hamiltonian we obtain

$$\begin{aligned} H_3^{(\text{fast})} = & \Omega \sum_{m=1}^6 |m\rangle\langle m+1| + \frac{\omega}{2} \frac{\mathbf{q}^2}{x_0^2} - 2\omega \\ & + P_1 \left[\frac{\sqrt{2}\kappa}{x_0} (q_1 - q_2) + \frac{\xi}{x_0^2} (q_1 - q_2)^2 + \frac{\nu\kappa}{\sqrt{2}x_0^2} (q_3 + q_{\text{t}})^2 \right] \\ & + P_3 \left[\frac{\sqrt{2}\kappa}{x_0} \left(q_1 + \frac{1}{2}q_2 - \frac{\sqrt{3}}{2}q_3 \right) + \frac{\xi}{x_0^2} \left(q_1 + \frac{1}{2}q_2 - \frac{\sqrt{3}}{2}q_3 \right)^2 + \frac{\nu\kappa}{\sqrt{2}x_0^2} \left(\frac{\sqrt{3}}{2}q_2 + \frac{1}{2}q_3 - q_{\text{t}} \right)^2 \right] \\ & + P_5 \left[\frac{\sqrt{2}\kappa}{x_0} \left(q_1 + \frac{1}{2}q_2 + \frac{\sqrt{3}}{2}q_3 \right) + \frac{\xi}{x_0^2} \left(q_1 + \frac{1}{2}q_2 + \frac{\sqrt{3}}{2}q_3 \right)^2 + \frac{\nu\kappa}{\sqrt{2}x_0^2} \left(-\frac{\sqrt{3}}{2}q_2 + \frac{1}{2}q_3 - q_{\text{t}} \right)^2 \right]. \end{aligned} \quad (\text{III.11})$$

To arrive at this result, the explicit expressions for $\epsilon_m^{\parallel/\perp}$, $m = 1, 3, 5$, Eq. (II.14), are plugged in. The energy shift $4 \cdot (-\omega/2) = -2\omega$ is a result of the *four* non-zero canonical commutators of the modes b_1, b_2, b_3 and b_{t} . As a remainder, the states $|1\rangle, \dots, |6\rangle$ are defined in Fig. II.3 and we use the projectors $P_m = |m\rangle\langle m|$. Odd-numbered states label the electronic states with two Rydberg excitations, such that electron-phonon entanglement appears. Moreover, the electronic Hilbert space is 6-d., opposed to the 2-d. space appearing for the dumbbell. Consequently, there are not two but six BOES. This time, we cannot give analytical expressions for all Ω , but we may still obtain them numerically by diagonalizing $H_3^{(\text{fast})}$. Also, we cannot directly plot the surfaces because they are, generally, dependent on four positions q_i . Interestingly, the BOES could be plotted in Ref. [37], because without the Hessian there are only three relevant modes, instead of four. Reason for this circumstance is that the twist mode q_{t} only couples through the contribution proportional to $\nu\kappa$. This is a manifestation of the vector structure of the Hessian, appearing for more than two atoms in the array. Nonetheless,

we will make use of the BOES and call them $E_1, \dots, E_6$, where $E_1 = E_{\text{GS}}$ shall be the ground state energy surface. Without loss of generality, we assume that the following chain of inequalities holds true for any $\mathbf{q} \in \mathbb{R}^4$: $E_1(\mathbf{q}) \leq E_2(\mathbf{q}) \leq \dots \leq E_6(\mathbf{q})$. In addition to the BOES the diagonalization process yields the unitary transformation $U(\hat{\mathbf{q}})$ into the frame of reference where the Hamiltonian of the fast electronic dynamics is diagonal. Using the properties of this transformation, we arrive at the (exact) Hamiltonian

$$U^\dagger(\hat{\mathbf{q}})H_3U(\hat{\mathbf{q}}) = \frac{\omega x_0^2}{2} \left(\hat{\mathbf{p}}^2 - \{i\hat{\mathbf{p}}, U^\dagger(\hat{\mathbf{q}})(\nabla U)(\hat{\mathbf{q}})\} - (U^\dagger(\hat{\mathbf{q}})(\nabla U)(\hat{\mathbf{q}}))^2 \right) + \sum_{n=1}^6 E_n(\hat{\mathbf{q}})\mathcal{P}_n. \quad (\text{III.12})$$

In this expression the projectors $\mathcal{P}_m$ project the electronic state down to the eigenvector corresponding to E_m and the gradient is taken with respect to $\mathbf{q}$, $\nabla \equiv \nabla_{\mathbf{q}}$. As a side remark, for the projectors this implies that they are non-local to the electronic subspace. In the position space representation this is directly visible,

$$\mathcal{P}_m = \int_{\mathbb{R}^8} d(\mathbf{q}, \mathbf{q}') \mathcal{P}_m(\mathbf{q}, \mathbf{q}') \otimes |\mathbf{q}\rangle\langle\mathbf{q}'| = \int_{\mathbb{R}^4} d\mathbf{q} \mathcal{P}_m(\mathbf{q}, \mathbf{q}) \otimes |\mathbf{q}\rangle\langle\mathbf{q}|, \quad (\text{III.13})$$

with the projectors $\mathcal{P}_m(\mathbf{q}, \mathbf{q}') = \langle \mathbf{q} | \mathcal{P}_m | \mathbf{q}' \rangle$, locally acting on the electronic subspace, exactly like the P_m . Generally speaking, this observation is important because it implies $[\hat{\mathbf{p}}, \mathcal{P}_m] \neq 0$.

Next, we want to obtain analytical expressions for the ground state energy in the BOA. To this end, we minimize the GSBOES, E_{GS} . Of course, for arbitrary Rabi frequencies Ω , this is not possible in an analytical fashion. However, for $\Omega = 0$ and $\Omega \rightarrow \infty$, it is again an option. In the limit $\Omega = 0$, one arrives at the energy

$$E_{\text{GS},3}^{(\text{BO}),\Omega=0} = -2\omega - \frac{2\kappa^2}{\omega} \frac{1}{1 - \xi/\xi_c}. \quad (\text{III.14})$$

On the other hand, we find for the large Ω asymptote,

$$E_{\text{GS},3}^{(\text{BO}),\Omega \rightarrow \infty} = -2\Omega - 2\omega - \frac{\kappa^2}{4\omega} \frac{1}{1 - \xi/(4\xi_c)}. \quad (\text{III.15})$$

There are some less important steps involved to obtain these results, not relevant for the aim of identifying quantum corrections to the BOA. Therefore, the derivation of these results is shifted to Supp. S.5 for brevity. As one might have foreseen from the results of the BOA before, the CC, cf. Eq. (II.42), lead to quantum corrections not grasped by the standard BO approach to find the ground state energy. Furthermore, the instability in ξ is pushed deeper into the unstable regime in the limit $\Omega \rightarrow \infty$. The new insight we get from these results is that the vector-structure contribution proportional to $\nu\kappa$ gives rise to even more quantum corrections to the BOA. The ground state energy in the BOA, presented in Eq. (III.14), does not seem to depend on $\nu\kappa$, on first sight. It is still evident though, because at $\xi, \Omega = 0$ the GSBOES for the triangle takes the form

$$E_{\text{GS}}^{\xi=0}(Q_\perp, Q_\parallel) = \frac{\omega}{2} Q_\parallel^2 + 2\kappa Q_\parallel + \frac{\omega}{2} (1 + 2\sqrt{2}\nu\kappa/\omega) Q_\perp^2 - 2\omega. \quad (\text{III.16})$$

$Q_\parallel$ and $Q_\perp$ describe (dimensionless) displacements along and perpendicular to a triangle edge, respectively, obtained by a rotation of $\mathbf{q}/x_0$. For details, the reader is referred to Supp. S.5. We clearly see that the energy minimum becomes negative infinity if $\kappa < \kappa_c = -\omega/(2\sqrt{2}\nu)$, because the corresponding parabola flips sign. This result demonstrates that the BOA is working well enough to even capture the second instability. Therefore, Eq. (III.14) is only complete with the knowledge that

if $\kappa < \kappa_c$, the ground state energy is not well defined any more. The reason for this instability to not make an explicit appearance in the energy is that all (finite, well defined) dependence on it is contained within the quantum corrections to the BOA. This circumstance is also clear: Only contributions proportional to $\nu\kappa$ couple to the perpendicular modes, describing perpendicular displacements of the atoms with respect to the triangles edges. Since there are no linear terms in these modes, the ground state is not a coherent state for this displacement signature, cf. Eq. (II.14). However, because there are quadratic terms in the perpendicular displacements, their ground state is a squeezed vacuum¹. Such a state generates additional contributions to the energy that are purely quantum, as squeezing has no classical analogy. From this we take that the quantum corrections to the energy are a result of squeezed bosonic states. For example, a discussion on squeezed states of light is given in Ref. [83]. Finally, we note that the quantum corrections to the BOA are shown in Fig. II.8 by the discrepancy of the dashed red and dash-dotted light red line, for the case of the large Ω asymptote. In all panels the contributions are rather small, however, their significance is of course strongly dependent on the specific parameter choices.

Our next goal is to examine the transition into and out of the JT regime and in particular to understand how this regime is linked to the instability in ξ . To achieve this goal we want to make use of the BOES, similar to Ref. [37]. To facilitate plotting of the BOES we neglect contributions of order $\nu\kappa$, such that q_t decouples and only q_1 , q_2 and q_3 are left, Eq. (III.11). We justify this approximation by only looking at parameter choices where $\kappa \sim \xi$ and $\kappa, \xi < \omega$, such that for $\nu = 0.1$, $|\kappa\nu| \ll \kappa, \xi, \omega$. In this special case, contributions of order Ω/Δ , which are neglected because we assume perfect Rydberg facilitation and blockade (Sec. I.2.2 and Ref. [73]), can be of the same magnitude as $\nu\kappa$. This would mean that we have to omit $\nu\kappa$ just for consistency. With this final approximation, the stage is now set to look at the BOES, and in particular the ground state energy. Numerically obtained results are shown in Fig. III.2. In panel (a) the ground state energy (black line) dependence on Ω is plotted for two choices of ξ . These lines host a non-differentiable point in their Ω dependence (the point where the lines abruptly change their slope). This point is examined using the numerical results for the value of $\mathbf{q}$ defining the energy minimum. We call the minimum $\mathbf{q}_{\min}$. Recall that the GSBOES has three global minima, Ref. [37]. One of the three degenerate ground state minima is chosen, in this case the one corresponding to $|1\rangle = |\uparrow\uparrow\downarrow\rangle$. It comes with the convenient property that only q_1 and q_2 are finitely displaced and $q_3 = 0$. Therefore, we have only two variables left which our GSBOES depends on. We show the dependence on q_1 and q_2 in panels (c-e) of Fig. III.2. The plots illustrate that *two* minima emerge and a transition between them is exactly what causes the non-differentiable point in Ω , visible in panel (a) marked by a yellow dot. At this specific critical value Ω_c , the system leaves the JT regime and no spontaneous symmetry breaking is evident when $\Omega > \Omega_c$. Note that in total there are 6 minima, 2 for each of the three degenerate ground state atom configurations. An even clearer picture is obtained by looking at the section through the surface plots (c-e) at which for every possible q_2 the specific q_1 is selected that minimizes the energy. This ensures that the resulting curve $q_{1,\min}(q_2)$ surpasses both the local and global minimum. The energy along the sections is shown in panel (b) of Fig. III.2. Here the two promised (see discussion on Fig. III.1) minima emerge and the transition process is indicated with the help of three (yellow markers) selected points. The left minimum describes the symmetrically displaced and the right one the JT regime. Note that the existence of two minima was not the case for the dumbbell, where no JT regime as for the triangle could be defined. From the two ground state energy curves (black) in panel Fig. III.1 (a) it becomes

¹ The squeezed vacuum is defined as the state $S(\beta)|0\rangle$, $\beta \in \mathbb{C}$, where $S(\beta) = \exp(1/2(\beta a^{\dagger 2} - \beta^* a^2))$ is the squeezing operator and $|0\rangle$ the vacuum of the bosonic mode a .

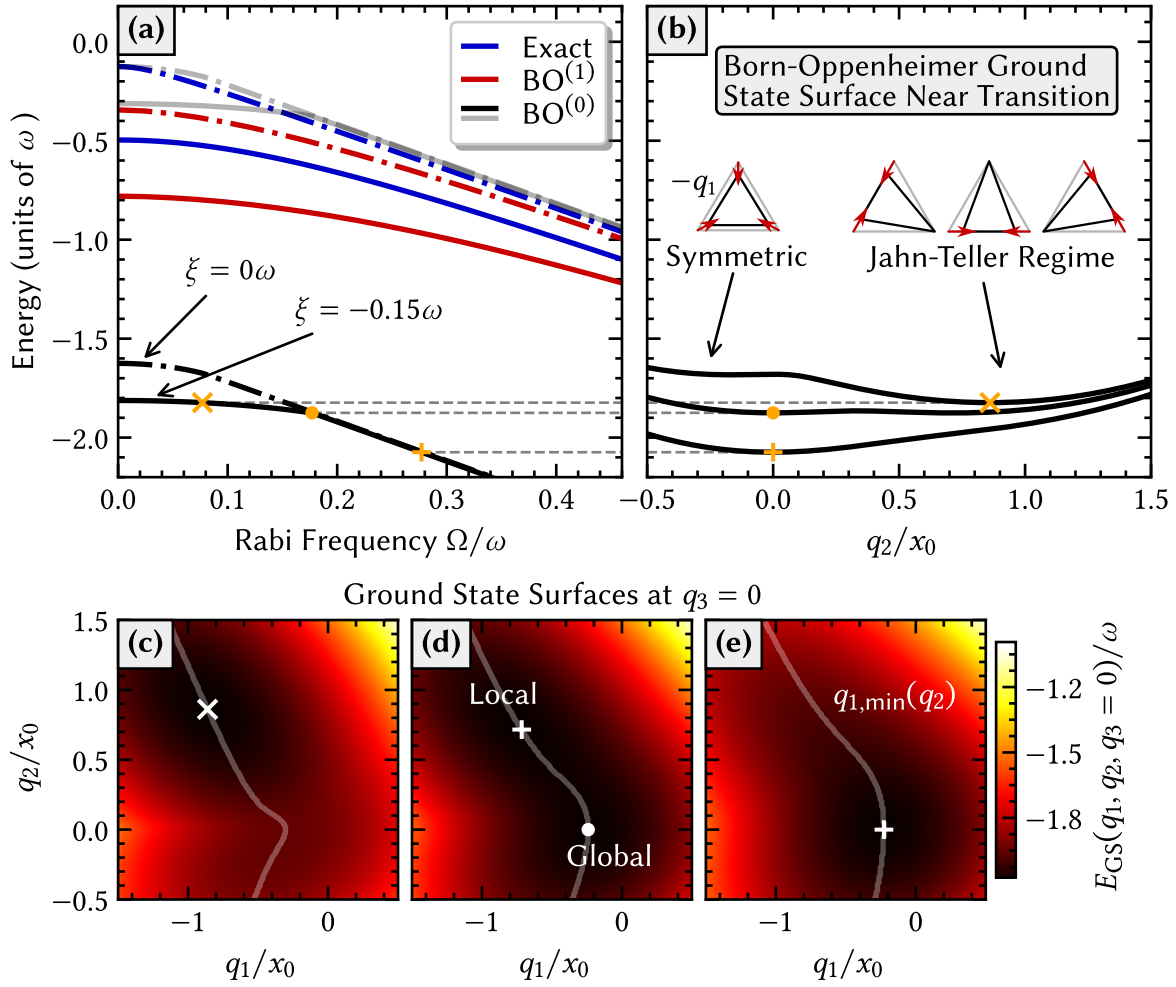


Figure III.2.: Born-Oppenheimer Approximation for Triangle. (a) Ground state energy dependence on Ω . Dash-dotted lines correspond to the choice $\xi = 0\omega$ and full lines to the ‘more critical’ value $\xi = -0.15\omega$. We use $\kappa = 0.25\omega$. Note that $\nu = 0$, because contributions of order $\nu\kappa$ are neglected. Blue lines show the ground state energy from exact diagonalization with maximally 20 phonons per mode. Black lines show the result obtained from minimizing the GSBOES $E_{\text{GS}} = E_1$, i.e. application of procedure BO⁽⁰⁾ (see p. 41). Light grey lines are equal to the black lines without the energy shift $-3\omega/2$. Red curves show the ground state energy defined by Eq. (III.17), which is the Hamiltonian for the refined version of the BOA, BO⁽¹⁾ (see p. 41). To obtain the ground state energy the time-independent Schrödinger equation is solved on a discrete position space with each $q_i/x_0 \in (-5, 5)$ ($i = 1, 2, 3$) and 100 points. (c-e) Plots of the GSBOES near the minimum corresponding to the subspace $|\uparrow\uparrow\downarrow\rangle$. To each marked $\Omega/\omega = 0.177, 0.127, 0.277$ (orange markers) in (a) there is one associated panel. Light white lines show the minimal energy position $q_{1,\min}$ for fixed q_2 . Minima are indicated with white markers. (b) Section through the BOES, where the energy surface is computed at position $(q_{1,\min}(q_2), q_2)$. The orange marker shows the minimum as well as which curve corresponds to which of the three selected Ω in (a). Note that $q_2 \neq 0$ indicates that the system is in the regime hosting spontaneous symmetry breaking.

evident why the JT regime in Fig. II.8 has become wider (in Ω) for $\xi \searrow \xi_c$. In the BOA there is a sharp transition point Ω_c , and we observe it to be shifted right as ξ approaches ξ_c . In conclusion, the BOA works well enough to explain the existence of a JT regime (see Ref. [37]) but also the, up until now in the literature unknown, linkage to the instability. Apart from this, we can also understand now why the limits $\xi \searrow \xi_c$ and $\Omega \rightarrow \infty$ do not commute, i.e. why the large Ω asymptotes yield the ‘wrong’ position of the instability in ξ ; see discussion on Eq. (II.51). If we first take the limit $\Omega \rightarrow \infty$, we basically assume that the left minimum in Fig. III.2 (b) is at negative infinity. This is what justifies the approximation of a factorized ground state into an electronic and a vibronic state. We give the following, mathematically imprecise but intuitive, explanation: Because we would have to surpass a minimum at ‘negative infinity’ (left minimum) with the right minimum (here the minimal value decreases with the distance of ξ to ξ_c) we are not able to do so. The impossibility of this is encoded in the fact that the ground state energy does not host an instability in ξ any more, because we would need to set ξ to a value in the critical regime, where our expression has no validity from the start. However, for finite Ω we expect that there are terms proportional to $1/\Omega$, yielding the correct ξ_c . We just do not see them any more, because we made a too crude assumption when we assumed $\Omega \rightarrow \infty$, i.e. a fully factorized ground state.

In the final paragraph of this chapter we want to examine what significance the quantum corrections to the BOA have. The answer is shown in panel (a) of Fig. III.2. Attention shall be directed to the blue lines. The dash-dotted line shows the exact quantum mechanical solution of the ground state energy for a vanishing Hessian, $\xi = 0$, and the full line the result for $\xi = -0.15\omega$. These curves are far separated from the BOA, because of the trivial energy shift $-3\omega/2$, caused by the *three* classically treated quadratures $\hat{q}_1, \hat{q}_2$ and $\hat{q}_3$. Recall that there are only three modes left (not four) to describe the system, because we assumed $\nu\kappa$ to be negligible, which resulted in the decoupling of q_t . Note that the energy offset $-3\omega/2$ is constantly there for all Ω , such that the form of the black curves in Fig. III.2 (a) does not depend on it. However, we can clearly see that the non-approximated result (blue lines) do not show a hard transition at some Ω_c as the black curves do. Further, we know that the trivial energy shift is not responsible for this qualitatively different form of the ground state energy band. Therefore, the reason for this quantum induced ‘washing-out’ of the transition must be linked to the quantum corrections to the BOA, which we could obtain in the limits $\Omega = 0$ and $\Omega \rightarrow \infty$ even analytically. This observation showcases that relevant, in the sense that they change the physics qualitatively as well as quantitatively, quantum effects are emergent in our simulation framework. Perhaps also interesting to see is how the refined version of the BOA performs, which we have named $\text{BO}^{(1)}$ on p. 41. The ground state energy is the one of a particle with position (q_1, q_2, q_3) in the potential defined by the GSBOES. To be concrete, the ground state energy is obtained from the Hamiltonian

$$H_{\text{BO}} = \frac{\omega x_0^2}{2} \hat{\mathbf{p}}^2 + E_{\text{GS}}(\hat{\mathbf{q}}). \quad (\text{III.17})$$

Because the position $\hat{\mathbf{q}}$ cannot be treated classically since $\hat{\mathbf{p}}$ makes an appearance, we expect quantum corrections to be evident again. This is exactly what we can see; the JT regime is not defined by a specific critical Ω_c , but the transition is ‘washed-out’ by the quantum corrections to the BOA. That we have neglected terms proportional to ∇U is evident from the fact that the blue and red lines do not match perfectly. However, we can see that they approach each other for larger Ω . The fact that the blue and red curves in Fig. III.2 (a) approach each other for increasing Ω underlines numerically how the BOA, in particular $\text{BO}^{(1)}$, is better justified for larger Rabi frequencies Ω , where coupling between the BOES becomes weaker: The time scales of atomic vibrations and electronic motion in the tight-binding lattice, Fig. II.3, have more discrepancy and thus the BOA becomes a well-justified approximation. As a final remark, we do not observe the trivial (global) energy shift $-3\omega/2$ for the

red curves in Fig. III.2 (a), because we have a quantum mechanical problem defined by Eq. (III.17). Again, it should be emphasized that this shift has no physical relevance here, because it is present for all energy levels equally, such that it merely constitutes a homogeneous shift of the full energy spectrum.

IV. State Preparation Protocol

Along the course of this thesis we have gained insight on the question, ‘If we can use optically trapped atom arrays as a simulator for phenomena linked to molecular physics?’. Our results demonstrate that such a simulator indeed is possible, at least on a theoretical basis. In Chap. I, some experimental aspects were discussed, giving us confidence to assume tunable coupling strengths κ and ξ . This circumstance, however, is not sufficient to observe phenomena like the instabilities and the transition into and out of the Jahn-Teller (JT) regime. In order to observe these effects, it is an indispensable task to be able to prepare molecular states within this framework. In the following we will mainly focus on the ground state, in order to link this chapter to the previous ones. As an example: To observe the transition into and out of the JT regime for the triangular array, one could prepare the system’s ground state for different Rabi frequencies Ω and measure how many atoms reside in their respective Rydberg state. Of course, one would need many measurements for each Ω , e.g. by using multiple triangles simultaneously. With such measurements at hand, one is able to determine the Ω dependence of the order parameter,

$$o_{\text{JT}} = 1 - 2 \langle P_2 + P_4 + P_6 \rangle_{\text{GS}} , \quad (\text{IV.1})$$

indicating the prominence of the JT regime on an experimental basis. Note that o_{JT} is equal to 0 in the limit of $\Omega \rightarrow \infty$, because the ground state factorizes into an electronic state proportional to $|1\rangle + |3\rangle + |5\rangle - |2\rangle - |4\rangle - |6\rangle$, cf. p. 20. For $\Omega \rightarrow 0$, the system is driven into the JT regime (for ξ sufficiently close to ξ_c) and the order parameter takes the value 1, as only the states $|1\rangle$, $|3\rangle$ and $|5\rangle$ are energetically favourable, since they allow atomic displacements. Therefore, measurement of o_{JT} in the ground state indicates how pronounced the JT regime is, without the need for experimentally accessing the displacement structure of the array. Take note of the fact that the observable being measured is solely acting on the electronic subspace. Moreover, this procedure motivates us to develop a protocol for preparing the ground state of the artificial molecular system, which is done in the following section. For concreteness, the section is centred around the example given here.

IV.1. Ground State Preparation

For the rest of this chapter we want to consider the triangular array. However, the ideas of this section are similarly applicable to other array shapes, such that this choice is not a hard restriction. In fact, other array shapes may have even better properties when it comes to the preparation of selected states, e.g. the tetrahedron yields flat bands in Ω , which allows for rapid increases in the Rabi frequency, Ω , without changing the electronic state, cf. p. 23. This may remove the necessity of slow, simulation time consuming, adiabatic ramps. Initially we assume the electronic state to be the state where all atoms are found in the ground state, i.e. $|\downarrow\downarrow\downarrow\rangle$. Further, we assume the bosonic modes to be in the vacuum. We choose this state because this is the typical initial state in an experiment (at zero temperature). Note that we will assume zero temperature throughout this section. For measurement of purely electronic observables, such as the order parameter o_{JT} , this should not be too much of a problem, if the electron-phonon coupling constants generate large, compared to the thermally induced, displacements. However, temperature is still problematic as black-body radiation is one of the main sources restricting coherence time, Ref. [42].

We split the protocol into two steps. First, we prepare the electronic ground state. In a second

step we use adiabatic ramps in the electron-phonon coupling constants to excite the phonon modes we need, in order to obtain the motional ground state of the triangle, i.e. the true ground state for finite κ and ξ . First step in our electronic ground state preparation protocol is to drive the state $|\downarrow\downarrow\downarrow\rangle$ into the manifold of states selected by the facilitation constraint, i.e. $|1\rangle, \dots, |6\rangle$, cf. p. 20. To do so, we first assume the electron-phonon coupling constants, κ, ξ , to be small compared to the frequency of the traps, ω , such that we may neglect them for the first steps of the protocol. In an experiment one would need to tune the dipole-dipole interaction potential to be almost flat in the vicinity of the equilibrium atom positions. In mathematical terms, this means that we consider the Hamiltonian, Eq. (I.48),

$$H = \sum_{k=1}^3 (\Delta n^{(k)} + \Omega \sigma_x^{(k)}) + \mu \sum_{k=1}^3 \sum_{k' < k} n^{(k)} n^{(k')}, \quad (\text{IV.2})$$

with notation adapted from Chap. I and sums running over atoms k, k' . It should be emphasized that we eventually want to make use of the non-local terms to implement the facilitation constraint $\Delta = -\mu$, where μ is the energy shift introduced by the zeroth order of the dipole-dipole interaction potential, cf. Eq. (I.48). Initially, we observe that the laser drive (the laser defining the Rabi frequency Ω) acts homogeneously on the array such that we must be able to drive the system directly into the 1-d. subspace defined by $|E\rangle \equiv (|2\rangle + |4\rangle + |6\rangle)/\sqrt{3}$ (E: Even-numbered), i.e. the symmetric linear combination of single excited electronic states. For the symmetric linear combination of double excited states, we introduce the shorthand $|O\rangle \equiv (|2\rangle + |4\rangle + |6\rangle)/\sqrt{3}$ (O: Odd-numbered). At the beginning of the protocol we apply a resonant Rabi pulse of duration $\pi/(2\sqrt{3}\Omega)$. Using the resonance condition $\Delta = 0$, the Hamiltonian is effectively given as

$$H^{(\text{ii})} = \Omega \sum_{k=1}^3 \sigma_x^{(k)} + \mathcal{O}(\mu/\Omega). \quad (\text{IV.3})$$

Note that we have already assumed contributions of order μ/Ω to be negligible on the time scale defined by the laser pulse (pulse in the time domain described through $\Omega(t)$), since we assume $|\mu| \gg \Omega$, cf. Ref. [73]. The superscript (ii) labels the time domain defined in Fig. IV.1, where an exemplary implementation of the protocol is run numerically, using the full Hamiltonian in Eq. (IV.2). In a more suitable basis, with first and second basis vector $|\downarrow\downarrow\downarrow\rangle$ and $|E\rangle$, the Hamiltonian reduces to

$$H^{(\text{ii})} \stackrel{|\mu|/\Omega \ll 1}{\approx} \langle \downarrow\downarrow\downarrow | H^{(\text{ii})} | E \rangle |\downarrow\downarrow\downarrow\rangle \langle E| + \text{h.c.} = \begin{bmatrix} & \sqrt{3}\Omega \\ \sqrt{3}\Omega & \end{bmatrix}, \quad (\text{IV.4})$$

as all other basis vectors host at least one of the spin (electronic) states with two excitations. Note that the exact form of the remaining basis vectors is irrelevant, it is only important that there exists an orthonormal basis containing $|\downarrow\downarrow\downarrow\rangle$ and $|E\rangle$, which is guaranteed by the *basis extension theorem* and the *Gram-Schmidt procedure* from linear algebra. Our Hamiltonian is a simple 2-level system with enhanced Rabi frequency $\sqrt{3}\Omega$. Thus, the Bloch vector has rotated 180° after a pulse duration $2\pi/(4\sqrt{3}\Omega)$. If the system is initialized in the state $|\downarrow\downarrow\downarrow\rangle$, the final state then is $|E\rangle$. So at this point (end of domain ii), the system is in the state $|E\rangle$. Next, the detuning is rapidly decreased from 0 to $\Delta + \delta_{\text{fe}}$, where $|\delta_{\text{fe}}/\Delta| \ll 1$. This means that we have almost implemented the facilitation condition, but left a small ‘facilitation error’, δ_{fe} . A rapid change in Δ may be achieved by switching off the driving laser and simultaneously turning on another one with the correct detuning. Suppose that we are in the nearly facilitated regime. The states $|\downarrow\downarrow\downarrow\rangle$ and $|\uparrow\uparrow\uparrow\rangle$ are effectively decoupled from the system dynamics from this point onward. Next, we continue the laser pulse from before. Remember that we currently are in the state $|E\rangle$. Because the laser pulse is applied homogeneously over the

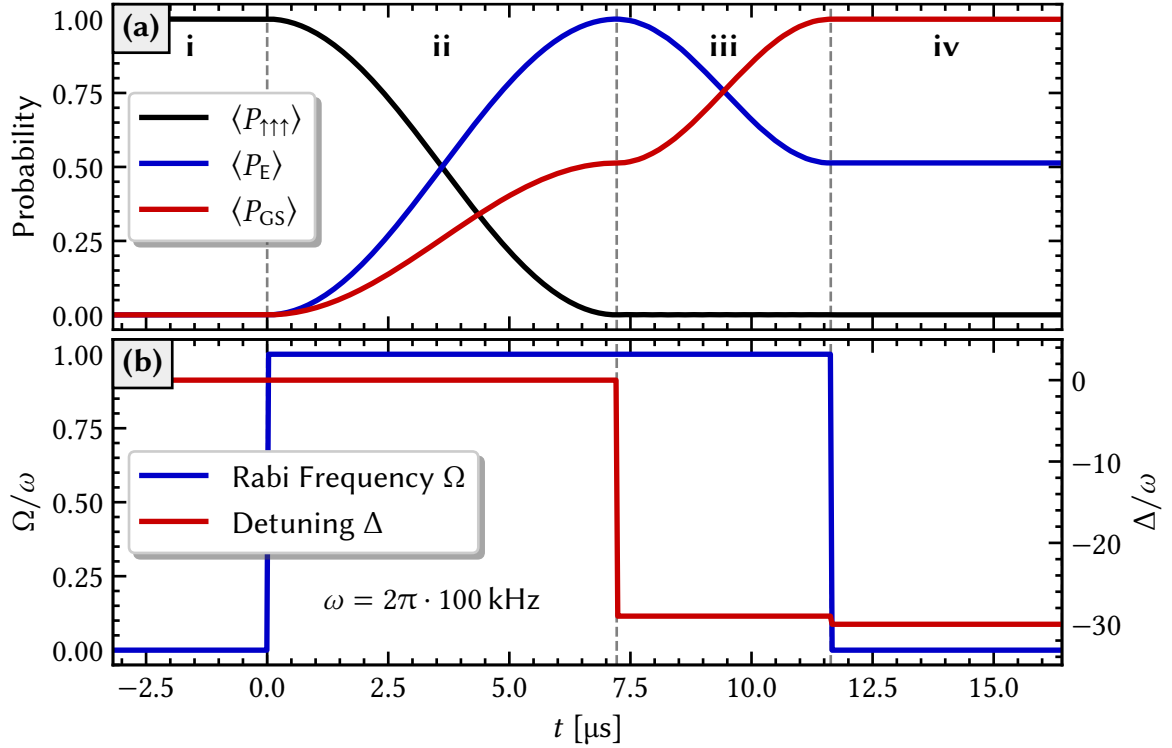


Figure IV.1.: Electronic Ground State Preparation. Example of electronic ground state preparation protocol. Time evolution is computed in the limit of zero temperature and sufficiently small electron-phonon coupling constants, idealized with $\kappa = \xi = 0$. For the trap frequency a value of $\omega = 2\pi \cdot 100 \text{ kHz}$ is assumed and for the detuning $\Delta = -2\pi \cdot 3 \text{ MHz}$. Note that these are exemplary selected values. With the Roman numerals (i-iv) the different time domains are labelled. For the whole duration of the protocol it is assumed that the 0th order dipole-dipole interaction energy has the value $\mu = 30\omega$. For the complete evolution time the full Hamiltonian, given in Eq. (IV.2), is used to evolve the initial state. In both (a) and (b), dashed light lines separate the different time domains. Panel (a) shows different probabilities: In black, the probability for being in the state $|\downarrow\downarrow\downarrow\rangle$ is shown. For the simulation we have set $\Omega = 0.2\omega$. In blue, the probability of being in $|E\rangle$ (symmetric linear combination of single excited electronic states) is plotted. Finally, in red, the expectation value of the projector to the electronic ground state, proportional to $|O\rangle - |E\rangle$, is provided. The quantity P_a is the projector to the state $|a\rangle = |\uparrow\uparrow\uparrow\rangle, |E\rangle, |GS\rangle$. (b) The blue curve describes the sequence of applied laser pulses and the red curve represents the time dependence of the laser detuning. Please note the two different vertical axis and that both panel (a) and (b) have the same time axis.

array, the system switches solely between $|E\rangle$ and $|O\rangle$. Mathematically this is seen by performing an appropriate rotation of the basis ($|1\rangle, \dots, |6\rangle$). As a result, we find the dynamics to be governed by a non-resonant two-level system of the form

$$H^{(\text{iii})} \big|_{\delta_{fe}/\Delta \ll 1} \approx \begin{bmatrix} 0 & 2\Omega \\ 2\Omega & \delta_{fe} \end{bmatrix}, \quad (\text{IV.5})$$

where we have chosen the basis ($|E\rangle, |O\rangle$) to represent the Hamiltonian. Note that the detuning is given by the facilitation error introduced earlier. Now its purpose is clear: If we want to prepare

the electronic ground state $(|O\rangle - |E\rangle)/\sqrt{2}$, we have to make sure that the correct phase is found between the two states $|O\rangle$ and $|E\rangle$, namely π . For $\delta_{fe} = 0$, we would be restricted to the phase 0. By varying δ_{fe} we accomplish a freely selectable phase, as δ_{fe} defines the direction of motion on the Bloch sphere. The general result for time evolution of the Bloch vector is given in Supp. S.6 as a reference. To obtain the electronic ground state, we set $\delta_{fe} = 4\Omega$, in agreement with the assumption $|\delta_{fe}/\Delta| \ll 1$, and apply a laser pulse of duration $\pi/(4\sqrt{2}\Omega)$. The factor $2\sqrt{2}$ is a result of the system evolving with the effective Rabi frequency $\sqrt{(2\Omega)^2 + \delta_{fe}^2}/4$, cf. Supp. S.6. At the turn-off (time) point of the laser pulse (transition to domain iv in Fig. IV.1), we rapidly decrease the facilitation error, δ_{fe} , to zero. Note that we do not need to have $\Omega = 0$ at the end of the pulse, in fact, we can set it to any desired value respecting the condition $\Omega/|\Delta| \ll 1$.

Before we continue, it should be emphasized what we have achieved. A protocol was developed which allows to prepare the electronic ground state for arbitrary (but still physical) Ω in the regime where the electron-phonon coupling strengths κ and ξ are assumed small (in magnitude), compared to the trapping frequency ω . It is important to notice that for any $\Omega > 0$, the correct ground state is given by the state we have prepared. Therefore, this state is adiabatically coupled to the motional ground state of the system, for finite κ and ξ . Next, we want to make use of this property to complete the protocol. Remember that our goal is to prepare the motional ground state of the triangle at finite κ and ξ .

In order to prepare the motional ground state of the triangular array, we consider linear adiabatic ramps of the form

$$\kappa(t) = \kappa_{\max} r(t), \quad \xi(t) = \xi_{\max} r(t), \quad (IV.6)$$

where we define $r : [0, t_{\max}] \rightarrow [0, 1]$, $r(t) = t/t_{\max}$. Such ramps may be implemented by appropriate time dependent changes of the, experimentally accessible, microwave parameters determining the values of κ and ξ ; cf. Subsec. I.2.1. The idea is to slowly increase the electron-phonon coupling strengths to the desired values. It is crucial that we know that the state $(|O\rangle - |E\rangle)/\sqrt{2}$ is the ground state for $\Omega > 0$ and $\kappa, \xi = 0$. Therefore, if we increase κ and ξ slowly, the adiabatic theorem, Ref. [80], ensures that we (approximately) stay in the ground state over time, even as the coupling parameters increase. Hence, we end up with the motional ground state of the triangle, i.e. the ground state of the triangle for finite electron-phonon coupling constants. To get some intuition on the necessary time scales, Fig. IV.2 shows results from simulation of the adiabatic ramps at the same trap frequency, $\omega = 100$ kHz, as used to create Fig. IV.1. Here we show two observables: In blue, we show the time evolution of the probability to be in the motional ground state at the maximal electron-phonon coupling constants, $\kappa_{\max}$ and $\xi_{\max}$, which the adiabatic ramps in panel (d) maximally reach. In lighter blue the optimal value, which we want to achieve, is indicated. For the ground state probability it is of course 1. If the blue curve reaches 1, we have achieved our goal of preparing the motional ground state. Errors are introduced because the ramps are not infinitely slow. As $t_{\max} \rightarrow \infty$, the blue curve is able to reach 1, however, after an infinite amount of time. The purpose of the plots is to show how well one can do on typical experimental time scales of order $\sim 10 \mu\text{s}$. Of course, the time it takes to prepare the ground state to some given precision is highly dependent on the parameters choices for $\kappa_{\max}$ and $\xi_{\max}$. For large $\kappa_{\max}$ and nearly critical $\xi_{\max}$, many phonons are needed to describe the large displacements. Therefore, it takes much slower ramps, because the ground state is made up of a greater mix of states. From this perspective, an increase in $\kappa_{\max}$ and $\xi_{\max}$ goes hand-in-hand with a decrease in Ω . Therefore, we expect that we need slower ramps in order to prepare the ground state in panel (a) as compared to panel (c), both located in Fig. IV.1. This is exactly what is evident in the simulations by the discrepancy of the blue curve to the dashed light blue line. In panels (a-c) also the order parameter o_{JT} is shown. The dashed light red line shows the expected value of this

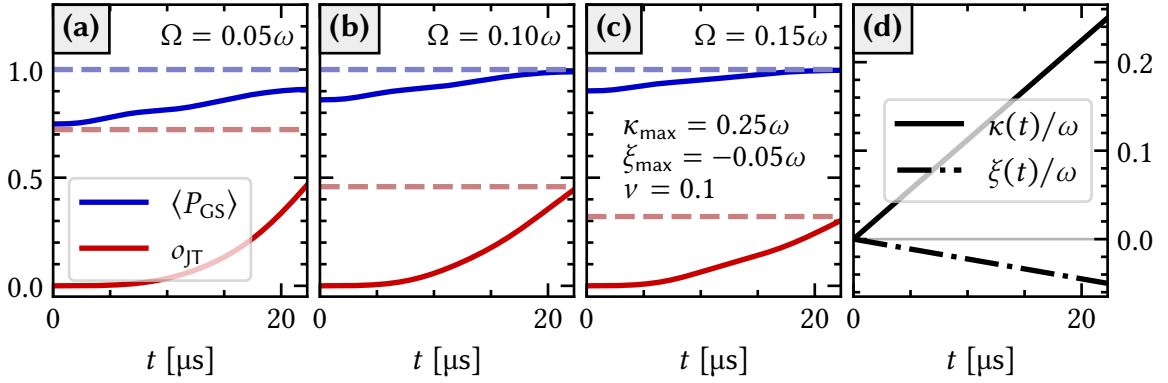


Figure IV.2.: Adiabatic State Preparation. Preparation of the motional ground state of the triangle using adiabatic ramps to increase the electron-phonon coupling constants κ and ξ . We assume a trapping frequency of $\omega = 2\pi \cdot 100$ kHz and start in the initial state $(|O\rangle - |E\rangle)/\sqrt{2}$, resulting from the electronic ground state preparation protocol. The ramps for κ and ξ are defined in Eq. (IV.6). (a-c) Time evolution for different Rabi frequencies Ω . Values are given in the top right of the corresponding panel. Panel (c) shows the maximal values of the adiabatic ramps, κ_{max} and ξ_{max} as well as $\nu = x_0/d$. In blue, the probability to be in the motional ground state, $\langle P_{\text{GS}} \rangle$, for $\kappa = \kappa_{\text{max}}$ and $\xi = \xi_{\text{max}}$ is plotted (P_{GS} is the projector the ground state). The red curves show time evolution of the order parameter o_{JT} , defined in Eq. (IV.1). Dashed lines in lighter colours represent the values the blue and red lines should approach, if the adiabatic ramps would be infinitely slow, i.e. the values the quantities would take if the exact motional ground would be prepared by our protocol. Finally, note that all panels (a-c) have the same x - and y -axis and maximally 5 excitations per mode were allowed. (d) Adiabatic ramps in κ and ξ , defined in Eq. (IV.6). The final parameters where the ramps arrive at time t_{max} are given in panel (a). The time-axis is the same as for panels (a-c).

observable at time t_{max} , where the ramps are finished. Red curves show the time evolution of o_{JT} , where the expectation value in Eq. (IV.1) is taken in the time-evolved initial state $(|O\rangle - |E\rangle)/\sqrt{2}$. For slow enough ramps, the final state should be the correct ground state, such that the dashed light red line should be reached. In panel (a) we can see that this is not the case, because the ramps are still too fast. For larger Ω in panels (b) and (c), we grasp the value of o_{JT} practically correctly. From this figure we can imagine the time scales of the ramps which are experimentally necessary in order to achieve reasonable results. Generally, it is desirable to keep the ramping times as short as possible in order to save costly coherent evolution time in the simulator. Because close to the instabilities found in Chap. II the ramps must be slow, it is very important to know where the critical coupling constants lay before approaching an experiment. In particular, it illustrates that although the critical value ξ_c may not be reached, because the atoms leave the traps beforehand, the knowledge of the value of ξ_c is still of practical relevance.

To conclude this chapter we stress that the here developed strategy for state preparation shall be seen as a suggestion. There may be other, better techniques, not relying on adiabatic ramps. As an example, such an option could be radio frequency spectroscopy, similar to the procedure applied in Ref. [53]. The difficulty in this approach would be to prepare a state which admits a large Franck-Condon factor with the desired motional ground state, Ref. [81]. Nevertheless, we have shown in this chapter that, in principle, we are able to prepare molecular states in our system, via standard experimental techniques, such as laser pulses and adiabatic ramps in Rydberg atom systems. Moreover, the simulated measurement of o_{JT} has demonstrated that we are able to use our

simulator to extract physically meaningful observables.

V. Conclusion and Outlook

In this thesis we examined the possibility to use optically trapped atom arrays for the purpose of simulating molecular dynamics in a quantum simulator. In the first chapter, we investigated how such a simulator could be approached conceptually. Making use of strong dipole-dipole interactions between Rydberg atoms, coupling between electronic and vibrational degrees of freedom is caused by the distance dependence of these interactions, Ref. [43]. We gave a brief overview on how this coupling can be tailored with the help of microwave fields, such that we are able to control the simulation environment. Finally, we arrived at a Hamiltonian describing our simulator, which is valid for any possible array shape, Ref. [11–13].

Motivated by the general result for the Hamiltonian, we tested if the simulator works as intended. To this end we took three simple array shapes. We reported unexpected vibrational signatures, unveiled by rewriting the Hamiltonians for the arrays in vibrational modes, customized to their symmetries. First simulations showed the significance of the various vibrational modes appearing in our systems. For all three array shapes we established emergence of instabilities, caused by electron-phonon coupling. In spite of dealing with the general form of the Hamiltonian, we managed to obtain analytical expressions for the critical coupling strengths. This we achieved using a multimode Bogoliubov transformation to diagonalize the bosonic Hamiltonians. Numerical considerations were employed to study dynamical manifestations of the unstable regime. Analytical expressions for the ground state energy in certain regimes showcased the instabilities in a direct fashion. Our work demonstrates that electron-phonon coupling gives rise to non-trivial behaviour in our simulation framework. As a consequence, we answer the question if our simulator is capable of reproducing non-trivial phenomena with a clear ‘Yes!’. The discussion on the tetrahedral array lead us to experience that only small arrays are feasible in a theory based approach. Hence, we could confirm the relevance and necessity of a quantum simulation framework in our setting. Finally, in order to link our simulator to molecular phenomena we made use of results from Ref. [37], where the emergence of the Jahn-Teller (JT) effect could be observed, however, without contributions stemming from the Hessian of the dipole-dipole interaction potential. Such contributions were included in our framework from the start. Hence, our work is a direct extension of this result from the literature. The spectrum of the triangle revealed that the instabilities, rooted solely in the contributions from the Hessian, affect the JT regime. We could see that the JT regime is more prominent near the unstable regime and established that the JT regime can also be effectively annihilated with contributions stemming from the Hessian. Both possibilities have not been studied before in this framework. In summary, it became clear that the Hessian carries important aspects of the system. We consider this as yet another motivation to build such a simulator, because experimentally all contributions of the dipole-dipole interaction potential would make an appearance.

After we proved the simulator to be a useful tool for its purpose, focus was directed towards the Born-Oppenheimer approximation (BOA). Our goal was to explore if the simulator is well-described by a BOA. The chain of thoughts is that if we build a quantum simulator for the problem of describing molecular systems, we should verify that well established methods from the field, like the BOA, yield reasonable results in the regimes where expected. In order to tackle this question, we computed analytical expression for the ground state energy in the same parameter limits as before. Comparison of these results to the full quantum ones, allowed us to identify quantum corrections to the BOA in an analytical fashion. We found that corrections do not appear if the Hessian of the interaction potential is neglected. To explain the missing terms, we associated them to energy contributions

connected to squeezed bosonic states. In a full numerical description, we illustrated that the BOA works as expected, in the limit of fast electronic dynamics. In fact, our findings demonstrated how the BOA is able to explain the JT regime's relationship to the instabilities. Finally, we wanted to determine the role of quantum effects in our simulator. Investigated was the transition out of the JT regime, using the systems ground state energy. In the BOA we found that the transition takes place at a particular value of the laser drive, Ω . In the full quantum picture, we learned that the transition becomes 'washed-out'. From this we deduct that our simulation framework, indeed, hosts practically relevant quantum effects caused by the vibrational degrees of freedom.

In the last chapter we developed a protocol, implemented by laser pulses and adiabatic ramps in the micro wave parameters, which allows to prepare the triangles motional ground state. Our protocol displays that it is possible to extract relevant observables not only on a theoretical basis.

A general task for future works would be to clarify how the results of this thesis, i.e. the instabilities and quantum corrections, are linked to real molecules. Perhaps it is possible to bridge dissociation energies of certain molecular shapes to their critical coupling strengths. With the emergent quantum corrections and their practical relevance, one could try to investigate the quantum-to-classical transition in an experiment, where the transition out of JT regime is inspected for increasingly large lattices of triangles. It would be interesting to see if the transition becomes more 'sudden' in larger lattices. Apart from simulating molecular dynamics, one could make use of the tetrahedron as a sensor. If one manages to couple the detuning of the driving laser to an external field (e.g. electric, magnetic, gravitational, ...), the presented ground state preparation protocol could be run while the field is active. Because each atom experiences a slightly different ground to Rydberg state excitation probability, the displacements in the symmetry broken JT regime could be evaluated to infer the external field's gradient in 3-d. space.

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S. Supplemental Material

S.1. Derivation of Solutions for x_{ph}

We first solve for *all* solutions of Eq. (II.33) from the main text and later determine which solutions are physical. As a reminder, we have already seen that we may assume $u > 0$, without loss of generality. Further, recall that $\alpha > 0$. Different regimes are distinguished. First assume $\gamma = 0$. Then we need to solve $-\alpha x = 0$. Trivially, this means $x = 0$ because $\alpha > 0$. The solution in this case is trivially given by $v = 0$ and thus $u = 1$, which is a physical solution, because the bosonic algebra is conserved as $|u|^2 - |v|^2 = 1^2 - 0^2 = 1$. We conclude $x_{\text{ph}}(0, \alpha) = 0$. Now let $\gamma \neq 0$. Therefore, we may divide by γ to obtain the equation

$$x^2 - 2(1 + \zeta)x + 1 = 0, \quad \zeta = \frac{\alpha}{2\gamma}. \quad (\text{S.1})$$

We distinguish two regimes, $(1 + \zeta)^2 > 1$ and $(1 + \zeta)^2 \leq 1$. This distinction allows to summarize *all* possible solutions, $x_{\pm}(\zeta)$, in the following expression:

$$x_{\pm}(\zeta) = \begin{cases} 1 + \zeta \pm \sqrt{(1 + \zeta)^2 - 1} & ; \quad (1 + \zeta)^2 > 1 \\ 1 + \zeta \pm i\sqrt{1 - (1 + \zeta)^2} & ; \quad (1 + \zeta)^2 \leq 1 \end{cases} \quad (\text{S.2})$$

We now need to determine whether these solutions are physical or not, i.e. allow results for u and v obeying the symplecticity relation $|u|^2 - |v|^2 = 1$. Clearly, if $|x_{\pm}| = u/|v| \geq 1$, it is not a valid solution, because $u^2(1 - |x_{\pm}|^2) \leq 0$. However, we have solutions whenever $x_{\pm}$ lays strictly within the unit circle. This criterion is much simpler to check after taking the absolute value of Eq. (S.2), resulting in

$$|x_{\pm}(\zeta)| = \begin{cases} |1 + \zeta \pm \sqrt{(1 + \zeta)^2 - 1}| & ; \quad (1 + \zeta)^2 > 1 \\ 1 & ; \quad (1 + \zeta)^2 \leq 1 \end{cases} \quad (\text{S.3})$$

Evidently, there are no physical solutions in case $(1 + \zeta)^2 \leq 1$, or, equivalently, $\zeta \in [-2, 0]$. Consequently, possible solutions lay solely in the regime $\zeta \in (-\infty, -2) \cup (0, \infty)$. Suppose $\zeta \in (-\infty, -2)$. Clearly the solution $|x_{-}(\zeta)| > 1$. However, the solution x_{+} is always a solution in this regime, as can be seen by taking the derivative with respect to ζ :

$$\frac{dx_{+}}{d\zeta}(\zeta) = 1 + \frac{1 + \zeta}{\sqrt{(1 + \zeta)^2 - 1}} = 1 - \sqrt{1 + \frac{1}{(1 + \zeta)^2 - 1}} < 0 \quad (\text{S.4})$$

So on the domain $(-\infty, -2)$, x_{+} is strictly decreasing. From the boundary relations $x_{+}(\zeta \rightarrow -\infty) = 0$ and $x_{+}(-2) = -1$, we find that all solutions are physical ones and unique. Consider now the case $\zeta \in (0, \infty)$. We observe that $x_{+}(\zeta) > 1$ for all ζ . However, from the derivative

$$\frac{dx_{-}}{d\zeta}(\zeta) = 1 - \frac{1 + \zeta}{\sqrt{(1 + \zeta)^2 - 1}} = 1 - \sqrt{1 + \frac{1}{(1 + \zeta)^2 - 1}} < 0, \quad (\text{S.5})$$

we close again strict monotonic decrease on the whole open interval. Since $x_{-}(0) = 1$ and the limit $x_{-}(\zeta \rightarrow \infty) = 0$, all solutions are physical and unique. On the domain where solutions exist, one can collect them in the expression

$$x_{\text{ph}}(\alpha, \gamma) = 1 + \zeta - \frac{\zeta}{|\zeta|} \sqrt{(1 + \zeta)^2 - 1}, \quad \zeta = \frac{\alpha}{2\gamma} \in (-\infty, -2) \cup (0, \infty). \quad (\text{S.6})$$

Note that the condition of ζ being in the union of open intervals $(-\infty, -2) \cup (0, \infty)$ is equivalent to the more simple condition $\zeta^{-1} \in (-1/2, \infty) \setminus \{0\}$. Writing ζ^{-1} is valid, because $\zeta = 0$ is excluded within the assumption $\alpha > 0$. Moreover, to give a criterion for the existence of physical solutions, the case gamma $\gamma = 0$ can be included by allowing 0 as well. Therefore, we have a unique physical solution iff $2\gamma/\alpha \in (-1/2, \infty)$. Finally eye that for each parameter choice the physical solution is unique (for $u > 0$), if it exists. In combination, the results of this paragraph prove the validity of Eq. (II.34) from the main text, which was our goal.

S.2. Rotation to Normal Modes for the Tetrahedral Array

In the main text, the normal modes $\mathbf{b} = T^t \mathbf{a}$ are defined for the tetrahedral array. A possible solution for the transformation T is the following matrix:

$$T = \begin{bmatrix} \frac{1}{\sqrt{6}} & 0 & -\frac{1}{2\sqrt{3}} & -\frac{1}{2\sqrt{6}} & -\frac{1}{2\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{3}} & 0 & 0 & 0 & \frac{1}{2} \\ -\frac{1}{3\sqrt{2}} & -\frac{1}{2\sqrt{6}} & -\frac{1}{6} & -\frac{\sqrt{2}}{3} & \frac{1}{6} & \frac{1}{6\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{3} & -\frac{1}{2\sqrt{3}} & 0 & \frac{1}{2} & 0 \\ -\frac{1}{6} & \frac{1}{2\sqrt{3}} & -\frac{\sqrt{2}}{3} & \frac{1}{6} & -\frac{1}{3\sqrt{2}} & -\frac{1}{6} & \frac{1}{2\sqrt{3}} & -\frac{1}{3\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{2} & 0 & 0 \\ -\frac{1}{\sqrt{6}} & 0 & -\frac{1}{2\sqrt{3}} & -\frac{1}{2\sqrt{6}} & \frac{1}{2\sqrt{3}} & -\frac{1}{\sqrt{6}} & -\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{3}} & 0 & 0 & 0 & \frac{1}{2} \\ -\frac{1}{3\sqrt{2}} & -\frac{1}{2\sqrt{6}} & \frac{1}{6} & \frac{\sqrt{2}}{3} & \frac{1}{6} & \frac{1}{6\sqrt{2}} & -\frac{1}{\sqrt{6}} & -\frac{1}{3} & \frac{1}{2\sqrt{3}} & 0 & \frac{1}{2} & 0 \\ -\frac{1}{6} & \frac{1}{2\sqrt{3}} & \frac{\sqrt{2}}{3} & -\frac{1}{6} & -\frac{1}{3\sqrt{2}} & -\frac{1}{6} & -\frac{1}{2\sqrt{3}} & -\frac{1}{3\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{4} & 0 & 0 & \frac{1}{2\sqrt{2}} & 0 & -\frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{\sqrt{2}}{3} & -\frac{1}{2\sqrt{6}} & 0 & 0 & -\frac{1}{3} & -\frac{5}{6\sqrt{2}} & 0 & \frac{1}{6} & 0 & 0 & \frac{1}{2} & 0 \\ -\frac{1}{6} & -\frac{1}{\sqrt{3}} & 0 & 0 & -\frac{1}{3\sqrt{2}} & \frac{1}{3} & 0 & \frac{\sqrt{2}}{3} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{3}} & -\frac{1}{2\sqrt{6}} & 0 & 0 & \frac{1}{2\sqrt{2}} & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{\sqrt{6}}{4} & 0 & 0 & 0 & \frac{1}{2\sqrt{2}} & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix} \quad (\text{S.7})$$

This matrix was obtained with *MATHEMATICA*.

S.3. Triangle Hamiltonian Projected to Electronic Ground State

To obtain Eq. (II.58) from the main text the Hamiltonian in Eq. (II.11) is written in the b -modes and projected to the atomic ground state, $|\text{GS}\rangle$, with the result:

$$\begin{aligned} \langle \text{GS} | H_3 | \text{GS} \rangle = & -2\Omega + \omega \mathbf{b}^\dagger \mathbf{b} + \frac{\kappa}{6} \left(b_1 - b_2 + b_1 + \frac{1}{2}b_2 - \frac{\sqrt{3}}{2}b_3 + b_1 + \frac{1}{2}b_2 + \frac{\sqrt{3}}{2}b_3 + \text{h. c.} \right) \\ & + \frac{\xi}{12} \left((b_1 - b_2 + \text{h. c.})^2 + \left(b_1 + \frac{1}{2}b_2 - \frac{\sqrt{3}}{2}b_3 + \text{h. c.} \right)^2 \right. \\ & \quad \left. + \left(b_1 + \frac{1}{2}b_2 + \frac{\sqrt{3}}{2}b_3 + \text{h. c.} \right)^2 \right) \\ & + \frac{\nu\kappa}{12\sqrt{2}} \left((b_3 + b_t + \text{h. c.})^2 + \left(\frac{\sqrt{3}}{2}b_2 + \frac{1}{2}b_3 - b_t + \text{h. c.} \right)^2 \right. \\ & \quad \left. + \left(-\frac{\sqrt{3}}{2}b_2 + \frac{1}{2}b_3 - b_t + \text{h. c.} \right)^2 \right), \quad (\text{S.8}) \end{aligned}$$

where the explicit form of the electronic ground state is $\sqrt{6}|\text{GS}\rangle = |1\rangle + |3\rangle + |5\rangle - |2\rangle + |4\rangle + |6\rangle$ and the associated energy -2Ω . Note that computing the projection is equivalent to replacing each projector with a factor of $1/6$. After some straight-forward algebraic steps, we can see that many terms cancel and we finally arrive at

$$\begin{aligned} \langle \text{GS} | H_3 | \text{GS} \rangle = & -2\Omega + \omega \mathbf{b}^\dagger \mathbf{b} + \frac{\kappa}{2}(b_1^\dagger + b_1) + \frac{\xi}{4}(b_1^\dagger + b_1)^2 \\ & + \left(\frac{\xi}{8} + \frac{\nu\kappa}{8\sqrt{2}} \right) ((b_2^\dagger + b_2)^2 + (b_3^\dagger + b_3)^2) + \frac{\nu\kappa}{4\sqrt{2}}(b_t^\dagger + b_t)^2. \end{aligned} \quad (\text{S.9})$$

Note that it is clear from the start that this must be transformable in a form where mixing of modes is removed, because we have defined the modes b exactly with this property. Let h_a be the Hamiltonian in Eq. (II.28) written in the bosonic mode a . With this definition we can write the projected Hamiltonian as the following sum:

$$\begin{aligned} \langle \text{GS} | H_3 | \text{GS} \rangle = & h_{b_1} \left(\omega, \frac{\kappa}{2}, \frac{\xi}{4}, -2\Omega \right) \\ & + h_{b_2} \left(\omega, 0, \frac{\xi}{8} + \frac{\nu\kappa}{8\sqrt{2}}, 0 \right) + h_{b_3} \left(\omega, 0, \frac{\xi}{8} + \frac{\nu\kappa}{8\sqrt{2}}, 0 \right) + h_{b_4} \left(\omega, 0, \frac{\nu\kappa}{4\sqrt{2}}, 0 \right) \end{aligned} \quad (\text{S.10})$$

With this identification Eq. (II.58) from the main text follows immediately with Eq. (II.42) in the special case $n = 0$.

S.4. Born-Oppenheimer for Dumbbell

In this appendix we want to derive the large Rabi frequency, Ω , limit of the ground state energy in the Born-Oppenheimer approximation (BOA). To do so, the relevant BO energy surface (BOES) is minimized. First step is to identify the relevant BOES. Because we are interested in the limit $\Omega \rightarrow \infty$, the ground state BOES is given by the asymptotic behaviour of the general expression, Eq. (III.3), from the main text:

$$E_{\text{GS}}(q) = E_-(q) = \frac{\omega}{2} \left[\frac{1}{2} \left(Q^2 + \frac{\bar{\omega}}{\omega} \bar{Q}^2 \right) + \bar{\delta} - \sqrt{8 \frac{\Omega^2}{\omega^2} + \left(\frac{1}{2} \left(Q^2 - \frac{\bar{\omega}}{\omega} \bar{Q}^2 \right) - \bar{\delta} \right)^2} - 1 \right], \quad Q = \frac{q}{x_0} \quad (\text{S.11})$$

For definitions of the quantities with a bar on top, $\bar{\omega}$, $\bar{Q}$ and $\bar{\delta}$ the reader is referred to Eq. (III.3). Note that it is sufficient to only consider the case where $E_-(q) < E_+(q)$, because for every parameter choice κ, ξ , there exists an Ω large enough, such that the above relation holds true for any $q \in \mathbb{R}$. A simple Taylor expansion of this expression in ω/Ω around zero, yields the following scaling behaviour:

$$\lim_{\Omega \rightarrow \infty} E_-(q) \sim -\sqrt{2}\Omega - \frac{\omega}{2} + \frac{\omega}{2} \left[\frac{1}{2} \left(Q^2 + \frac{\bar{\omega}}{\omega} \bar{Q}^2 \right) - \bar{\delta} \right] + \mathcal{O}(\omega/\Omega) \quad (\text{S.12})$$

The right-hand-side is the potential energy surface the ‘particle’ with position q experiences if the Rabi frequency is set to large values with respect to ω . Therefore, to obtain the lowest possible energy, we minimize this expression in Q . Observe that $\bar{Q}$ depends linearly on Q , such that the BOES – as for the case $\Omega = 0$ – is a simple quadratic polynomial in the variable Q . Exactly this is the reason why we are able to give simple analytical expression only for $\Omega = 0$ and $\Omega \rightarrow \infty$. For arbitrary Ω , one would need to find the roots of a polynomial of degree 4. While technically possible, because

the degree is not 5 (cf. Galois Theory), the result is not really a feasible expression and definitely not an instructive one. For the simple BOES in the limit $\Omega \rightarrow \infty$, the minimum, $Q_{\min}$, is given by the relation

$$Q + \frac{2\bar{\omega}}{\omega} \bar{Q} = Q_{\min} \left(1 + \frac{2\bar{\omega}}{\omega}\right) + \frac{2\kappa}{\omega} = 0. \quad (\text{S.13})$$

In the second step, the explicit expression for $\bar{Q}$ was substituted. As a result of this equation, we find upon substitution of $\bar{\omega}$,

$$\frac{q_{\min}}{x_0} = Q_{\min} = -\frac{2\kappa}{\omega} \frac{1}{1 + \bar{\omega}/\omega} = -\frac{\kappa}{\omega} \frac{1}{1 + \xi/(2\xi_c)}, \quad \xi_c = -\frac{\omega}{4}. \quad (\text{S.14})$$

Evident from this expression is that the critical coupling strength is pushed into the unstable regime, such that the system does not take note of it anymore. For the energy we find the desired result,

$$E_{\text{GS},2}^{(\text{BO}),\Omega \rightarrow \infty} = -\sqrt{2}\Omega - \frac{\omega}{2} - \frac{\kappa^2}{2\omega} \frac{1}{1 + 2\xi/\omega}, \quad (\text{S.15})$$

which is Eq. (III.9) from the main text. For $\Omega \rightarrow \infty$, this quantity is approached asymptotically in the BOA.

S.5. Born-Oppenheimer for Triangle

First goal is to arrive at Eq. (III.14) from the main text. We assume $\Omega = 0$. In this limit the unitary transformation U , used to diagonalize the fast electronic dynamics, Eq. (III.11), is the identity – there are no off-diagonal entries to begin with. Therefore, the BO energy surfaces (BOES), $E_1, \dots, E_6$, are simply given by the diagonal entries of this Hamiltonian in the electronic basis ($|1\rangle, \dots, |6\rangle$). Another consequence of $\Omega = 0$ is that the ground state energy has degeneracy 3, Ref. [37]. The 3 BOES on the subspaces $|2\rangle, |4\rangle$ and $|6\rangle$ all have minimum energy 0, because they consist of a simple harmonic potential. On the odd-numbered subspaces the energy minimum must also be equal, so we are free to pick between the three surfaces if we are solely interested in the ground state energy. Here we choose the surface corresponding to the subspace defined by $|3\rangle$ and call it E . Note that we should not call it ‘ground state energy surface’, until we have proven that the energy is always less or equal to 0. Otherwise, the ground state surface, E_{GS} , is given by one of the surfaces that represent an undisturbed harmonic potential. From Eq. (III.11) in the main text, we copy and rewrite the following expression:

$$E(\mathbf{q}) = \frac{\omega}{2} \frac{\mathbf{q}^2}{x_0^2} - 2\omega + \sqrt{2}\kappa \left(q_1 + \frac{1}{2}q_2 - \frac{\sqrt{3}}{2}q_3\right) + \xi \left(q_1 + \frac{1}{2}q_2 - \frac{\sqrt{3}}{2}q_3\right)^2 + \frac{\nu\kappa}{\sqrt{2}} \left(\frac{\sqrt{3}}{2}q_2 + \frac{1}{2}q_3 - q_t\right)^2 \quad (\text{S.16})$$

We rewrite this object in a more convenient form:

$$E(\mathbf{q}) = \frac{\omega}{2} Q_3^{\parallel 2} + \frac{\omega}{2} Q_3^{\perp 2} + 2\kappa Q_3^{\parallel} + 2\xi Q_3^{\parallel 2} + \sqrt{2}\nu\kappa Q_3^{\perp 2} + \frac{\omega}{2} ((Q_3^1)^2 + (Q_3^2)^2) - 2\omega \quad (\text{S.17})$$

$$= \frac{\omega(1 + 4\xi/\omega)}{2} Q_3^{\parallel 2} + 2\kappa Q_3^{\parallel} + \frac{\omega(1 + 2\sqrt{2}\nu\kappa/\omega)}{2} Q_3^{\perp 2} - 2\omega \quad (\text{S.18})$$

Analogously to the operators $b_3^{\parallel,\perp}$ in Eq. (II.14), the displacements $Q_3^{\parallel,\perp}$ are defined on this subspace, completely eliminating mixing of modes. The displacements Q_3^1 and Q_3^2 are analogous to c_3^1 and c_3^2 in

Eq. (II.19). They do not play a role in the following and therefore are omitted. We see two harmonic potentials emerging, both with frequencies, dressed by electron-phonon coupling. Note that we find the correct values for ξ_c and κ_c . These values are determined by the point for which the constants in front of the parabolas in $Q_3^{\parallel,\perp}$ vanish. In the stable regime, the minimal energy is obtained by simply minimizing this quadratic polynomial. To obtain the displacement vector at the minimum, we solve

$$\nabla_{(Q_3^{\parallel}, Q_3^{\perp})} E(Q_3^{\parallel}, Q_3^{\perp}) = [\omega(1 - \xi/\xi_c)Q_3^{\parallel} + 2\kappa, \omega(1 - \kappa/\kappa_c)Q_3^{\perp}] = \mathbf{0}^t, \quad (\text{S.19})$$

with the result

$$Q_{3,\min}^{\parallel} = -\frac{2\kappa}{\omega} \frac{1}{1 - \xi/\xi_c}, \quad Q_{3,\min}^{\perp} = 0. \quad (\text{S.20})$$

Here we have once more made use of the definitions $\xi_c = -\omega/4$ and $\kappa_c = -\omega/(2\sqrt{2}\nu)$. Substituting the minimum positions $(Q_{3,\min}^{\parallel}, Q_{3,\min}^{\perp})$ back into the expression for the energy surface, E , yields the desired expression from the main text:

$$E_{\text{GS},3}^{(\text{BO}),\Omega=0} = -2\omega - \frac{2\kappa^2}{\omega} \frac{1}{1 - \xi/\xi_c} \quad (\text{S.21})$$

As a final note, we can see that this quantity is always less or equal to zero, such that we actually have $E = E_{\text{GS}}$.

A similar procedure is applied to arrive at the large Ω asymptote, Eq. (III.15). In the limit of $\Omega \rightarrow \infty$ the ground state must factorize into an electronic and vibronic factor. In fact, we should have this also for the remaining electronic states. Therefore, we may assume ∇U to become more and more irrelevant as Ω approaches infinity. One can think of it in the following way: If Ω is very large compared to all other energy scales, the digitalization procedure does not ‘care’ about the diagonal (small) contributions. As a result, we define the ground state BOES, E_{GS} , as the projection of the fast electronic Hamiltonian in Eq. (III.11) to the ground state of the Rabi Hamiltonian, i.e.

$$|\text{GS}\rangle = \frac{1}{\sqrt{6}} (|1\rangle + |3\rangle + |5\rangle - |2\rangle + |4\rangle + |6\rangle). \quad (\text{S.22})$$

After projection to the 1-d. subspace defined by $|\text{GS}\rangle$, one is left with

$$\lim_{\Omega \rightarrow \infty} E_{\text{GS}}(\mathbf{q}) \sim -2\Omega + \frac{\omega}{2} \frac{\mathbf{q}^2}{x_0^2} - 2\omega + \frac{\kappa}{\sqrt{2}} \frac{q_1}{x_0} + \frac{\xi}{4x_0^2} (2q_1^2 + q_2^2 + q_3^2) + \frac{\nu\kappa}{4\sqrt{2}x_0^2} (q_2^2 + q_3^2 + 2q_4^2) \quad (\text{S.23})$$

$$\begin{aligned} &= -2\Omega - 2\omega + \frac{\omega}{2} (1 + \xi/\omega) \frac{q_1^2}{x_0^2} + \frac{\kappa}{\sqrt{2}} \frac{q_1}{x_0} + \frac{\omega}{2} (1 + \nu\kappa/(\sqrt{2}\omega)) \frac{q_1^2}{x_0^2} \\ &\quad + \frac{\omega}{2} (1 + \xi/(2\omega) + \nu\kappa/(2\sqrt{2}\omega)) \frac{q_2^2 + q_3^2}{x_0^2} \end{aligned} \quad (\text{S.24})$$

Observe the correspondence of each term to one of the square roots in the correct result in Eq. (II.58). Minimization of this quadratic function yields only non-zero displacements in q_1 , because of the shifted critical values. In particular we find

$$\frac{q_{1,\min}}{x_0} = -\frac{\kappa}{\sqrt{2}\omega} \frac{1}{1 + \xi/(4\xi_c)}. \quad (\text{S.25})$$

Using this result, the in the main text stated expression for the energy is obtained:

$$E_{\text{GS},3}^{\text{BO},\Omega \rightarrow \infty} = -2\Omega - 2\omega - \frac{\kappa^2}{4\omega} \frac{1}{1 - \xi/\xi_c} \quad (\text{S.26})$$

S.6. Bloch Vector Dynamics

Here we give the general result for the dynamics of the Bloch vector, described by the most general two-level Hamiltonian

$$H_{2\text{-lvl.}} = \begin{bmatrix} 0 & \Omega \\ \Omega^* & \Delta \end{bmatrix}. \quad (\text{S.27})$$

In this expression $\Delta \in \mathbb{R}$ and $\Omega \in \mathbb{C}$ are arbitrary constants. The solution for the time-evolution operator U , evolving a state from time t_0 to t , is given by the expression

$$\begin{aligned} e^{i\Delta(t-t_0)/2} U(t, t_0) = & \cos\left(\sqrt{1 + \frac{\phi^2}{4}} |\Omega|(t - t_0)\right) \sigma_0 + \frac{i\phi}{2} \frac{\sin\left(\sqrt{1 + \frac{\phi^2}{4}} |\Omega|(t - t_0)\right)}{\sqrt{1 + \frac{\phi^2}{4}}} \sigma_z \\ & - i \frac{\sin\left(\sqrt{1 + \frac{\phi^2}{4}} |\Omega|(t - t_0)\right)}{\sqrt{1 + \frac{\phi^2}{4}}} \frac{1}{|\Omega|} (\text{Re}(\Omega)\sigma_x - \text{Im}(\Omega)\sigma_y), \quad \phi = \frac{\Delta}{|\Omega|}. \end{aligned}$$

For $\Omega = 0$ and/or $\Delta = 0$ the appropriate limits yield a well-defined results. Pauli matrices are represented by the following matrices:

$$\sigma_0 = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}, \quad \sigma_x = \begin{bmatrix} & 1 \\ 1 & \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} & -i \\ i & \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$$

We define the Bloch vector $\mathbf{b}(t) = (u(t), v(t), w(t))$, with the most general initial condition $\mathbf{b}(t_0) = (u_0, v_0, w_0)$, essentially as the coordinate vector of the density matrix ρ with respect to the basis $(\sigma_0, \sigma_x, \sigma_y, \sigma_z)$:

$$\rho(t) = \frac{1}{2}(1 + u(t)\sigma_x + v(t)\sigma_y + w(t)\sigma_z). \quad (\text{S.28})$$

Note that at time t_0 the initial condition for the density matrix yields the initial value $\mathbf{b}(t_0)$ of the Bloch vector. From $\rho(t) = U(t, t_0)\rho(t_0)U^\dagger(t, t_0)$ the evolution of the Bloch vector is inferred directly. The general result is given by the time evolution of u, v and w . For the sake of short expressions we define

$$S(t, t_0) = \sin\left(\sqrt{1 + \frac{\phi^2}{4}} |\Omega|(t - t_0)\right), \quad C(t, t_0) = \cos\left(\sqrt{1 + \frac{\phi^2}{4}} |\Omega|(t - t_0)\right). \quad (\text{S.29})$$

The first component of the Bloch vector, $u(t)$, is given by

$$\begin{aligned} u(t) = u_0 & \left[C^2(t, t_0) + \frac{S^2(t, t_0)}{1 + \frac{\phi^2}{4}} \left(\frac{\text{Re}^2(\Omega)}{|\Omega|^2} - \frac{\text{Im}^2(\Omega)}{|\Omega|^2} - \frac{\phi^2}{4} \right) \right] \\ & - v_0 \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} \left[2 \frac{\text{Re}(\Omega)}{|\Omega|} \frac{\text{Im}(\Omega)}{|\Omega|} \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} - \phi C(t, t_0) \right] \\ & - w_0 \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} \left[2 C(t, t_0) \frac{\text{Im}(\Omega)}{|\Omega|} + \phi \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} \frac{\text{Re}(\Omega)}{|\Omega|} \right]. \quad (\text{S.30}) \end{aligned}$$

The second component, $v(t)$, is found to be

$$\begin{aligned}
 v(t) = & -u_0 \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} \left[\phi C(t, t_0) + 2 \frac{\text{Re}(\Omega)}{|\Omega|} \frac{\text{Im}(\Omega)}{|\Omega|} \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} \right] \\
 & + v_0 \left[C^2(t, t_0) - \frac{S^2(t, t_0)}{1 + \frac{\phi^2}{4}} \left(\frac{\phi^2}{4} + \frac{\text{Re}^2(\Omega)}{|\Omega|^2} - \frac{\text{Im}^2(\Omega)}{|\Omega|^2} \right) \right] \\
 & - w_0 \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} \left[2C(t, t_0) \frac{\text{Re}(\Omega)}{|\Omega|} - \phi \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} \frac{\text{Im}(\Omega)}{|\Omega|} \right]. \quad (\text{S.31})
 \end{aligned}$$

Finally, for the last component one arrives at

$$\begin{aligned}
 w(t) = & u_0 \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} \left[\frac{\text{Im}(\Omega)}{|\Omega|} 2C(t, t_0) - \frac{\text{Re}(\Omega)}{|\Omega|} \phi \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} \right] \\
 & + v_0 \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} \left[\frac{\text{Re}(\Omega)}{|\Omega|} 2C(t, t_0) + \frac{\text{Im}(\Omega)}{|\Omega|} \phi \frac{S(t, t_0)}{\sqrt{1 + \frac{\phi^2}{4}}} \right] \\
 & + w_0 \left(1 - 2 \frac{S^2(t, t_0)}{1 + \frac{\xi^2}{4}} \right). \quad (\text{S.32})
 \end{aligned}$$

With these results, the time evolution of the Bloch vector is fully known and the steps from the main text may be extracted by appropriate choices for $\mathbf{b}(t_0)$, Ω and Δ .

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